MILITARY SPECIFICATION

MICROCIRCUITS, DIGITAL, ADVANCED LOW-POWER SCHOTTKY, TTL, DECADE COUNTERS, MONOLITHIC SILICON

This specification is approved for use by the Department of the Air Force and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for monolithic silicon, advanced low power Schottky TTL, decade counters. Two product assurance classes and a choice of case outlines and lead finishes are provided for each type and are reflected in the complete part number.

1.2 Part number. The part number shall be in accordance with MIL-M-38510, and as specified herein.

1.2.1 Device type. The device type shall be as follows:

Device type	Circuit
01	Synchronous 4-Bit Decade Counter (Asynchronous Clear)
02	Synchronous 4-Bit Decade Counter (Synchronous Clear)
03	Synchronous 4-Bit UP/DOWN Decade Counter
04	Synchronous 4-Bit Decade Counter with 3-State Outputs
05	Synchronous 4-Bit UP/DOWN Decade Counter with 3-State Outputs

1.2.2 Device class. The device class shall be the product assurance level as defined in MIL-M-38510.

1.2.3 Case outline. The case outline shall be designated as follows:

Outline letter	Case outline (see MIL-M-38510, appendix C)
E	D-2 (16-lead, 1/4" x 7/8"), dual-in-line package
F	F-5 (16-lead, 1/4" x 3/8"), flat package
R	D-8 (20-lead, 1/4" x 1 1/16"), dual-in-line package
S	F-9 (20-lead, 1/4" x 1/2"), flat package
2	C-2 (20-terminal, .350" x .350"), square chip carrier package

1.3 Absolute maximum ratings.

Supply voltage range	-0.5 V dc to +7.0 V dc
Input voltage range	-1.5 V dc at -18 mA to +7.0 V dc
Storage temperature range	-65°C to +150°C
Maximum power dissipation (P_D) 1/:	
Device types 01 and 02	115 mW
Device type 03	137.5 mW
Device type 04	198 mW
Device type 05	176 mW
Lead temperature (soldering, 10 seconds)	+300°C
Thermal resistance, junction-to-case (θ_{JC}):	
Case R	40°C/W
Case E	50°C/W
Cases S and 2	60°C/W
Case F	70°C/W
Junction temperature (T_J)	+175°C

1/ Must withstand added P_D due to short circuit test (e.g., I_O).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Rome Air Development Center (RBE-2), Griffiss AFB, NY 13441, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:34Z
Check the source to verify that this is the current version before use.

1.4 Recommended operating conditions.

Maximum low-level output current (I_{OL}):	
Device types 01, 02, and 03	4.0 mA
Device types 04 and 05	12.0 mA
Supply voltage (V_{CC})	4.5 V dc minimum to 5.5 V dc maximum
Minimum high-level input voltage (V_{IH})	2.0 V dc
Maximum low-level input voltage (V_{IL})	0.8 V dc
Normalized fanout (each output):	
Low-level	10 maximum
High-level	20 maximum
Input clock frequency (f_{MAX}):	
+25°C:	
Device types 01, 02, and 03	30 MHz
Device types 04 and 05	20 MHz
-55/+125°C:	
Device types 01, 02, and 03	25 MHz
Device types 04 and 05	18 MHz
Width of clock pulse (t_p Clock):	
+25°C:	
Device types 01, 02, and 03	15 ns
Device types 04 and 05	25 ns
-55/+125°C	
Device types 01, 02, and 03	20 ns
Device types 04 and 05	27.5 ns
Width of Asynchronous clear pulse (t_p ACLR):	
Device types 01, 04, and 05	20 ns
Width of Asynchronous load pulse (t_pALOAD):	
Device type 04	20 ns
Setup times before clock ↑:	
DATA:	
Device types 01, 02, and 03	20 ns
Device types 04 and 05	25 ns
Synchronous clear:	
(Low):	
Device types 02, 04, and 05	20 ns
(Inactive):	
Device type 02	10 ns
Device types 04 and 05	35 ns
Asynchronous clear:	
(Inactive):	
Device types 01, 04, and 05	10 ns
Asynchronous load:	
(Inactive):	
Device type 04	10 ns
Synchronous load:	
(Low):	
Device types 03, 04, and 05	20 ns
(Inactive):	
Device types 04 and 05	35 ns
Enable P/Enable T:	
(Low):	
Device types 01, 02, 03, 04, and 05	25 ns
(High):	
Device types 01, 02, 03, and 04	25 ns
Device type 05	35 ns
UP/Down:	
Device type 03	20 ns
Device type 05	35 ns
Hold times (t_{HOLD}):	
Device types 01, 02, 03, 04, and 05	0 ns
Case operating temperature range (T_C)	-55°C to +125°C

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:34Z
Check the source to verify that this is the current version before use.

2. APPLICABLE DOCUMENTS

2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-38510 - Microcircuits, General Specification for.

STANDARD

MILITARY

MIL-STD-883 - Test Methods and Procedures for Microelectronics.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Detail specification. The individual item requirements shall be in accordance with MIL-M-38510, and as specified herein.

3.2 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-M-38510 and herein.

3.2.1 Terminal connections and logic diagrams. The terminal connections and logic diagrams shall be as specified on figures 1 and 2, respectively.

3.2.2 Truth table. The truth tables shall be as specified on figure 3.

3.2.3 Typical operational sequencs. The typical operational sequences shall be as specified on figure 4.

3.2.4 Schematic circuits. The schematic circuits shall be submitted to the preparing activity prior to inclusion of a manufacturer's device in this specification and shall be submitted to the qualifying activity as a prerequisite for qualification. All manufacturers' schematics shall be maintained and available upon request.

3.2.5 Case outlines. The case outlines shall be as specified in 1.2.3.

3.3 Lead material and finish. The lead material and finish shall be in accordance with MIL-M-38510 (see 6.5).

3.4 Electrical performance characteristics. The electrical performance characteristics are as specified in table I, and apply over the full recommended case operating temperature range, unless otherwise specified.

3.5 Electrical test requirements. The electrical test requirements for each device class shall be the subgroups specified in table II. The electrical tests for each subgroup are described in table III.

3.6 Marking. Marking shall be in accordance with MIL-M-38510.

3.7 Microcircuit group assignment. The devices covered by this specification shall be in microcircuit group number 12 (see MIL-M-38510, appendix E).

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:34Z
Check the source to verify that this is the current version before use.

TABLE I. Electrical performance characteristics.

Test	Symbol	Conditions 1/ $-55°C \le T_C \le +125°C$		Device type	Limits Min	Limits Max	Unit
High-level output voltage	V_{OH1}	V_{CC} = 4.5 V, I_{OH} = -400 µA V_{IL} = 0.8 V V_{IH} = 2.0 V		01,02,03	2.5		V
			RCO/CCO	04,05	2.5		
	V_{OH2}	V_{CC} = 4.5 V V_{IL} = 0.8 V V_{IH} = 2.0 V	I_{OH} = -1 mA Q oututs	04,05	2.4		V
Low-level output voltage	V_{OL1}	V_{CC} = 4.5 V V_{IL} = 0.8 V V_{IH} = 2.0 V	I_{OL} = 4 mA All outputs	01,02,03		0.4	V
			I_{OL} = 4 mA RCO/CCO	04,05		0.4	V
	V_{OL2}		I_{OL} = 12 mA Q outputs	04,05		0.4	V
Input clamp voltage	V_{IC}	V_{CC} = 4.5 V I_{IN} = -18 mA	T_C = +25°C	All		-1.5	V
Low-level input current	I_{IL}	V_{CC} = 5.5 V V_{IN} = 0.4 V		All	0	-200	µA
High-level input current	I_{IH1}	V_{CC} = 5.5 V V_{IN} = 2.7 V	$\overline{LOAD}$, CLK or ENT	01,02		40	µA
			ENT/$\overline{ENP}$	04		40	
			All other inputs	01,02, 03,04,05		20	
	I_{IH2}	V_{CC} = 5.5 V V_{IN} = 7.0 V	$\overline{LOAD}$, CLK or ENT	01,02		200	µA
			ENT/$\overline{ENP}$	04		200	
			All other inputs	01,02, 03,04,05		100	
Short circuit output current	I_O 1/	V_{CC} = 5.5 V V_O = 2.25 V	Q outputs	All	-30	-112	mA
			RCO/CCO outputs	All	-15	-70	mA
Output current, low-level outputs off	I_{OZL}	V_{CC} = 5.5 V V_O = 0.4 V	Q outputs	04,05		-20	µA
Output current, high-level outputs off	I_{OZH}	V_{CC} = 5.5 V V_O = 2.7 V	Q outputs	04,05		20	µA
Supply current	I_{CC} 2/	V_{CC} = 5.5 V		01,02,03		25	mA
Supply current, high-level	I_{CCH}	V_{CC} = 5.5 V		04		27	mA
				05		26	

See footnotes at end of table.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:34Z
Check the source to verify that this is the current version before use.

TABLE I. Electrical performance characteristics - Continued.

Test	Symbol	Conditions 1/ $-55°C \leq T_C \leq +125°C$	Device type	Limits Min	Limits Max	Unit
Supply current, low-level	I_{CCL}	V_{CC} = 5.5 V	04		33	mA
			05		32	
Supply current, outputs disabled	I_{CCZ}	V_{CC} = 5.5 V	04		36	mA
			05		32	
Maximum input A, clock, or count up frequency	f_{MAX}	V_{CC} = 5.0 V C_L = 50 pF R_L = 500Ω	01,02,03	25		MHz
			04,05	18		
Propagation delay time low-to-high, CLK - Q	t_{PLH1}		01,02	4	18	ns
			03	5	19	
			04	4	15	
			05	4	17	
Propagation delay time high-to-low, CLK - Q	t_{PHL1}		01,02	6	20	ns
			03	5	20	
			04	5	21	
			05	7	18	
Propagation delay time low-to-high, CLK - RCO	t_{PLH2}		01,02	8	30	ns
			03	10	32	
			04	9	35	
			05	12	31	
Propagation delay time high-to-low, CLK - RCO	t_{PHL2}		01,02	7	25	ns
			03	6	22	
			04	8	29	
			05	10	22	

See footnotes at end of table.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:34Z
Check the source to verify that this is the current version before use.

TABLE I. Electrical performance characteristics - Continued.

Test	Symbol	Conditions 1/ $-55°C \leq T_C \leq +125°C$	Device type	Limits Min	Limits Max	Unit
Propagation delay time low-to-high, CLK - CCO	t_{PLH3}	V_{CC} = 5.0 V C_L = 50 PF R_L = 500Ω	04	8	31	ns
			05	5	15	
Propagation delay time high-to-low, CLK - CCO	t_{PHL3}		04	5	20	ns
			05	6	30	
Propagation delay time low-to-high, ALOAD - Q	t_{PLH4}		04	10	38	ns
Propagation delay time high-to-low, ALOAD - Q	t_{PHL4}		04	7	27	ns
Propagation delay time low-to-high, ALOAD - RCO	t_{PLH5}		04	15	55	ns
Propagation delay time high-to-low, ALOAD - RCO	t_{PHL5}		04	12	35	ns
Propagation delay time low-to-high, ALOAD - CCO	t_{PLH6}		04	25	65	ns
Propagation delay time high-to-low, ALOAD - CCO	t_{PHL6}		04	12	42	ns
Propagation delay time low-to-high, DATA - Q	t_{PLH7}		04	8	35	ns
Propagation delay time high-to-low, DATA - Q	t_{PHL7}		04	7	27	ns
Propagation delay time low-to-high, ENT - RCO	t_{PLH8}		01,02,04	5	20	ns
			03	5	19	
			05	6	17	
Propagation delay time high-to-low, ENT - RCO	t_{PHL8}		01,02	4	16	ns
			03	3	16	
			04	4	18	
			05	4	17	

See footnotes at end of table.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:34Z
Check the source to verify that this is the current version before use.

TABLE I. Electrical performance characteristics - Continued.

Test	Symbol	Conditions 1/ $-55°C \le T_C \le +125°C$	Device type	Limits Min	Limits Max	Unit
Propagation delay time low-to-high, ENT - CCO	t_{PLH9}	V_{CC} = 5.0 V C_L = 50 pF R_L = 500Ω	04	12	35	ns
			05	5	15	
Propagation delay time high-to-low, ENT - CCO	t_{PHL9}		04	4	15	ns
			05	9	28	
Propagation delay time low-to-high, ENP - CCO	t_{PLH10}		04	5	22	ns
			05	4	14	
Propagation delay time high-to-low, ENP - CCO	t_{PHL10}		04	4	14	ns
			05	5	17	
Propagation delay time low-to-high, U/D - RCO	t_{PLH11}		03	5	25	ns
			05	9	25	
Propagation delay time high-to-low, U/D - RCO	t_{PHL11}		03	5	23	ns
			05	9	23	
Propagation delay time high-to-low, $\overline{ACLR}$ - Q	t_{PHL12}		01	8	28	ns
			04	7	28	
Propagation delay time high-to-low, $\overline{CLR}$ - RCO	t_{PHL13}		05	9	22	ns
			01	11	31	
ENABLE time to high-level output	t_{PZH}	V_{CC} = 5.0 V C_L = 50 pF $R_1 = R_L$ = 500Ω	04	5	24	ns
			05	6	21	
ENABLE time to low-level output	t_{PZL}		04	8	28	ns
			05	11	29	

See footnotes at end of table.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:34Z
Check the source to verify that this is the current version before use.

TABLE I. Electrical performance characteristics - Continued.

Test	Symbol	Conditions 1/ $-55°C \leq T_C \leq +125°C$	Device type	Limits Min	Limits Max	Unit
Disable time from high-level	t_{PHZ}	V_{CC} = 5.0 V C_L = 50 pF $R_1 = R_L$ = 500Ω	04	2	15	ns
			05	1	8	
Disable time from low-level	t_{PLZ}		04	4	20	ns
			05	3	19	

1/ The output conditions have been chosen to produce a current that closely approximates one half of the true short-circuit output current, I_{OS}.

2/ I_{CCL} is measured: (a) With the clock input high; and (b) Then again with the clock input low with all other inputs low and all outputs open.

TABLE II. Electrical test requirements.

MIL-STD-883 test requirements	Subgroups (see table III) Class S devices	Class B devices
Interim electrical parameters (pre burn-in) (method 5004)	1	1
Final electrical test parameters (method 5004)	1*,2,3,7,9,10,11	1*,2,3,7,9,10,11
Group A test requirements (method 5005)	1,2,3,7,8,9,10,11	1,2,3,7,9
Group B test requirements (method 5005) subgroup 5	1,2,3,9,10,11	N/A
Group C end-point electrical parameters (method 5005)	N/A	1,2,3
Additional electrical subgroups for group C periodic inspections	N/A	N/A
Group D end-point electrical parameters (method 5005)	1,2,3	1,2,3

*PDA applies to subgroup 1 (see 4.2c).

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:34Z
Check the source to verify that this is the current version before use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection procedures shall be in accordance with MIL-M-38510 and methods 5005 and 5007, as applicable, of MIL-STD-883, except as modified herein.

4.2 Screening. Screening shall be in accordance with method 5004 of MIL-STD-883, and shall be conducted on all devices prior to qualification and quality conformance inspection. The following additional criteria shall apply:

a. Burn-in test (method 1015 of MIL-STD-883).

(1) Test condition D, using the circuit shown on figure 5, or equivalent.

(2) T_A = +125°C minimum.

b. Interim and final electrical test parameters shall be as specified in table II, except interim electrical parameters test prior to burn-in is optional at the discretion of the manufacturer.

c. The percent defective allowable (PDA) shall be as specified in MIL-M-38510.

4.3 Qualification inspection. Qualification inspection shall be in accordance with MIL-M-38510. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and heren for groups A, B, C, and D inspections (see 4.4.1 through 4.4.4).

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-M-38510. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, and D inspections (see 4.4.1 through 4.4.4).

4.4.1 Group A inspection. Group B inspection shall be in accordance with table I of method 5005 of MIL-STD-883 and as follows:

a. Tests shall be as specified in table II herein.

b. Subgroups 4, 5, and 6 shall be omitted.

4.4.2 Group B inspection. Group B inpection shall be in accordance with table II of method 5005 of MIL-STD-883. Electrical parameters shall be as specified in table II herein.

4.4.3 Group C inspection. Group C inspection shall be in accordance with table III of method 5005 of MIL-STD-883 and as follows:

a. End-point electrical parameters shall be as specified in table II herein.

b. Steady-state life test (method 1005 of MIL-STD-8830 conditions:

(1) Test condition D, using the circuit shown on figure 5, or equivalent.

(2) T_A = +125°C minimum.

(3) Test duration: 1,000 hours, except as permitted by appendix B of MIL-M-38510 and method 1005 of MIL-STD-883.

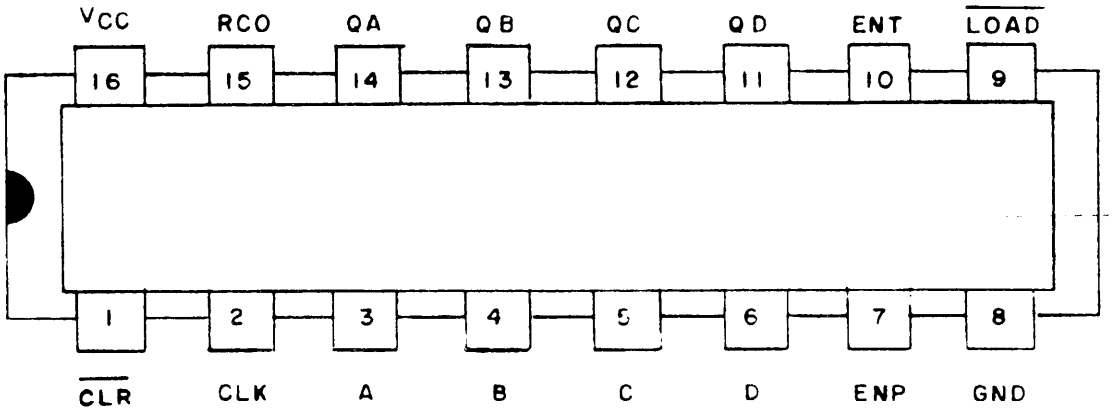
4.4.4 Group D inspection. Group D inspection shall be in accordance with table IV of method 5005 of MIL-STD-883. End-point electrical parameters shall be as specified in table II herein.

4.5 Methods of inspection. Methods of inspection shall be specified as follows:

4.5.1 Voltage and current. All voltages given are referenced to the microcircuit ground terminal. Currents given are conventional and positive when flowing into the reference terminal.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:34Z
Check the source to verify that this is the current version before use.

Device types 01 and 02
Cases E and F



Device type 03
Cases E and F

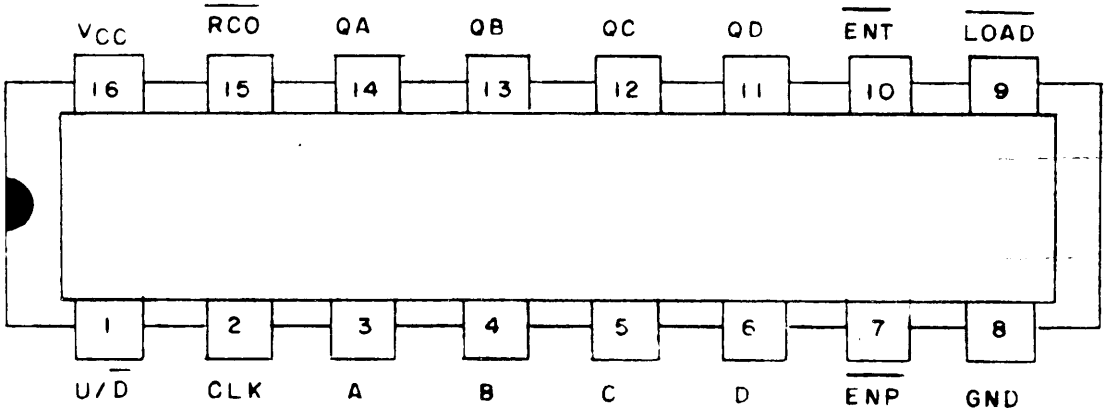


FIGURE 1. Terminal connections (top view).

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:34Z
Check the source to verify that this is the current version before use.

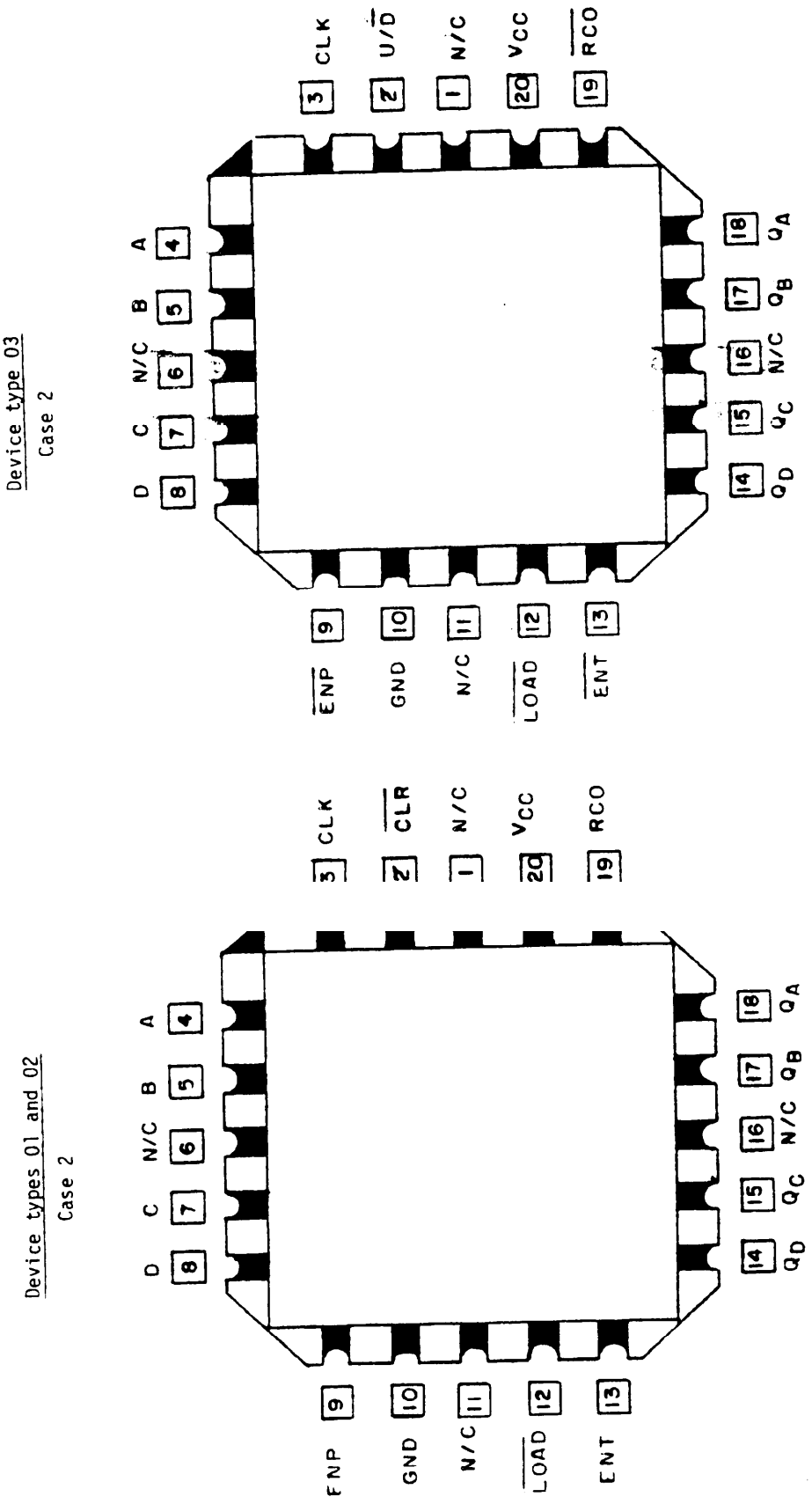


FIGURE 1. Ter 1 connections (top view) - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:34Z
Check the source to verify that this is the current version before use.

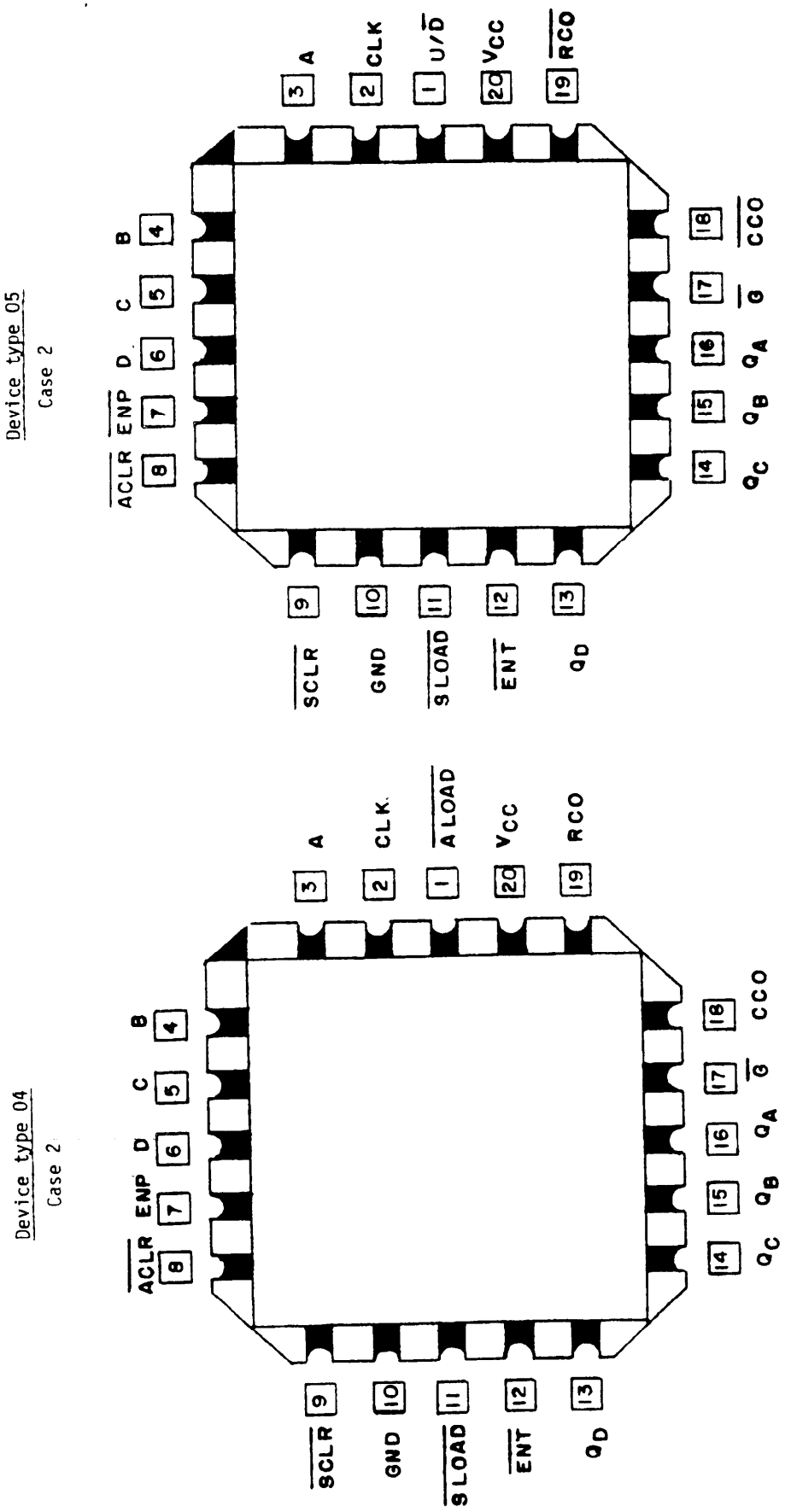
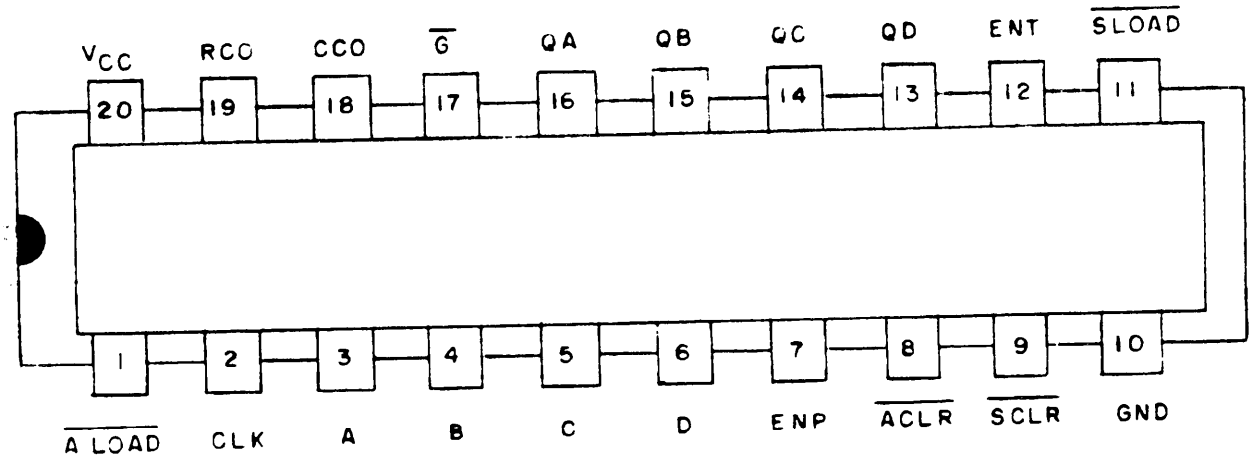


FIGURE 1. Terminal connections (top view) - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:34Z
Check the source to verify that this is the current version before use.

Device type 04
Cases E and F



Device type 05
CAses E and F

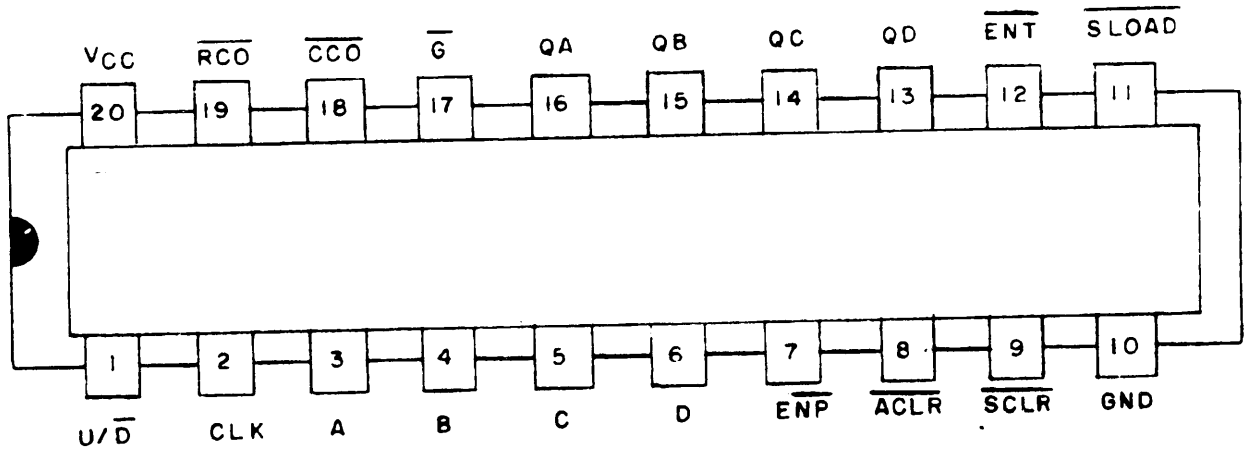


FIGURE 1. Terminal connections (top view) - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:34Z
Check the source to verify that this is the current version before use.

Device type 01

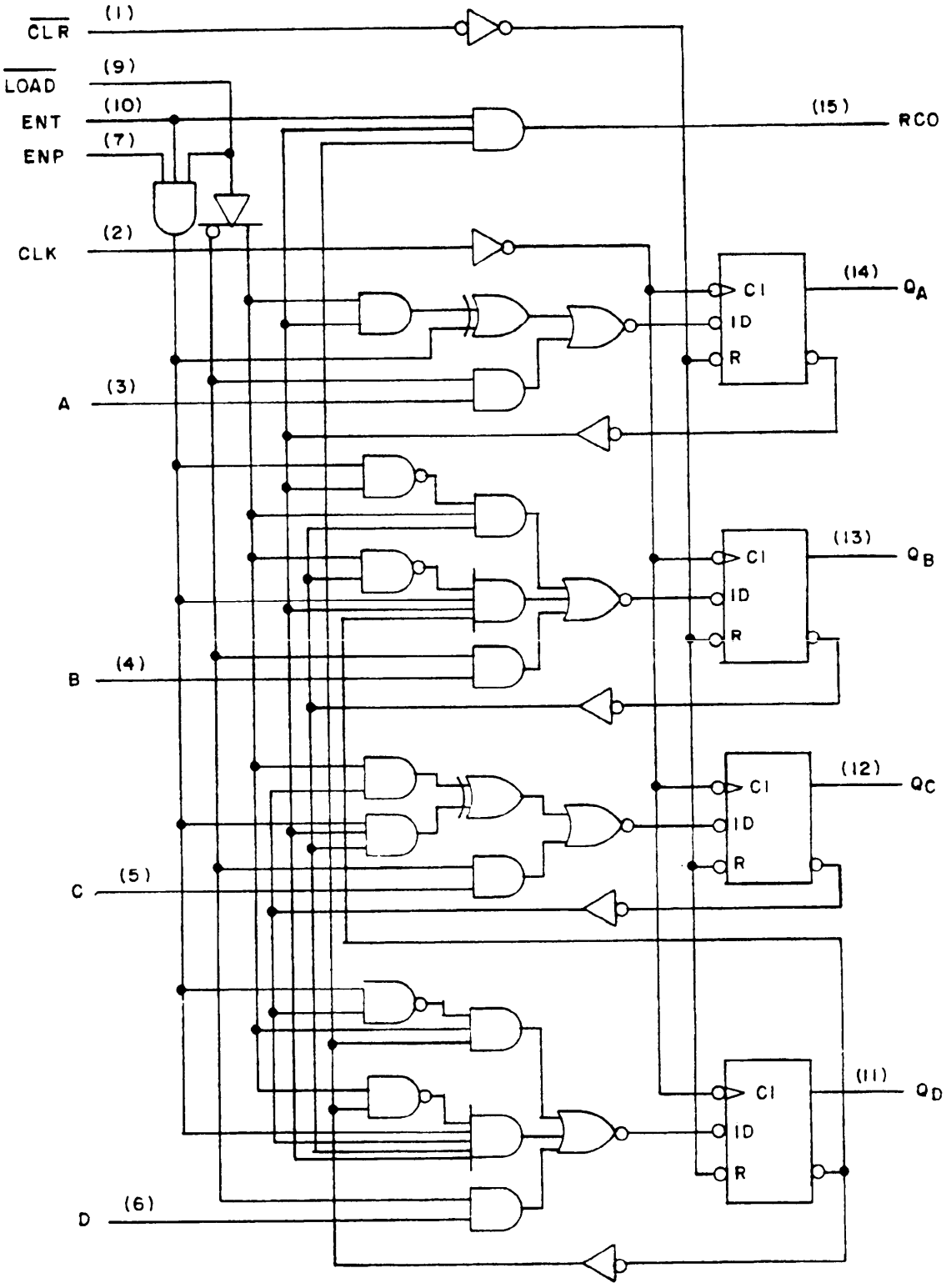


FIGURE 2. Logic diagrams.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:34Z
Check the source to verify that this is the current version before use.

Device type 02

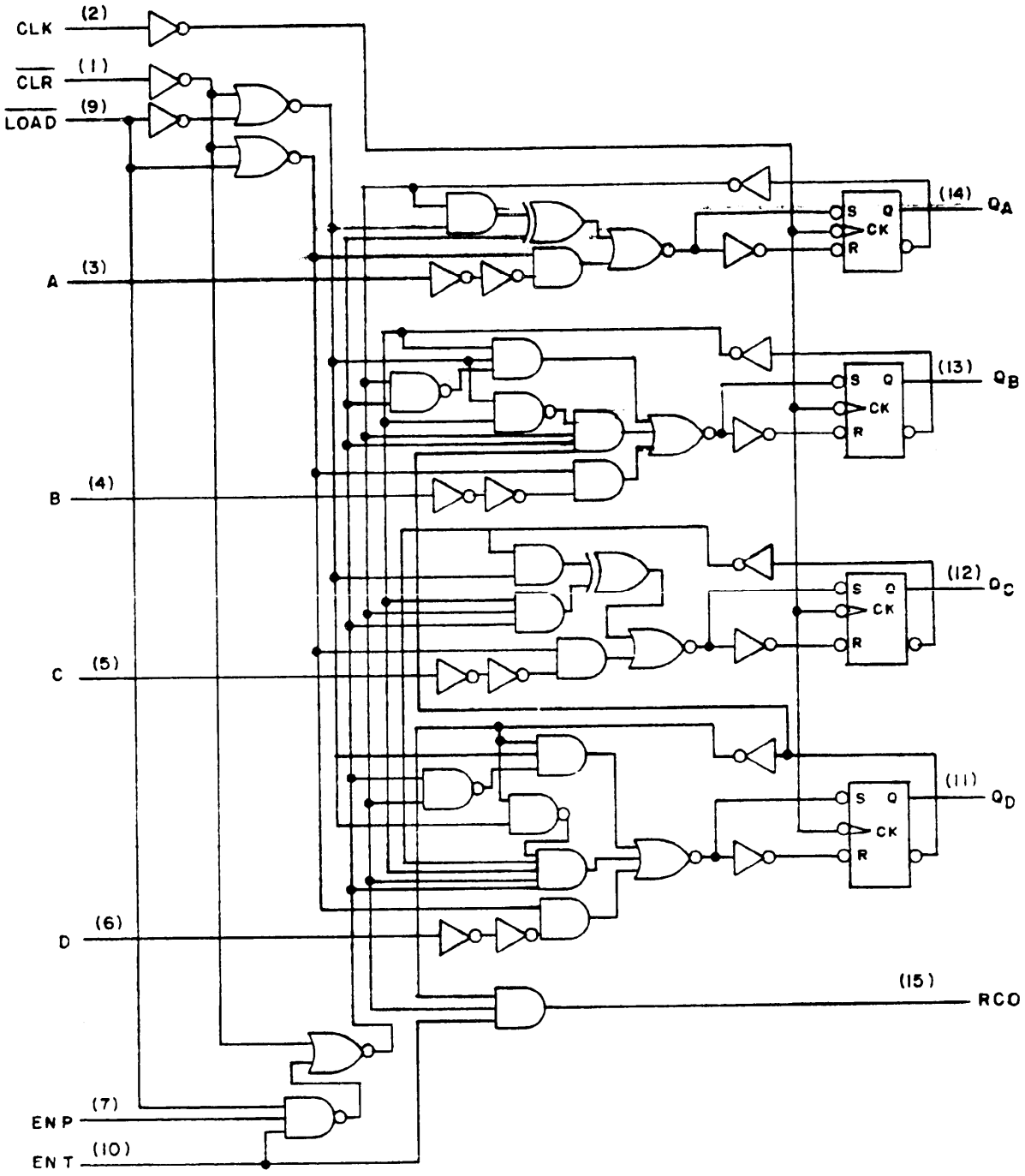


FIGURE 2. Logic diagrams - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:34Z
Check the source to verify that this is the current version before use.

Device type 03

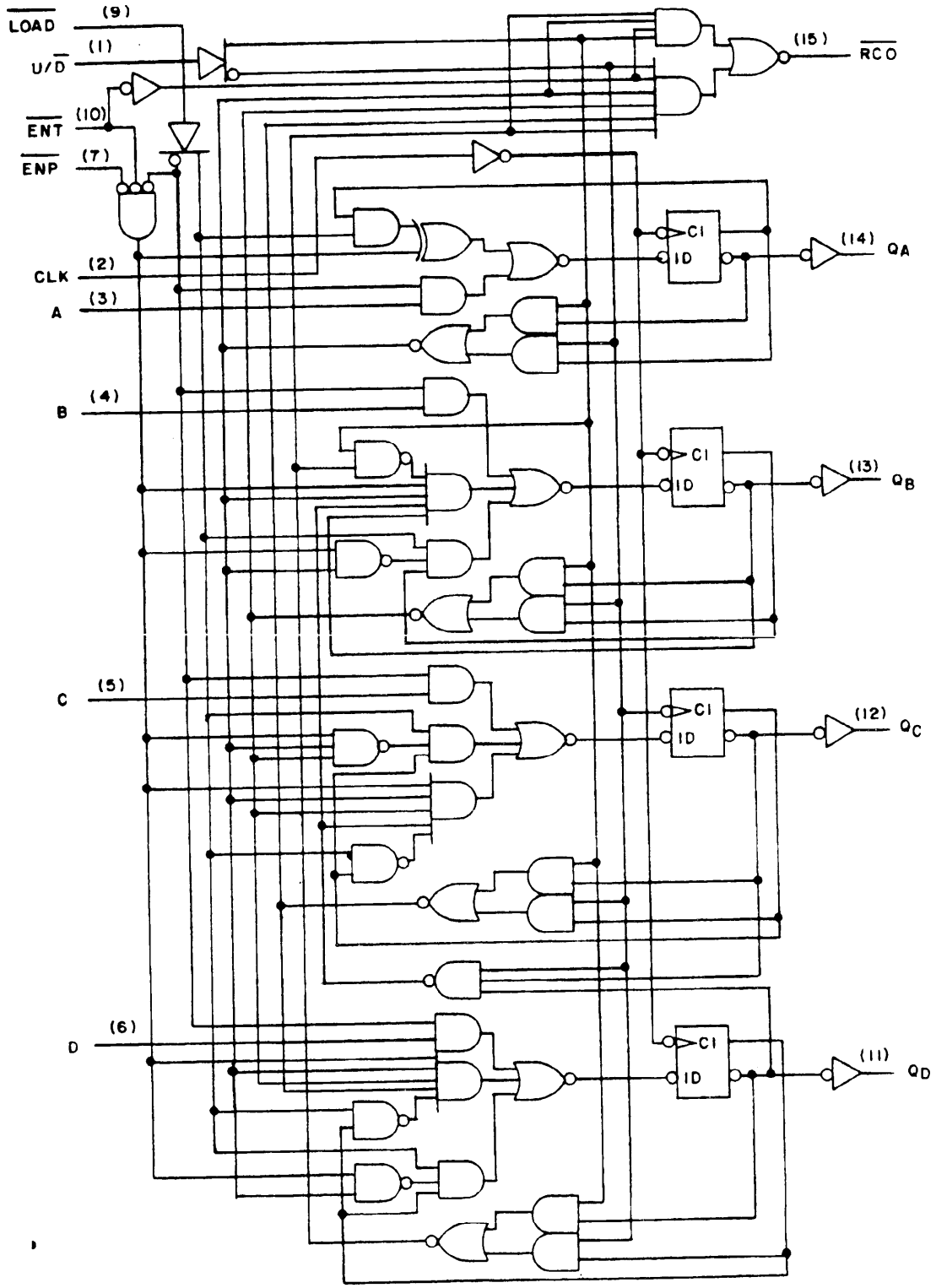


FIGURE 2. Logic diagrams - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:34Z
Check the source to verify that this is the current version before use.

Device type 04

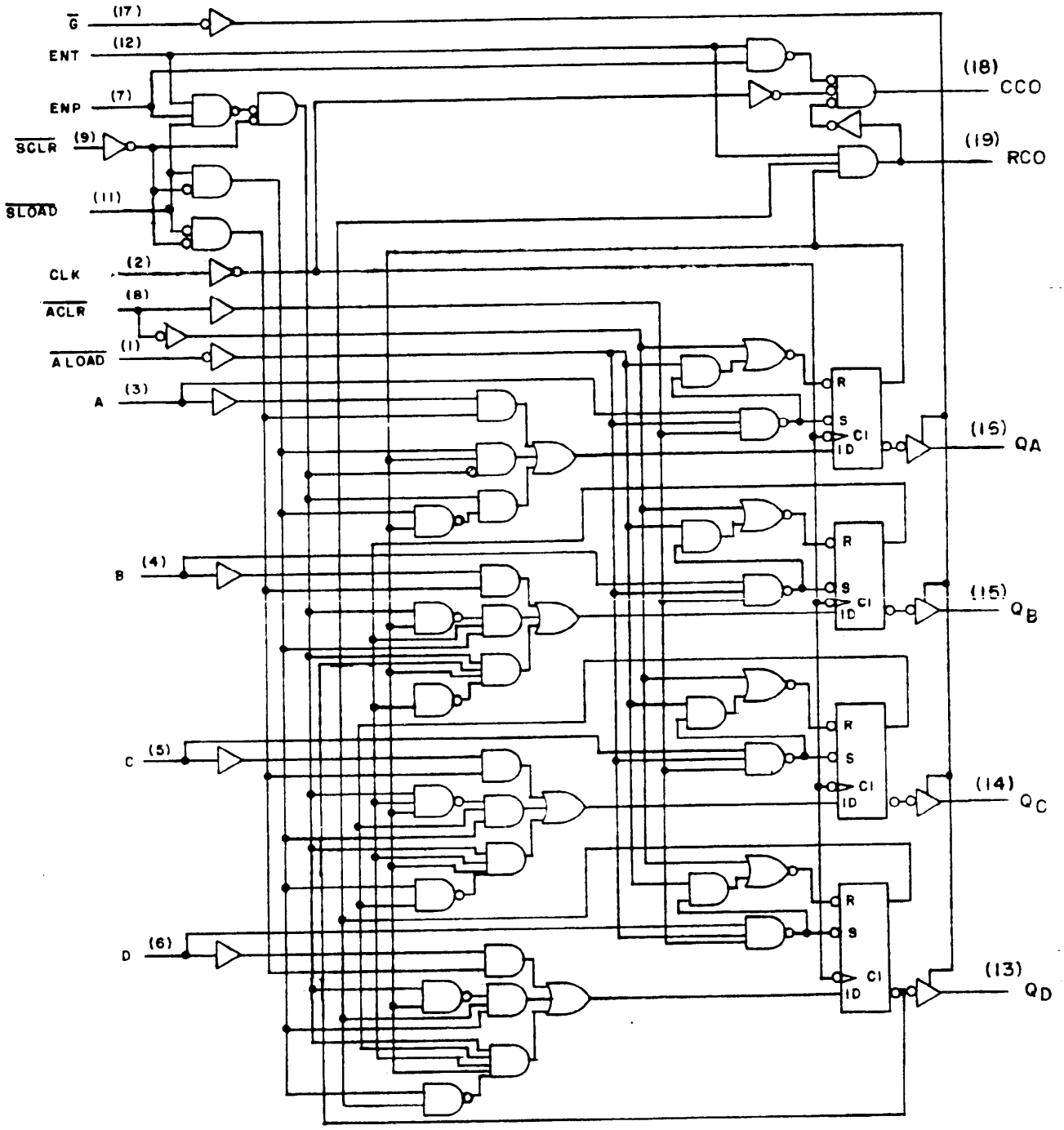


FIGURE 2. Logic diagrams - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:34Z
Check the source to verify that this is the current version before use.

Device type 05

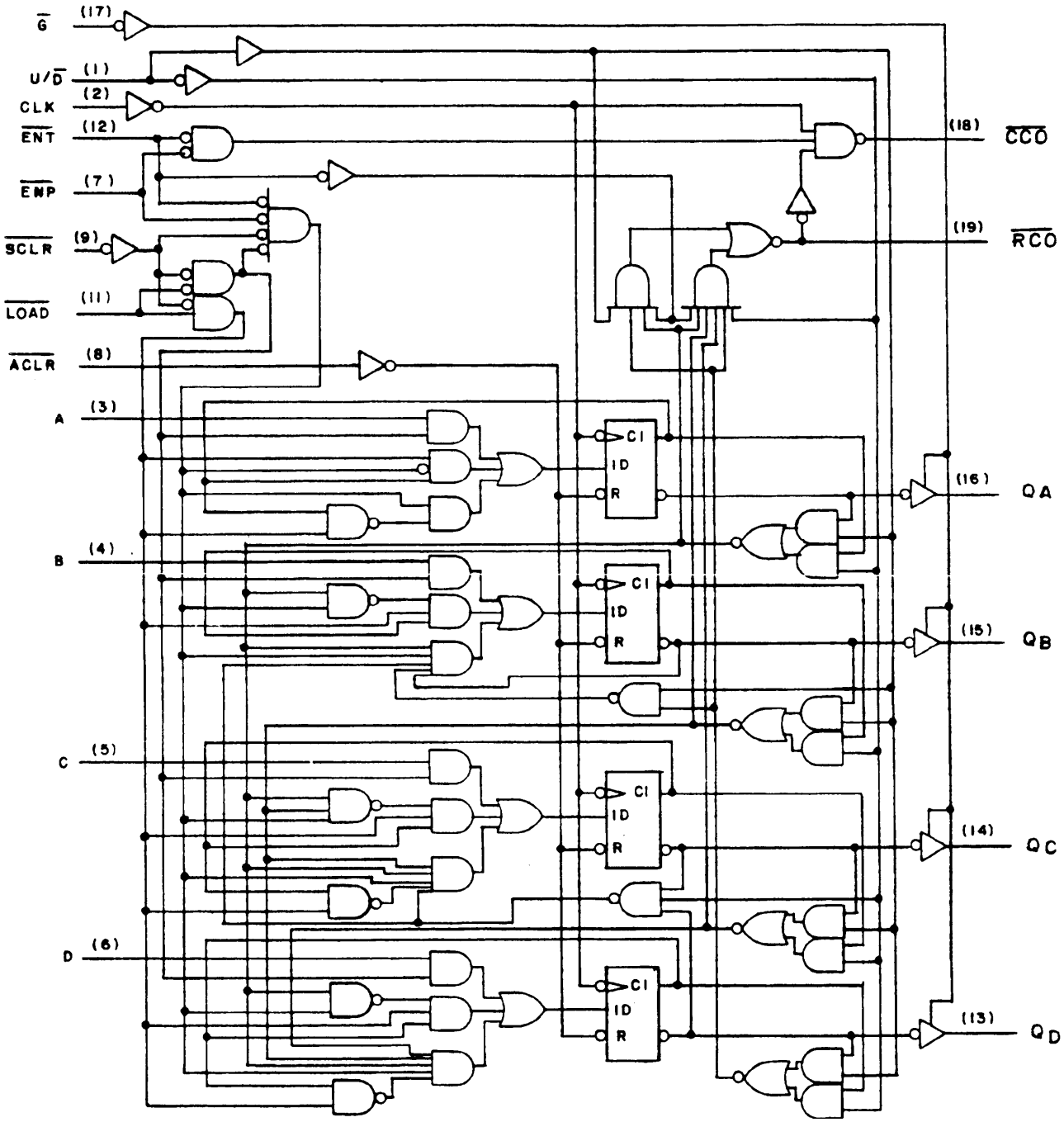


FIGURE 2. Logic diagrams - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:34Z
Check the source to verify that this is the current version before use.

SYNCHRONOUS TRUTH TABLE, DEVICE TYPES 01 AND 02

Inputs at time t_n									Outputs at time t_{n+1}				
Clock	Enable P	Enable T	$\overline{\text{Load}}$	A	B	C	D	$\overline{\text{Clear}}$	Q_A	Q_B	Q_C	Q_D	Carry output
CP	L	X	H	X	X	X	X	H	NC	NC	NC	NC	NC
CP	X	L	H	X	X	X	X	H	NC	NC	NC	NC	L
CP	H	H	H	X	X	X	X	H	Previous count plus 1 (note 1)				H if count = 9 L if count < 9
CP	X	H	L	X	X	X	X	H	A	B	C	D	H if count = 9 L if count < 9
CP	X	L	L	X	X	X	X	H	A	B	C	D	L
CP	X	X	X	X	X	X	X	L	L	L	L	L	L

ASYNCHRONOUS TRUTH TABLE, DEVICE TYPE 01

Inputs at time t_n									Outputs at time t_{n+1}				
Clock	Enable P	Enable T	$\overline{\text{Load}}$	A	B	C	D	$\overline{\text{Clear}}$	Q_A	Q_B	Q_C	Q_D	Carry output
X	X	X	X	X	X	X	X	L	L	L	L	L	L

UP COUNT SEQUENCE TABLE

Q_A (LSB)	Q_B	Q_C	Q_D (MSB)
L	L	L	L
H	L	L	L
L	H	L	L
H	H	L	L
L	L	H	L
H	L	H	L
L	H	H	L
H	H	H	L
L	L	L	H
H	L	L	H

NOTES:
1. See up count sequence table.
2. L = V_{IL} for inputs, V_{OL} for outputs.
3. H = V_{IH} for inputs, V_{OH} for outputs.
4. X = V_{IH} or V_{IL}.
5. CP = Clock pulse.
6. NC = No change.

FIGURE 3. TRUTH TABLES.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:34Z
Check the source to verify that this is the current version before use.

DEVICE TYPE 03

	Inputs at time t_n									Outputs at time t_{n+1}				
	Clock	$\overline{EN\ P}$	$\overline{EN\ T}$	$\overline{Load}$	A	B	C	D	U/$\overline{D}$	A	B	C	D	$\overline{RCO}$
Load	CP	L	L	L	X	X	X	X	X	A	B	C	D	L if count = 9 H if count ≠ 9
Count Up	CP	L	L	H	X	X	X	X	H	Previous count plus 1 (note 1)				L if count = 9 H if count ≠ 9
Count Down	CP	L	L	H	X	X	X	X	L	Previous count minus 1 (note 2)				L if count = 0 H if count ≠ 0
Inhibit	CP	H	L	H	X	X	X	X	X	No change				No change
	CP	L	H	H	X	X	X	X	X					L
	CP	H	H	H	X	X	X	X	X					L

UP COUNT SEQUENCE TABLE

Q_A (LSB)	Q_B	Q_C	Q_D (MSB)	Carry
L	L	L	L	H
H	L	L	L	H
L	H	L	L	H
H	H	L	L	H
L	L	H	L	H
H	L	H	L	H
L	H	H	L	H
H	H	H	L	H
L	L	L	H	H
H	L	L	H	L

DOWN COUNT SEQUENCE TABLE

Q_A (LSB)	Q_B	Q_C	Q_D (MSB)	Borrow
H	L	L	H	H
L	L	L	H	H
H	H	H	L	H
L	H	H	L	H
H	L	H	L	H
L	L	H	L	H
H	H	L	L	H
L	H	L	L	H
H	L	L	L	H
L	L	L	L	L

H = High voltage level
L = Low voltage level
X = Don't care

FIGURE 3. Truth tables - continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:34Z
Check the source to verify that this is the current version before use.

OPERATION	Inputs at time t_n												Outputs at time t_{n+1}					
	$\overline{G}$	$\overline{ACLR}$	$\overline{ALOAD}$	$\overline{SCLR}$	$\overline{SLOAD}$	ENT	ENP	CLK	D_A	D_B	D_C	D_D	Q_A	Q_B	Q_C	Q_D	RCO	CCO
OUTPUTS DISABLED	H	X	X	X	X	X	X	X	X	X	X	X	Z	Z	Z	Z	H if cnt = 9 L if cnt ≠ 9	note 1/
ASYNCHRONOUS CLEAR	L	L	X	X	X	X	X	X	X	X	X	X	L	L	L	L	L	L
ASYNCHRONOUS LOAD	L	H	L	X	X	X	X	X	X	X	X	X	D_A	D_B	D_C	D_D	H if cnt = 9 L if cnt ≠ 9	note 1/
SYNCHRONOUS CLEAR	L	H	H	L	X	X	X	↑	X	X	X	X	L	L	L	L	L	L
SYNCHRONOUS LOAD	L	H	H	H	L	X	X	↑	X	X	X	X	D_A	D_B	D_C	D_D	H if cnt = 9 L if cnt ≠ 9	note 1/
COUNT	L	H	H	H	H	H	H	↑	X	X	X	X	Previous count plus 1				H if cnt = 9 L if cnt ≠ 9	note 1/
INHIBIT COUNTING	L	H	H	H	H	L	X	X	X	X	X	X	No Change (hold)				H if cnt = 9 L if cnt ≠ 9	note 1/
	L	H	H	H	H	X	L	X	X	X	X	X						
	L	H	H	H	H	L	L	X	X	X	X	X						

NOTES:

1/ CCO produces a high level pulse for a duration equal to that of the low level of the clock when RCO is high and the counter is enabled, otherwise CCO is low.

COUNT UP SEQUENCE					
Q_D	Q_C	Q_B	Q_A	RCO	CCO
L	L	L	L	L	L
L	L	L	H	L	L
L	L	H	L	L	L
L	L	H	H	L	L
L	H	L	L	L	L
L	H	L	H	L	L
L	H	H	L	L	L
L	H	H	H	L	L
H	L	L	L	L	L
H	L	L	H	H	1/

FIGURE 3. Truth tables - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-02-21T14:34Z
Check the source to verify that this is the current version before use.

OPERATION	Inputs at time t_n												Outputs at time t_{n+1}					
	$\overline{G}$	$\overline{ACLR}$	$\overline{SCLR}$	$\overline{LOAD}$	$\overline{ENT}$	$\overline{ENP}$	U/$\overline{D}$	CLK	A	B	C	D	Q_A	Q_B	Q_C	Q_D	$\overline{RCO}$	$\overline{CCO}$
Q OUTPUTS DISABLED 3/	H	X	X	X	X	X	X	X	X	X	X	X	Z	Z	Z	Z	L if cnt = 9 H if cnt ≠ 9	note 1/
ASYNCHRONOUS CLEAR	L	L	X	X	X	X	X	X	X	X	X	X	L	L	L	L	H	H
SYNCHRONOUS CLEAR	L	H	L	X	X	X	X	↑	X	X	X	X	L	L	L	L	H	H
LOAD	L	H	H	L	X	X	X	↑	X	X	X	X	A	B	C	D	L if cnt = 9 H if cnt ≠ 9	note 1/
COUNT UP	L	H	H	H	L	L	H	↑	X	X	X	X	Previous count plus 1				L if cnt = 9 H if cnt ≠ 9	note 1/
COUNT DOWN	L	H	H	H	L	L	L	↑	X	X	X	X	Previous count minus 1				L if cnt = 0 H if cnt ≠ 0	note 1/
INHIBIT COUNT	L	H	H	H	X	H	X	X	X	X	X	X	No Change (HOLD)				note 2/	note 1/
	L	H	H	H	H	X	X	X	X	X	X	X						
	L	H	H	H	H	H	X	X	X	X	X	X						

NOTES:

1/ CCO produces a low level pulse for a duration equal to that of the low level of the clock when RCO is low and the counter is enabled, otherwise CCO is high.
2/ RCO produces a low level pulse while the count is 0 when counting down, or when the count is 9 when counting up.
3/ Counting continues.

COUNT UP SEQUENCE					
Q_D	Q_C	Q_B	Q_A	$\overline{RCO}$	$\overline{CCO}$
L	L	L	L	H	H
L	L	L	H	H	H
L	L	H	L	H	H
L	L	H	H	H	H
L	H	L	L	H	H
L	H	L	H	H	H
L	H	H	L	H	H
L	H	H	H	H	H
H	L	L	L	H	H
H	L	L	H	L	1/

COUNT DOWN SEQUENCE					
Q_D	Q_C	Q_B	Q_A	$\overline{RCO}$	$\overline{CCO}$
H	L	L	H	H	H
H	L	L	L	H	H
L	H	H	H	H	H
L	H	H	L	H	H
L	H	L	H	H	H
L	H	L	L	H	H
L	L	H	H	H	H
L	L	H	L	H	H
L	L	L	H	H	H
L	L	L	L	L	1/

FIGURE 3. Truth tables - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-02-21T14:34Z
Check the source to verify that this is the current version before use.

Device type 01 and 02

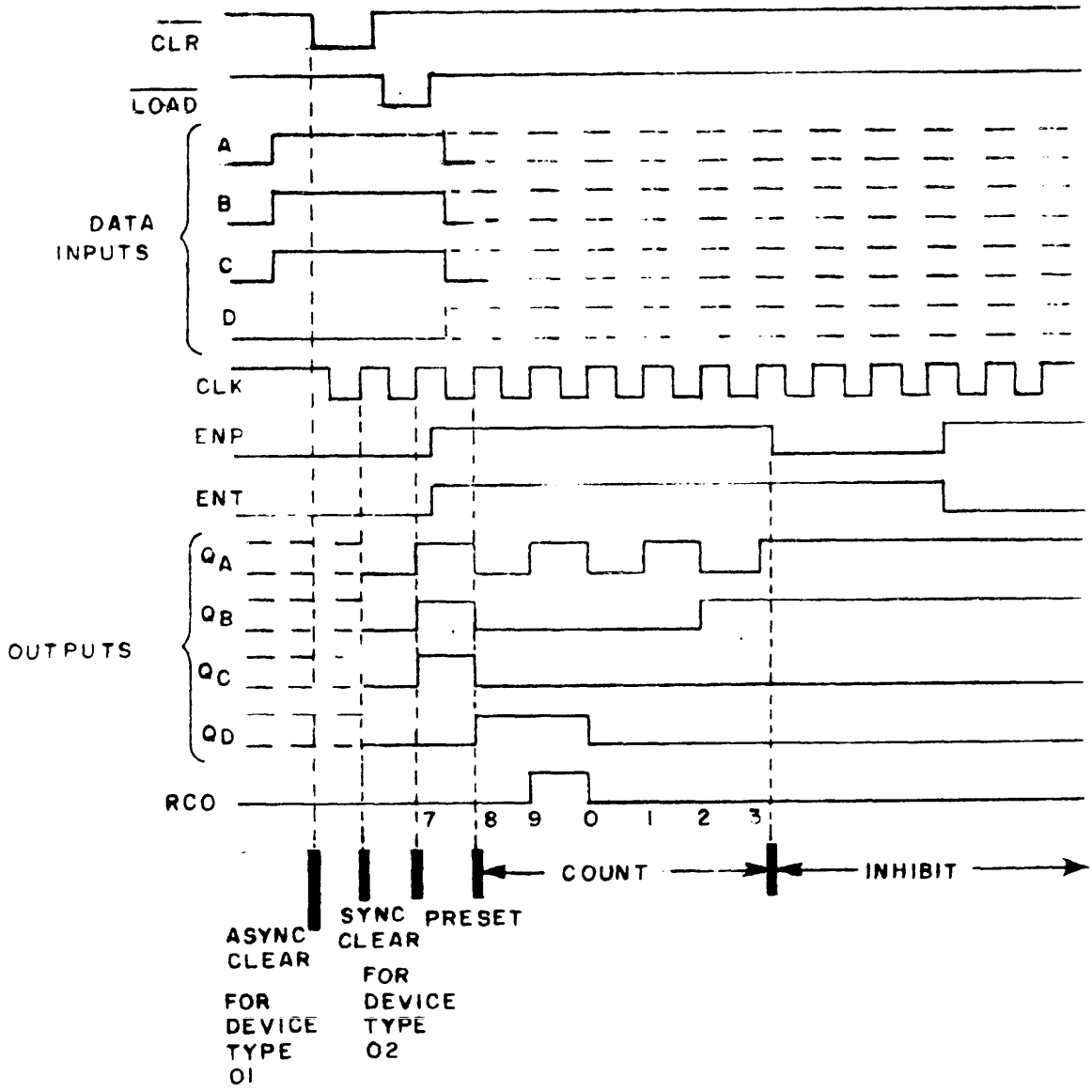


FIGURE 4. Typical operational sequences.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:34Z
Check the source to verify that this is the current version before use.

Device type 03

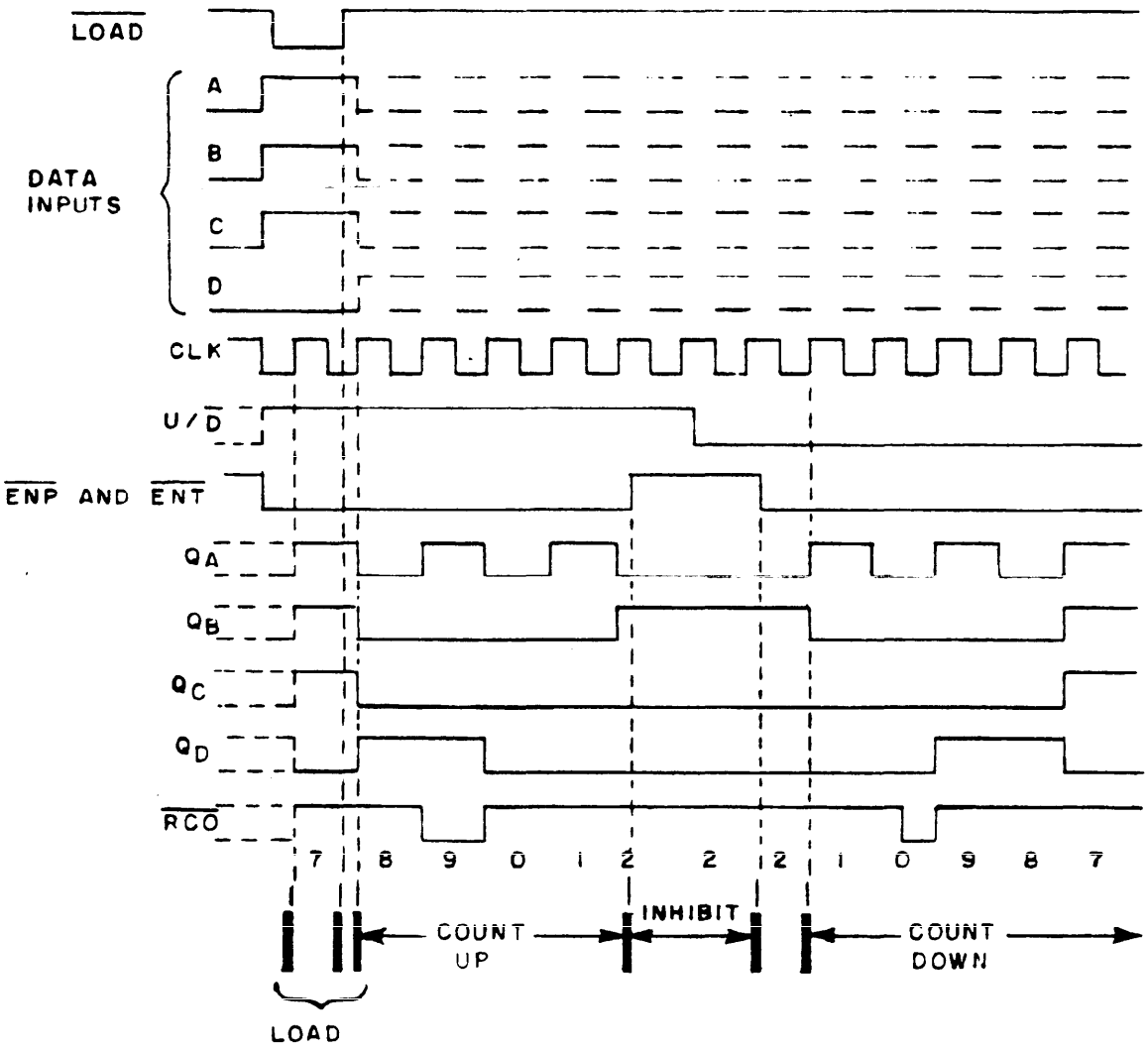


FIGURE 4. Typical operational sequences - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:34Z
Check the source to verify that this is the current version before use.

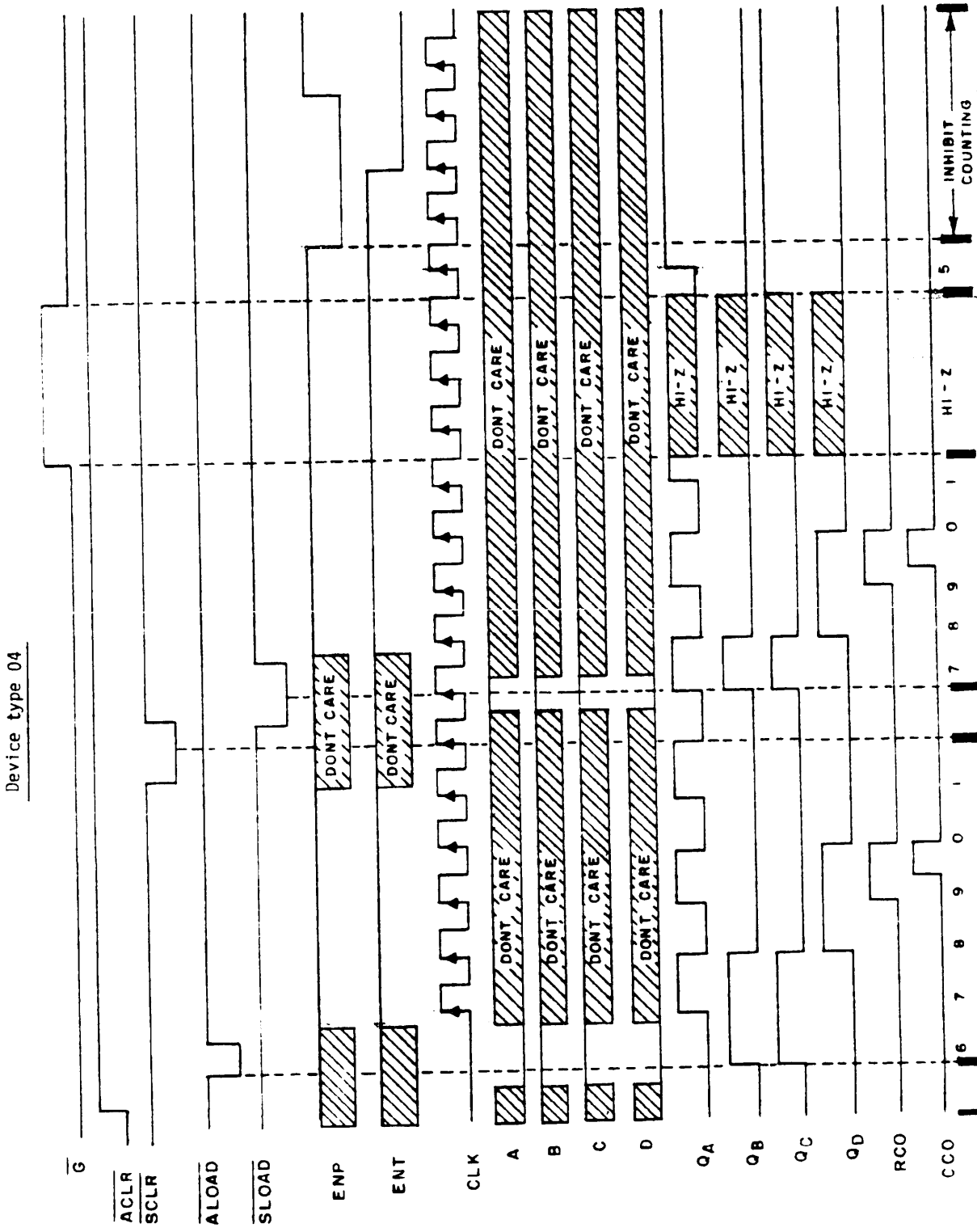


FIGURE 4. Typical operational sequences - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:34Z
Check the source to verify that this is the current version before use.

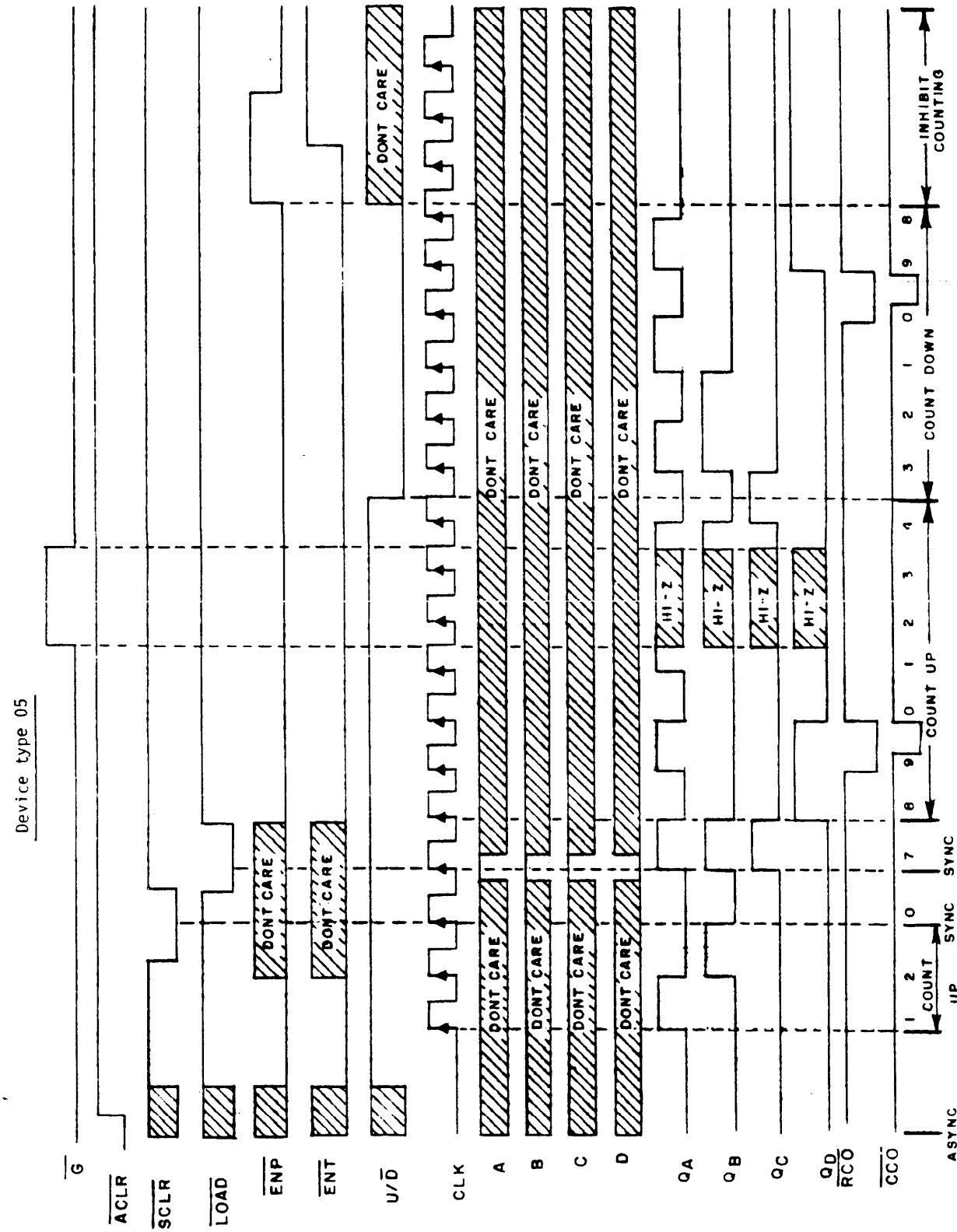
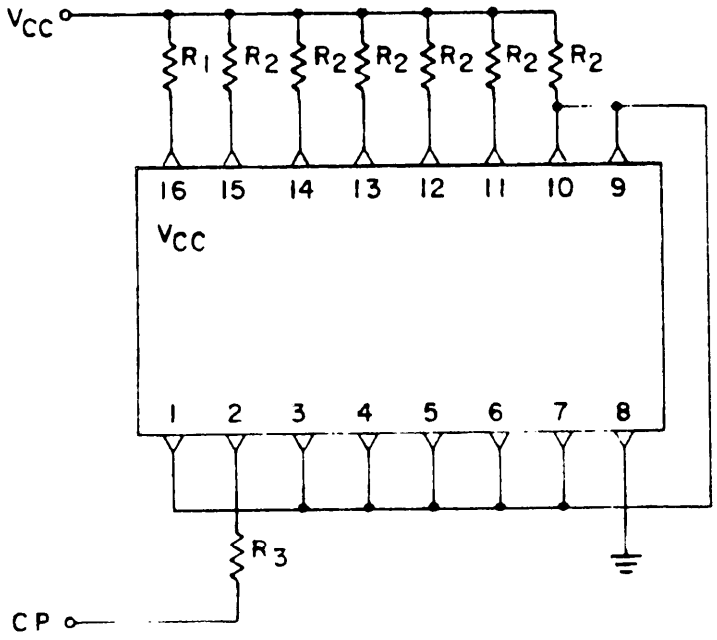


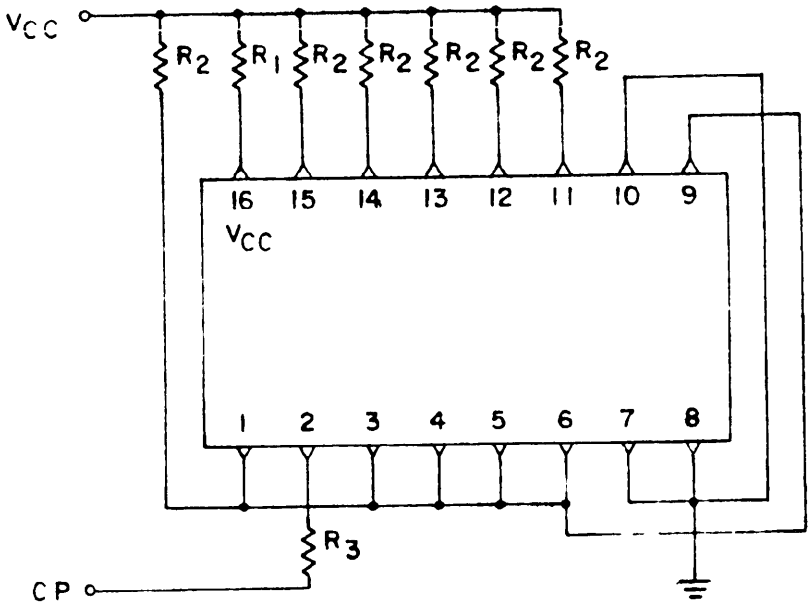
FIGURE 4. Typical operational sequences - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:34Z
Check the source to verify that this is the current version before use.

Device types 01 and 02



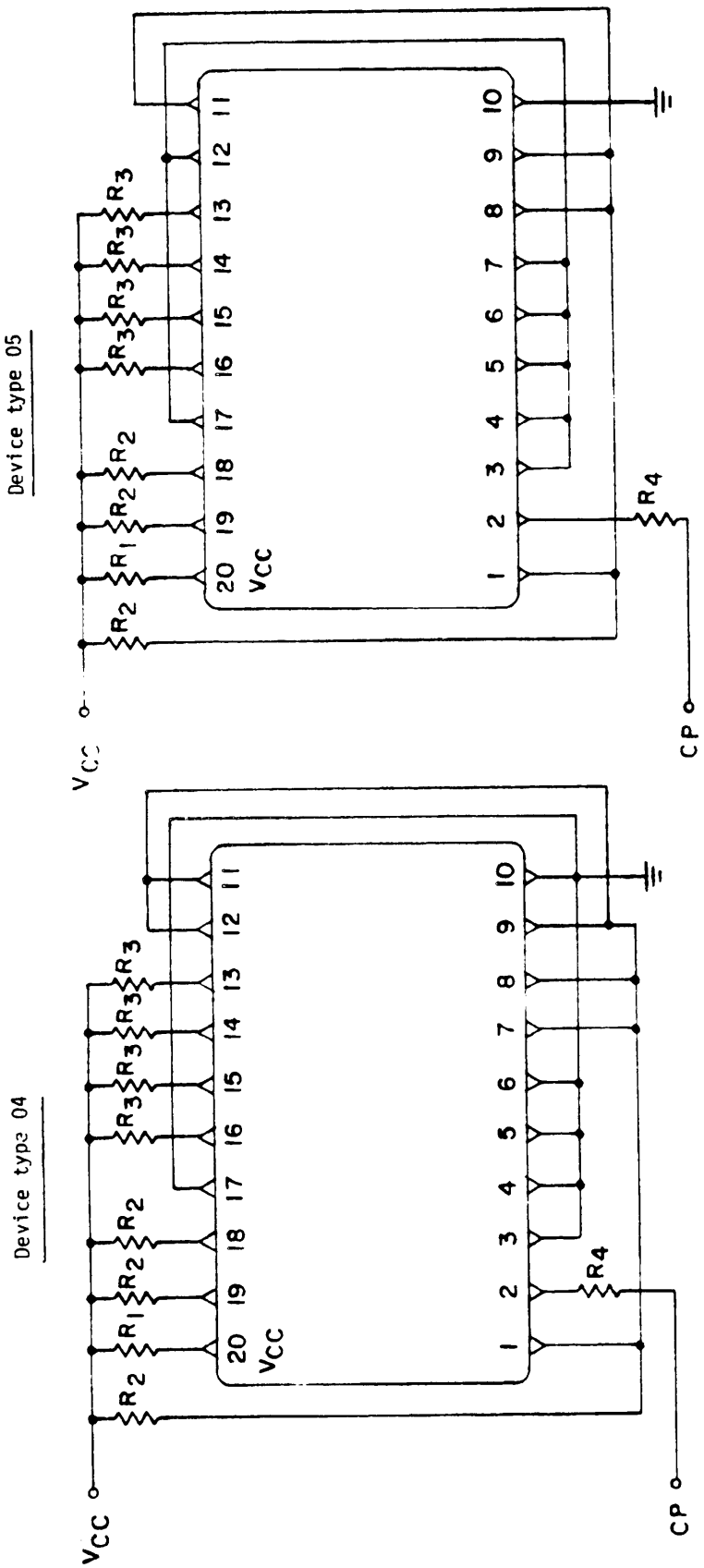
Device type 03



NOTES:
1. CP or $\overline{CP}$ = 100 kHz ±50% square wave; duty cycle = 50% ±15%; V_{IL} = -0.5 V minumum to 0.8 V maximum; V_{IH} = 2.0 V minimum to 5.5 V maximum.
2. V_{CC} and R_1 shall be chosen to insure 5.0 V minimum is present at device V_{CC} terminal, pin (16).

FIGURE 5. Burn-in and life test circuit.

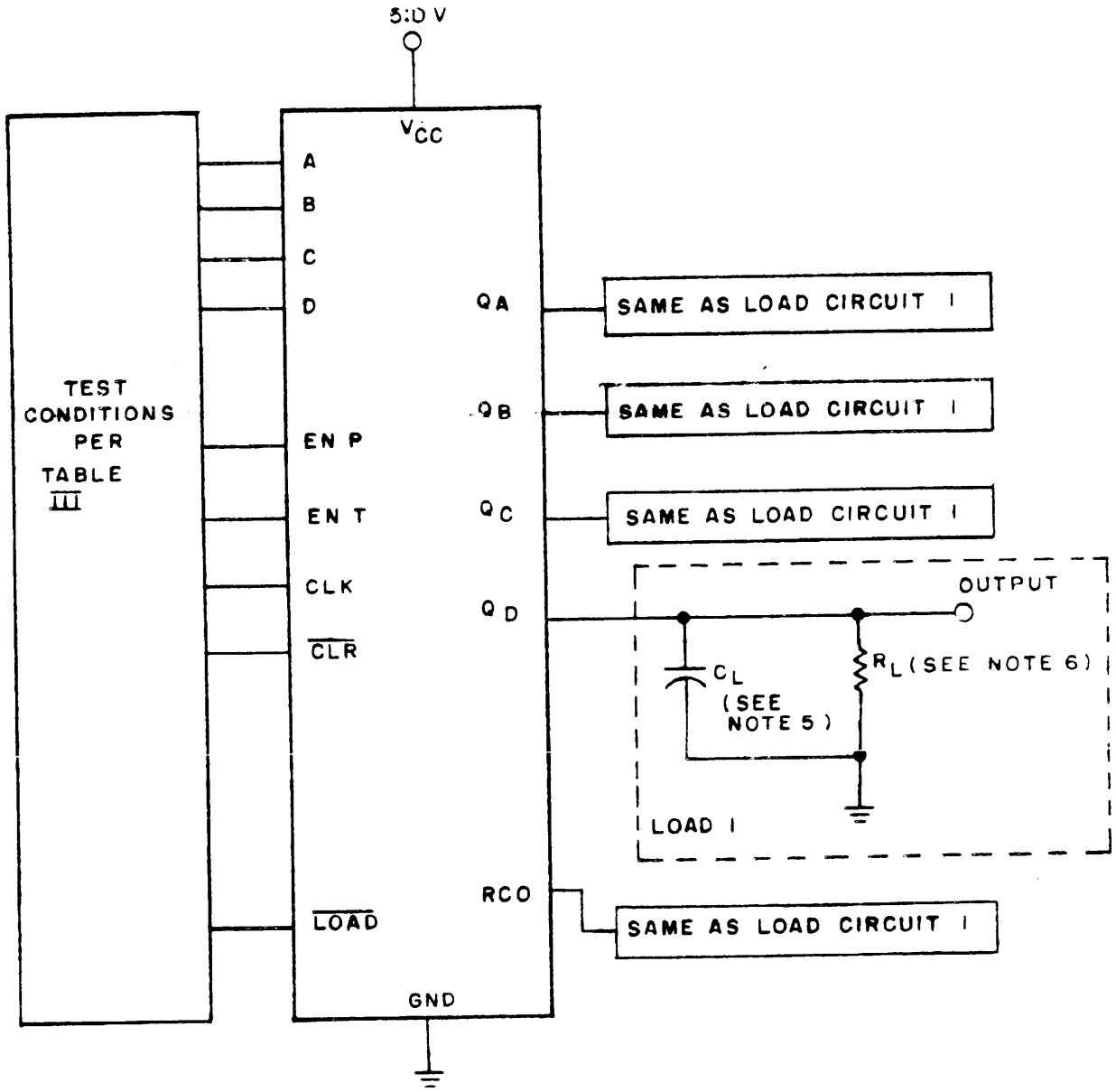
Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:34Z
Check the source to verify that this is the current version before use.



NOTES:
1. R_2 = 1 kΩ; R_3 = 470Ω; R_4 = 27Ω. All resistances ±5%.
2. CP = 100 kHz ±50% square wave; duty cycle = 50% ±15%, V_{IL} = -0.5 V minimum to 0.8 V maximum, V_{IH} = 2.0 V minimum to 5.5 V maximum.
3. V_{CC} and R_1 shall be chosen to insure 5.0 V minimum is present at device V_{cc} terminal (pin 20).

FIGURE 5. Burn-in and life test circuit - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:34Z
Check the source to verify that this is the current version before use.



NOTES:
1. Clock input pulse characteristics: $t_1 = t_0 = 6 \pm 1.5$ ns; $t_{p(clock)} = 20$ ns; PRR $\leq$ 1 MHz.
2. Clear input pulse characteristics: $t_1 = t_0 = 6 \pm 1.5$ ns; $t_{p(clear)} = 20$ ns.
3. For f_{MAX}, the clock input pulse are as follows: $t_1 = t_0 = 6 \pm 1.5$ ns; for 25°C, $t_{p(clock)} = 15$ ns; PRR = 30 MHz; for -55/125°C, $t_{p(clock)} = 20$ ns; PRR = 25 MHz.
4. Enable input pulse characteristics: $t_1 = t_0 = 6 \pm 1.5$ ns; for type 01, $t_{SETUP} = 25$ ns; $t_{HOLD} = 0$ ns; $t_{p(enable)} = 25$ ns; for type 02, $t_{SETUP} = 30$ ns; $t_{HOLD} = 0$ ns; $t_{p(enable)} = 30$ ns.
5. $C_L = 50$ pF ±10% (includes jig and probe capacitance).
6. $R_L = 499\Omega$ ±1%.

FIGURE 6. Switching time test circuit and waveforms for device types 01 and 02.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:34Z
Check the source to verify that this is the current version before use.

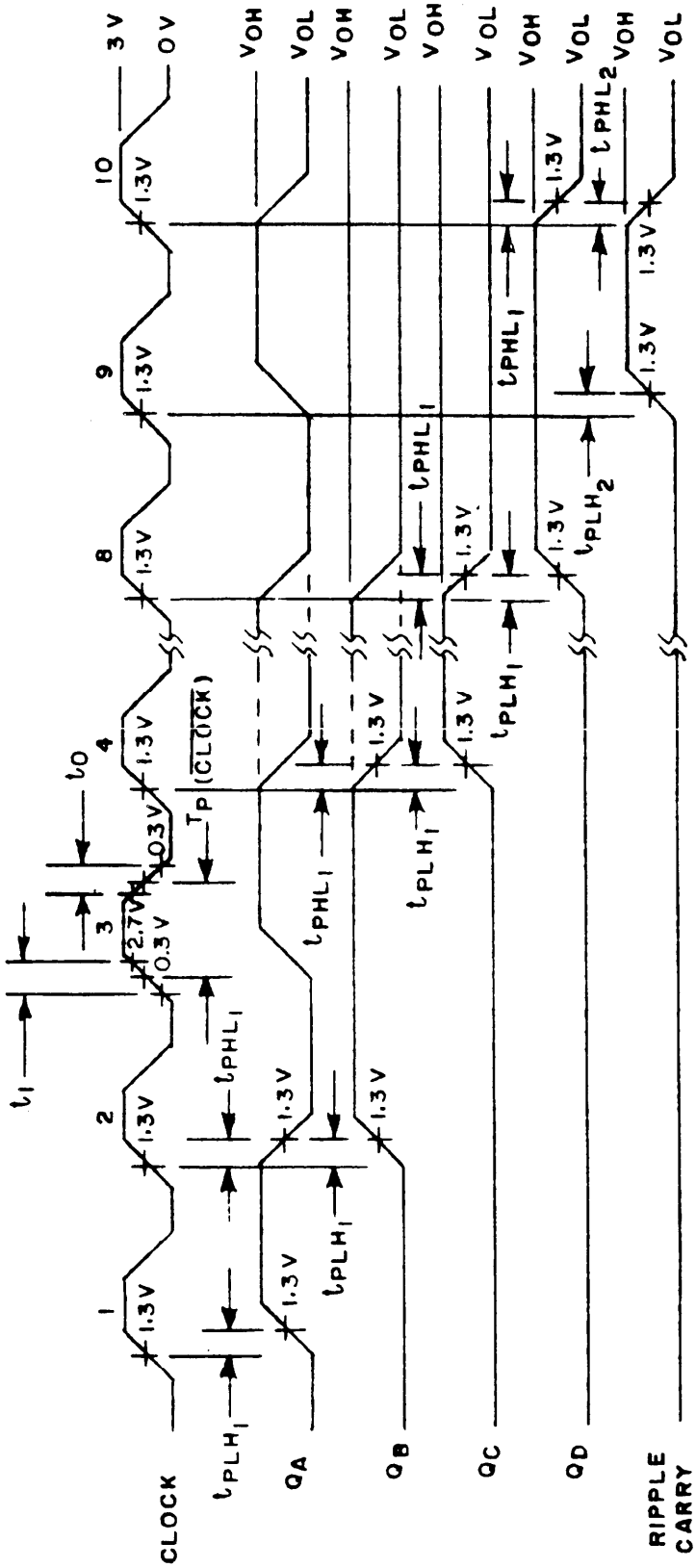


FIGURE 6. Switching time test circuit and waveforms for device types 01 and 02 - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:34Z
Check the source to verify that this is the current version before use.

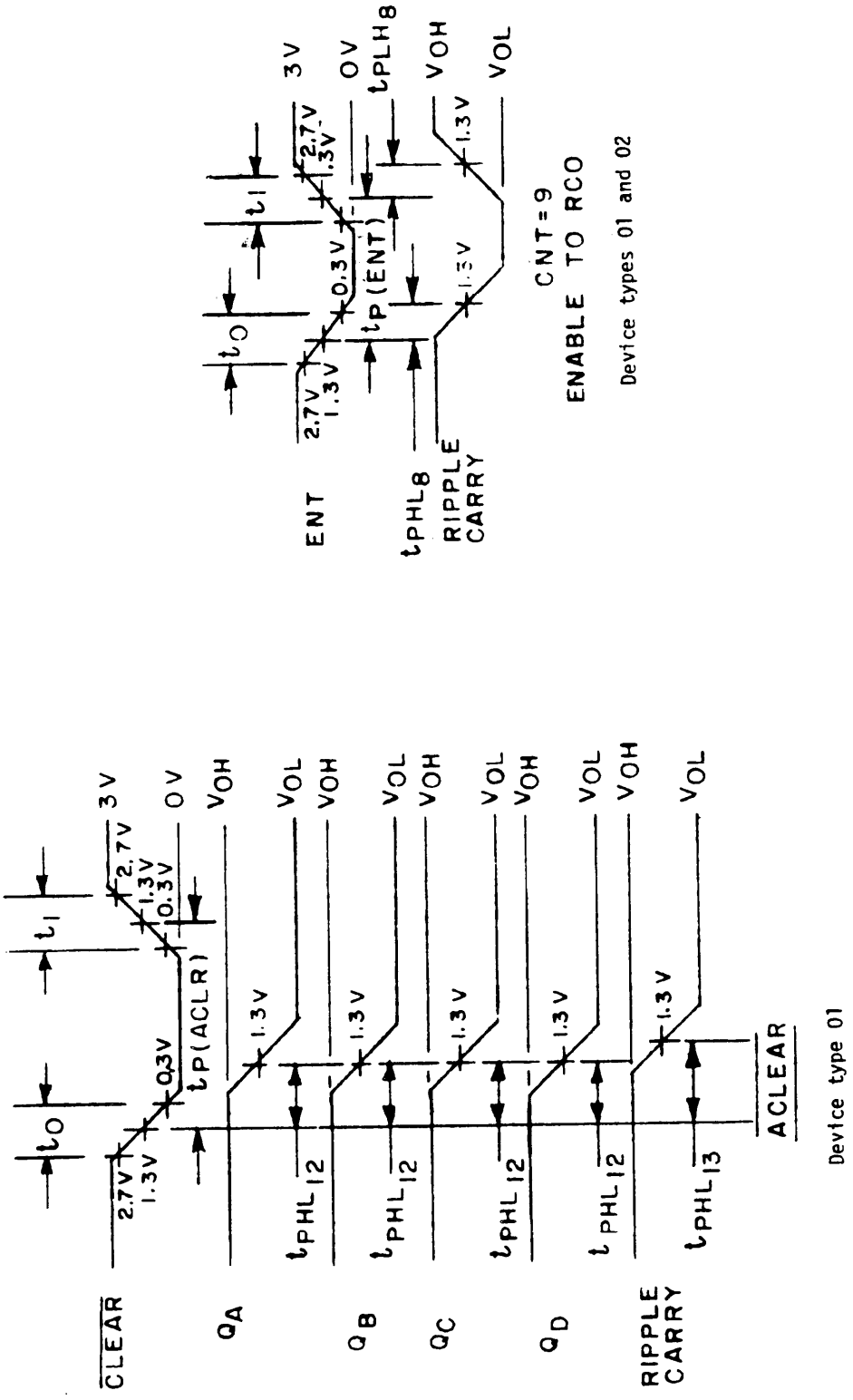
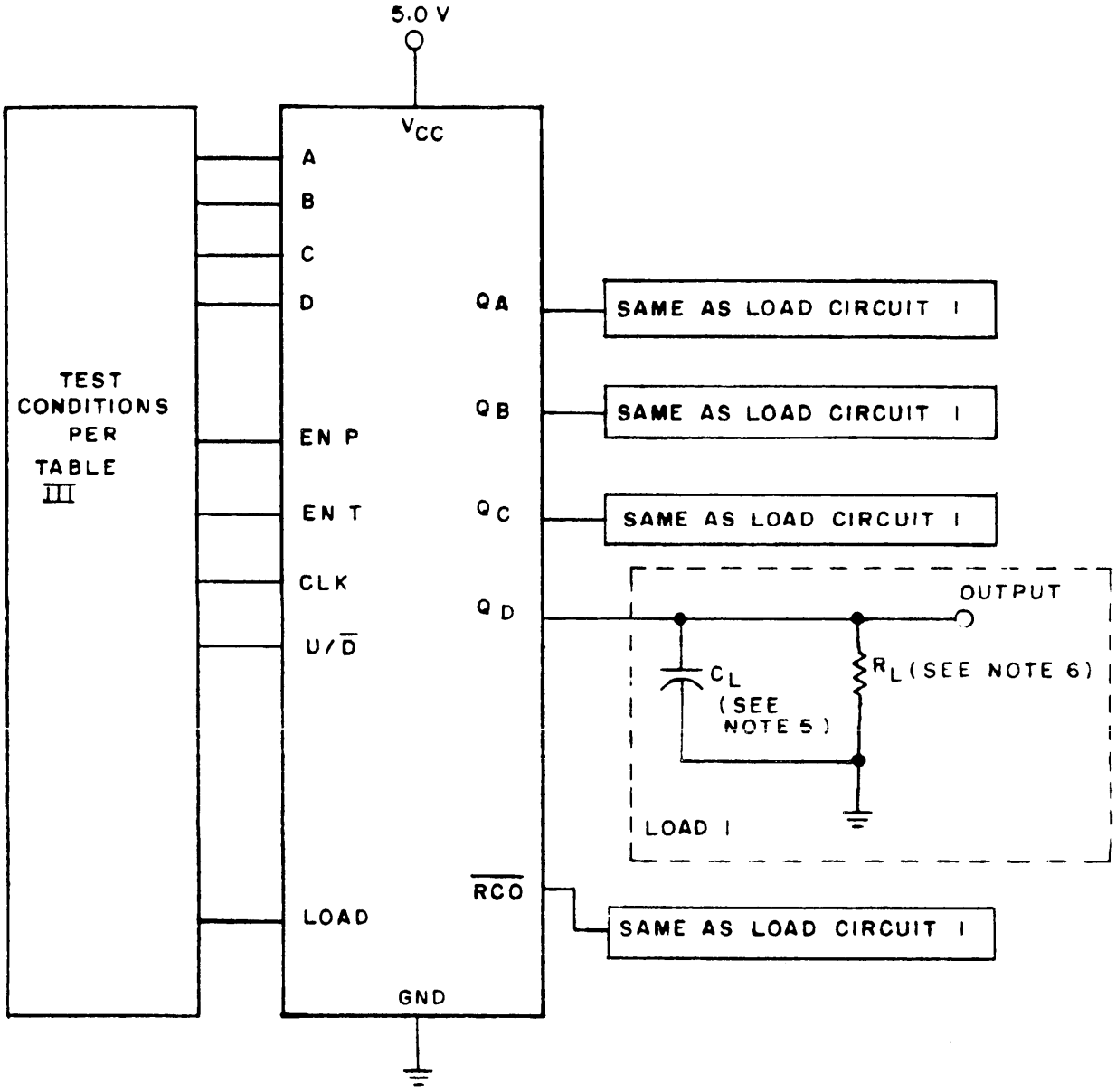


FIGURE 6. Switching time test circuit and waveforms for device types 01 and 02 - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:34Z
Check the source to verify that this is the current version before use.



NOTES:
1. Clock input pulse characteristics: $t_1 = t_0 = 6 \pm 1.5$ ns; $t_{p(clock)} = 20$ ns; PRR $\leq$ 1 MHz.
2. For f_{MAX}, the clock input pulse characteristics are as follows: $t_1 = t_0 = 6 \pm 1.5$ ns; for 25°C, $t_{p(clock)} = 15$ ns; PRR = 30 MHz; for -55/125°C, $t_{p(clock)} = 20$ ns; PRR = 25 MHz.
3. Enable input pulse characteristics: $t_1 = t_0 = 6 \pm 1.5$ ns; $t_{SETUP} = 25$ ns; $t_{HOLD} = 0$ ns; $t_{p(enable)} = 25$ ns.
4. Up/Down input pulse characteristics: $t_1 = t_0 = 6 \pm 1.5$ ns; $t_{SETUP} = 20$ ns; $t_{HOLD} = 0$ ns; $t_{p(up/down)} = 20$ ns.
5. $C_L = 50$ pF ±10% (includes jig and probe capacitance).
6. $R_L = 499\Omega \pm 1\%$.

FIGURE 6. Switching time test circuit and waveforms for device type 03.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:34Z
Check the source to verify that this is the current version before use.

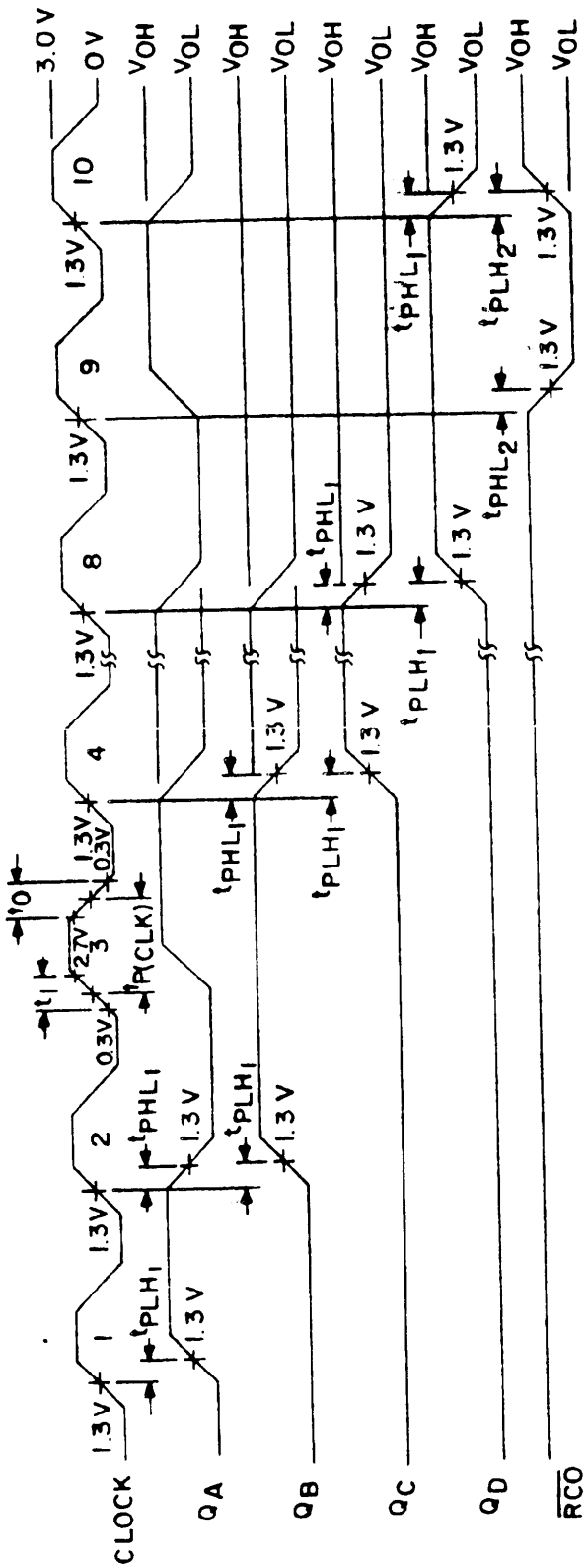
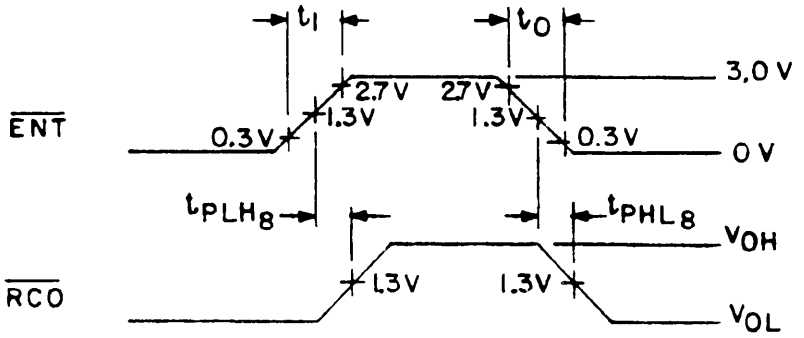
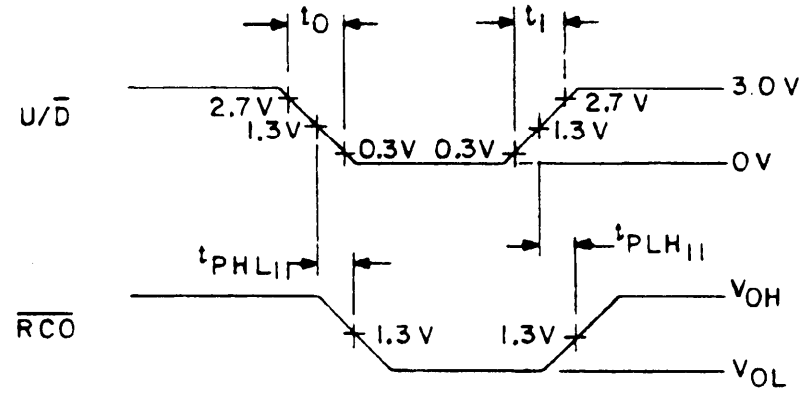


FIGURE 6. Switching time test circuit and waveforms for device type 03 - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:34Z
Check the source to verify that this is the current version before use.



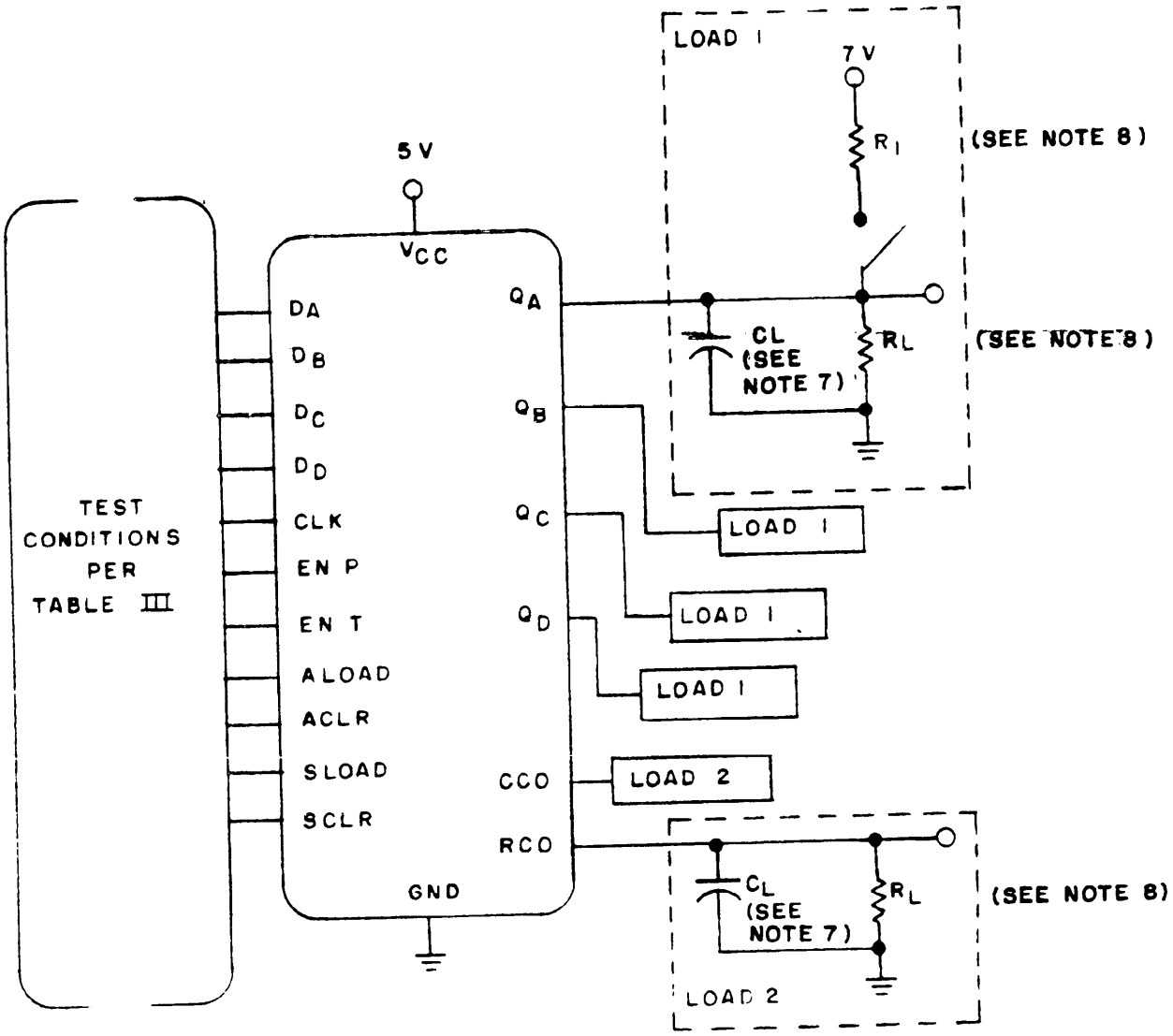
ENABLE TO RCO



UP/$\overline{DOWN}$ TO RCO

FIGURE 6. Switching time test circuit and waveforms for device type 03 - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:34Z
Check the source to verify that this is the current version before use.



NOTES:
1. Clock input pulse characteristics: $t_1 = t_0 = 6 \pm 1.5$ ns; $t_{p(clock)} = 27.5$ ns; PRR $\leq$ 1 MHz.
2. For f_{MAX}, the clock input pulse characteristics are as follows: $t_1 = t_0 = 6 \pm 1.5$ ns; for 25°C, $t_{p(clock)} = 25$ ns; PRR = 20 MHz; for -55/125°C, $t_{p(clock)} = 27.5$ ns; PRR = 18 MHz.
3. Asynchronous Clear/Load input pulse characteristics: $t_1 = t_0 = 6 \pm 1.5$ ns; $t_{p(ACLR/ALOAD)} = 20$ ns.
4. Enable input pulse characteristics: $t_1 = t_0 = 6 \pm 1.5$ ns; $t_{SETUP} = 25$ ns; $t_{HOLD} = 0$ ns; $t_{p(enable)} = 25$ ns.
5. Data input pulse characteristics: $t_1 = t_0 = 6 \pm 1.5$ ns; $t_{SETUP} = 25$ ns; $t_{HOLD} = 0$ ns; $t_{p(data)} = 25$ ns.
6. Output control ($\overline{G}$) input pulse characteristics: $t_1 = t_0 = 6 \pm 1.5$ ns; $t_p = 200$ ns.
7. $C_L = 50$ pF ±10% (including jig and probe capacitance).
8. $R_1 = R_L = 499\Omega$ ±1%.

FIGURE 6. Switching time test circuit and waveforms for device type 04.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:34Z
Check the source to verify that this is the current version before use.

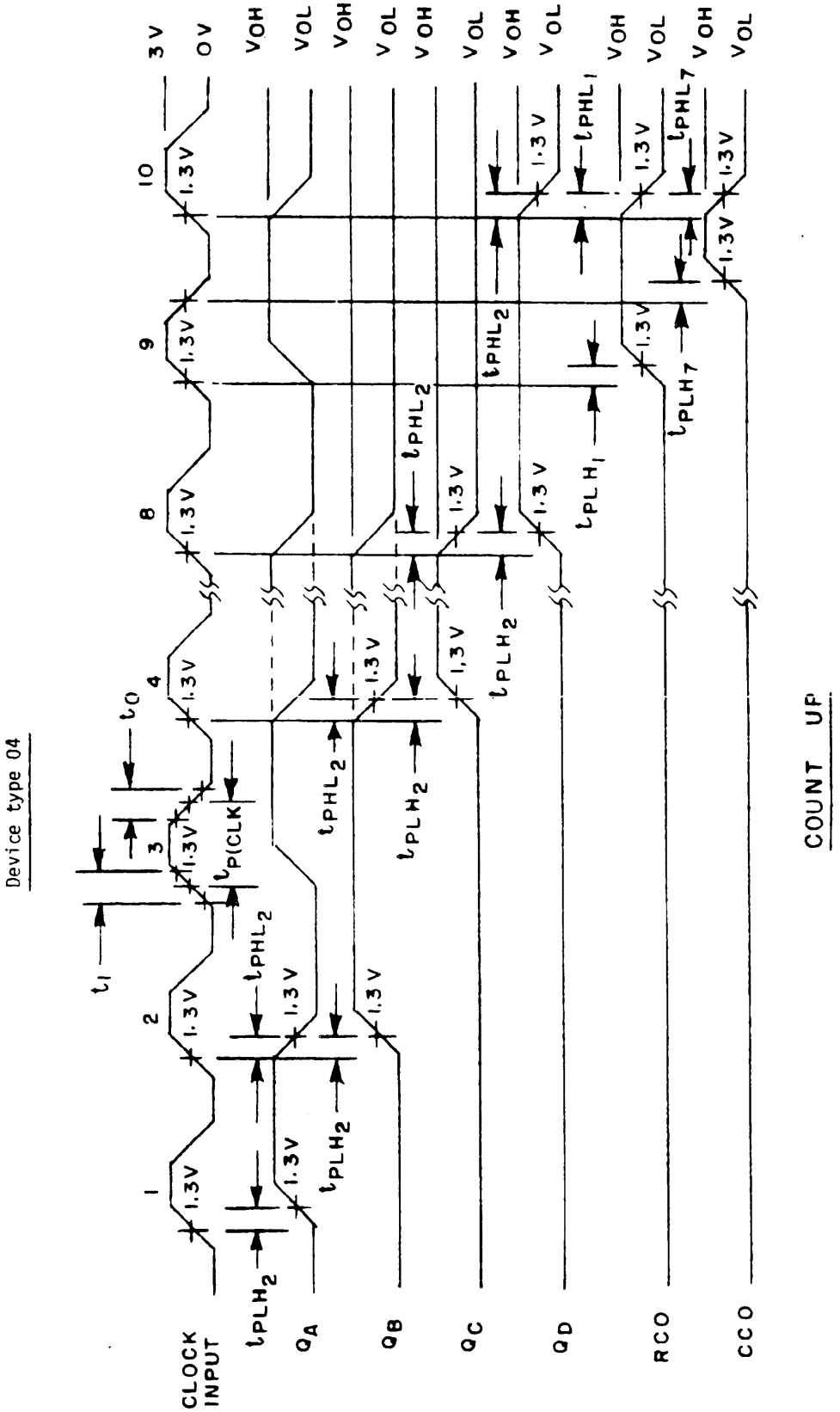


FIGURE 6. Switching time test circuit and waveforms for device type 04 - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:34Z
Check the source to verify that this is the current version before use.

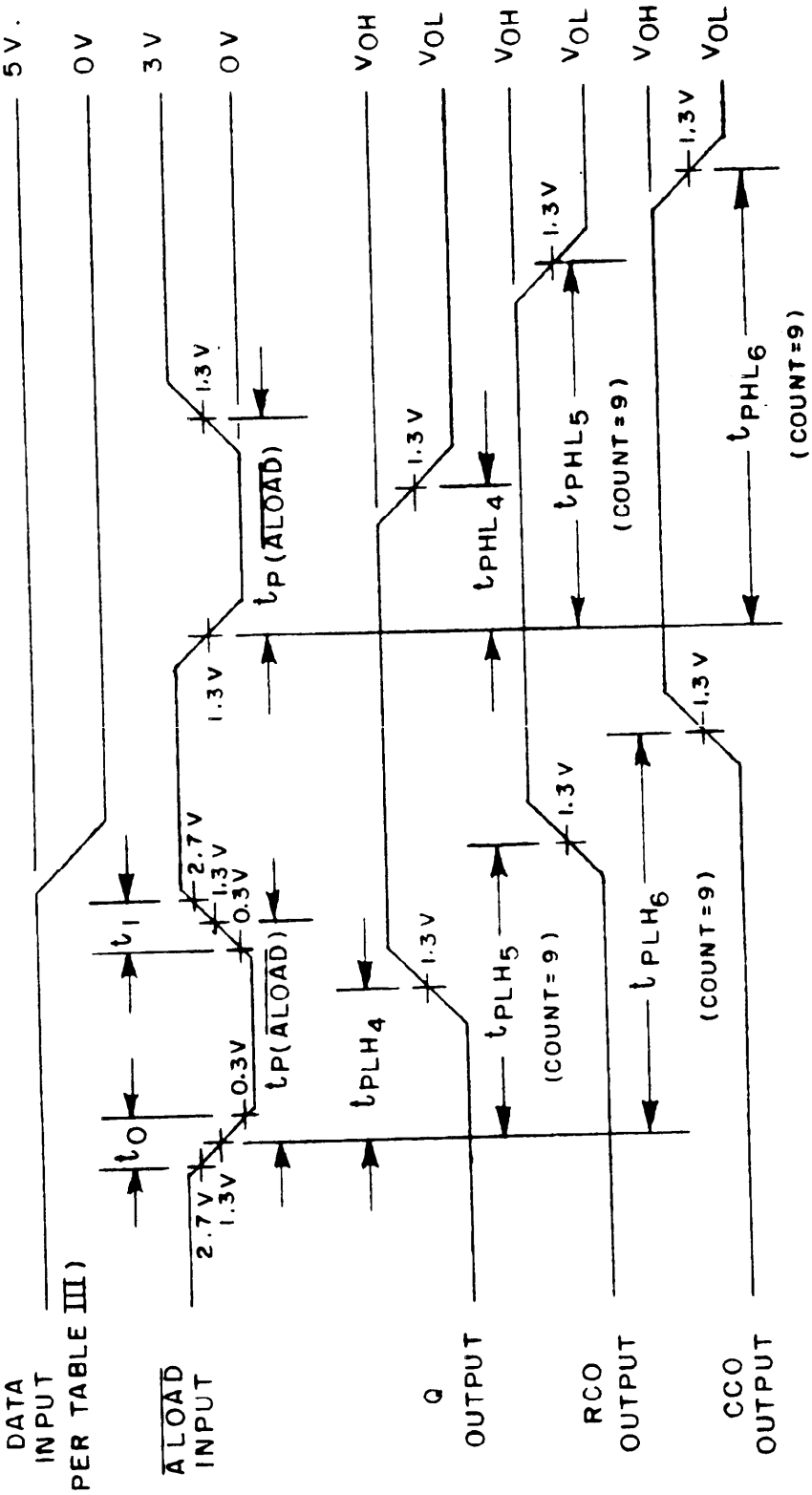


FIGURE 6. Switching time test circuit and waveforms for device type 04 - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:34Z
Check the source to verify that this is the current version before use.

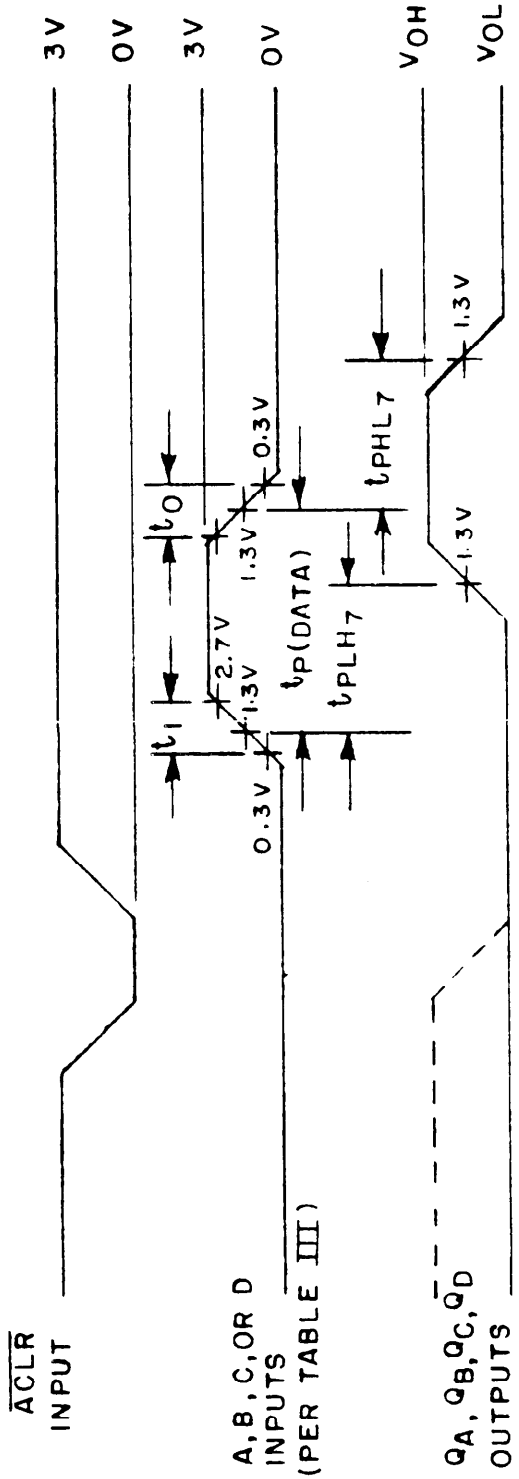


FIGURE 6. Switching time test circuit and waveforms for device type 04 - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:34Z
Check the source to verify that this is the current version before use.

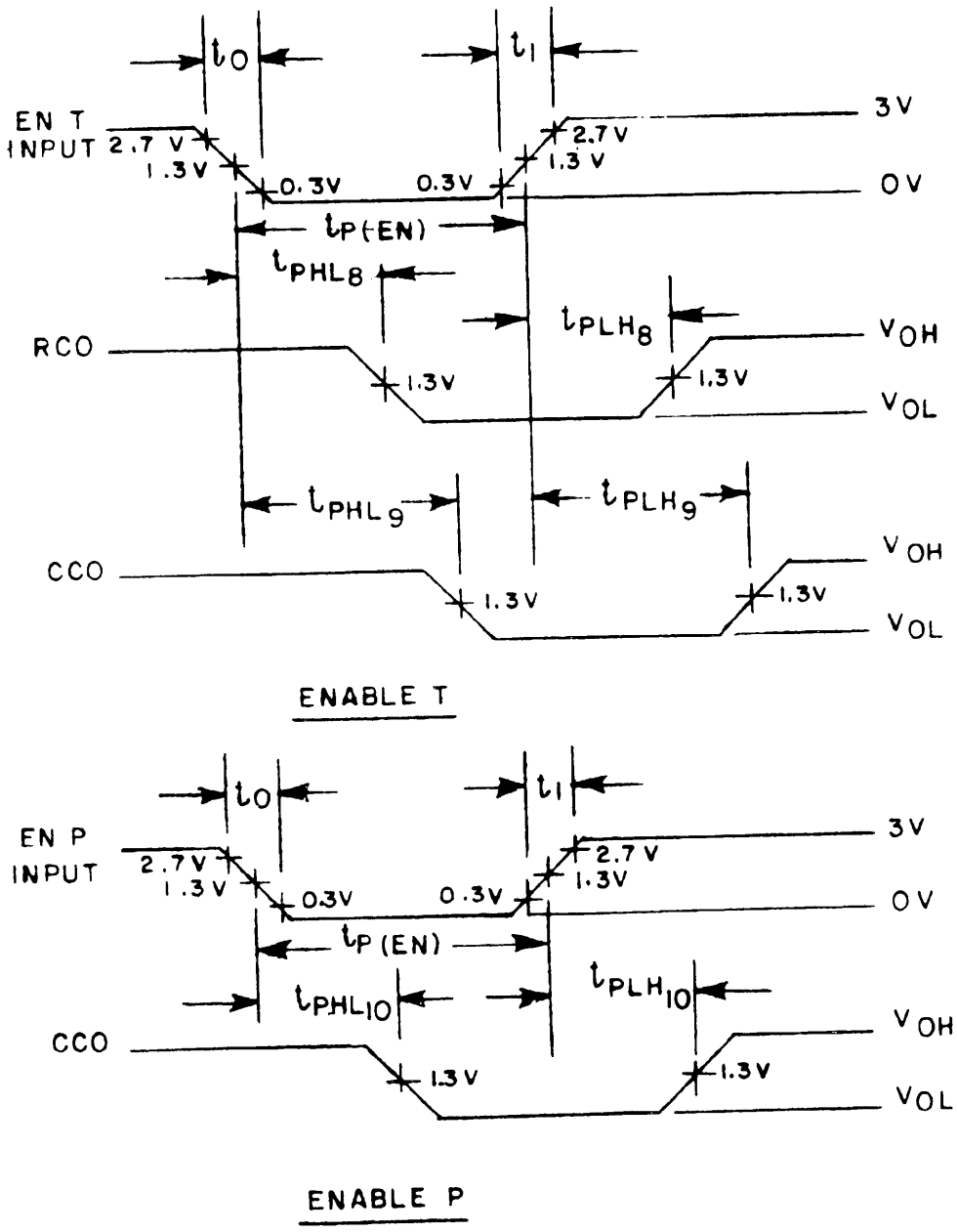
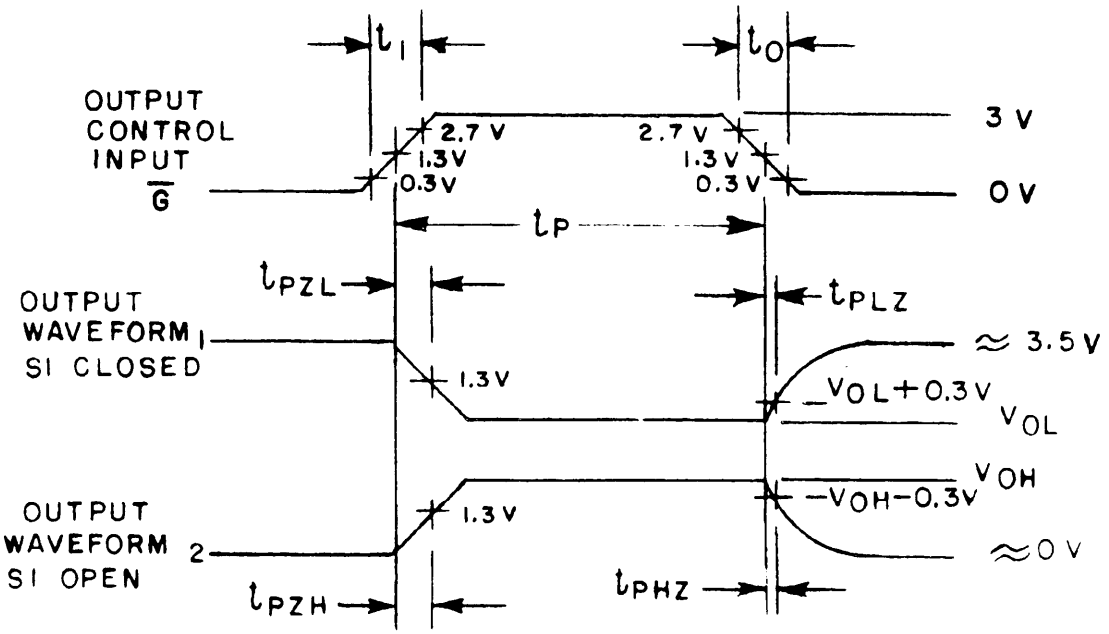


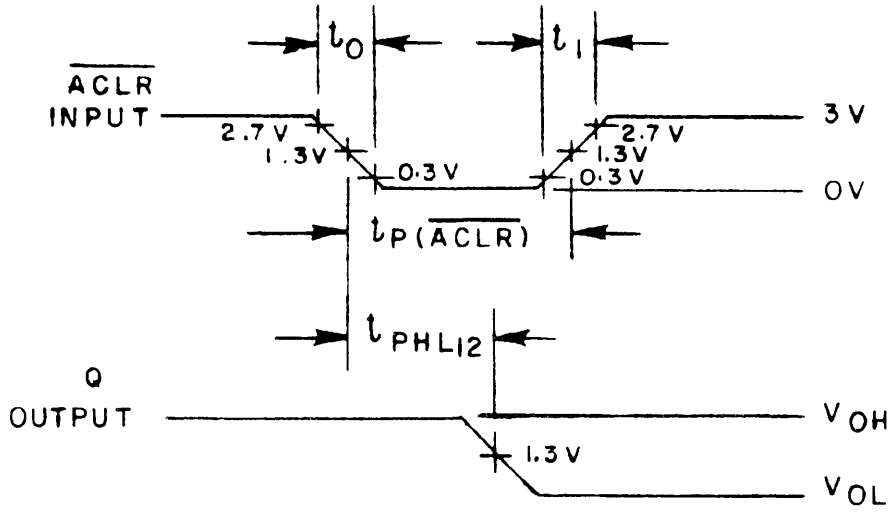
FIGURE 6. Switching time test circuit and waveforms for device type 04 - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:34Z
Check the source to verify that this is the current version before use.

Device type 04



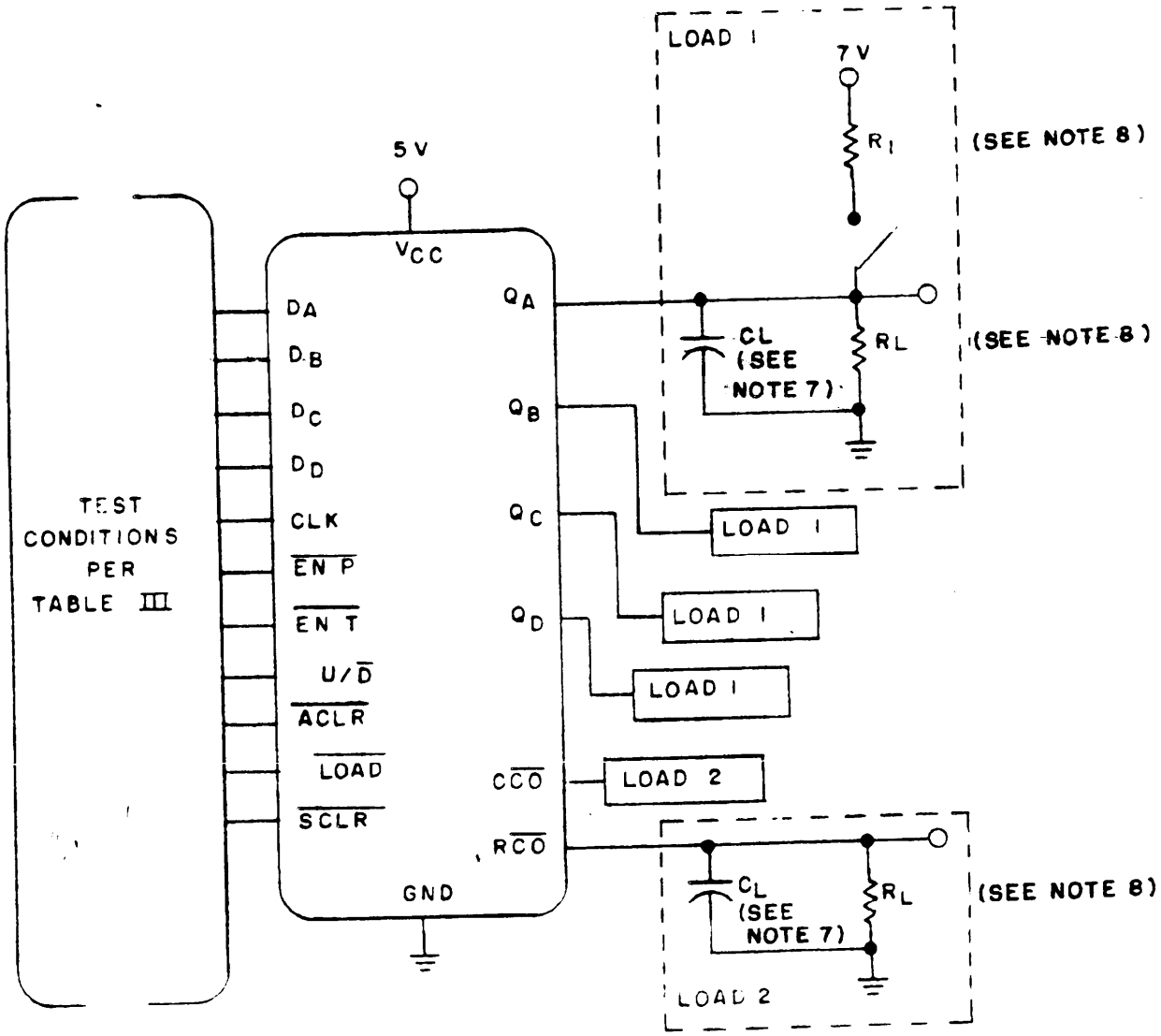
TRI-STATE SWITCHING



ASYNCHRONOUS CLEAR

FIGURE 6. Switching time test circuit and waveforms for device type 04 - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:34Z
Check the source to verify that this is the current version before use.



NOTES:
1. Clock input pulse characteristics: $t_1 = t_0 = 6 \pm 1.5$ ns; $t_{p(clock)} = 27.5$ ns; PRR $\leq$ 1 MHz.
2. For f_{MAX}, the clock input pulse characteristics are as follows: $t_1 = t_0 = 6 \pm 1.5$ ns; for 25°C, $t_{p(clock)} = 27$ ns; PRR = 20 MHz; for -55/125°C, $t_{p(clock)} = 27.5$ ns; PRR = 18 MHz.
3. Up/Down input pulse characteristics: $t_1 = t_0 = 6 \pm 1.5$ ns; $t_{SETUP} = 35$ ns; $t_{HOLD} = 0$ ns; $t_{p(up/down)} = 35$ ns.
4. Enable input pulse characteristics: $t_1 = t_0 = 6 \pm 1.5$ ns; Pulse High, $t_{SETUP} = 35$ ns; $t_{HOLD} = 0$ ns; $t_{p(enable)} = 35$ ns. Pulse Low, $t_{SETUP} = 25$ ns; $t_{HOLD} = 0$ ns; $t_{p(enable)} = 25$ ns.
5. Asynchronous Clear input pulse characteristics: $t_1 = t_0 = 6 \pm 1.5$ ns; $t_{p(ACLR)} = 20$ ns.
6. Output control ($\overline{G}$) input characteristics: $t_1 = t_0 = 6 \pm 1.5$ ns; $t_p = 200$ ns.
7. $C_L = 50$ pF ±10% (including jig and probe capacitance).
8. $R_1 = R_L = 499\Omega \pm 1\%$.

FIGURE 6. Switching time test circuit and waveforms for device type 05.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:34Z
Check the source to verify that this is the current version before use.

Device type 05

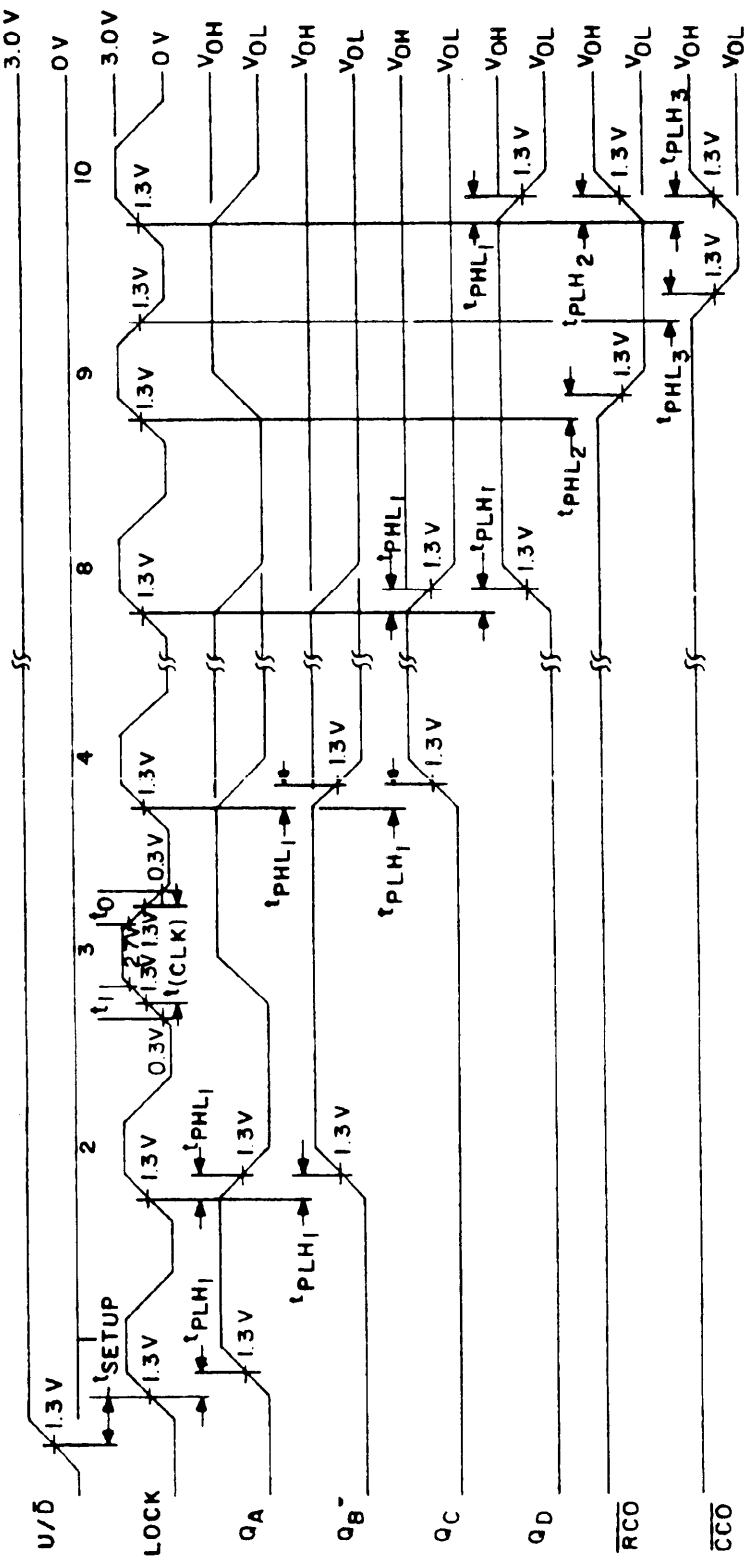
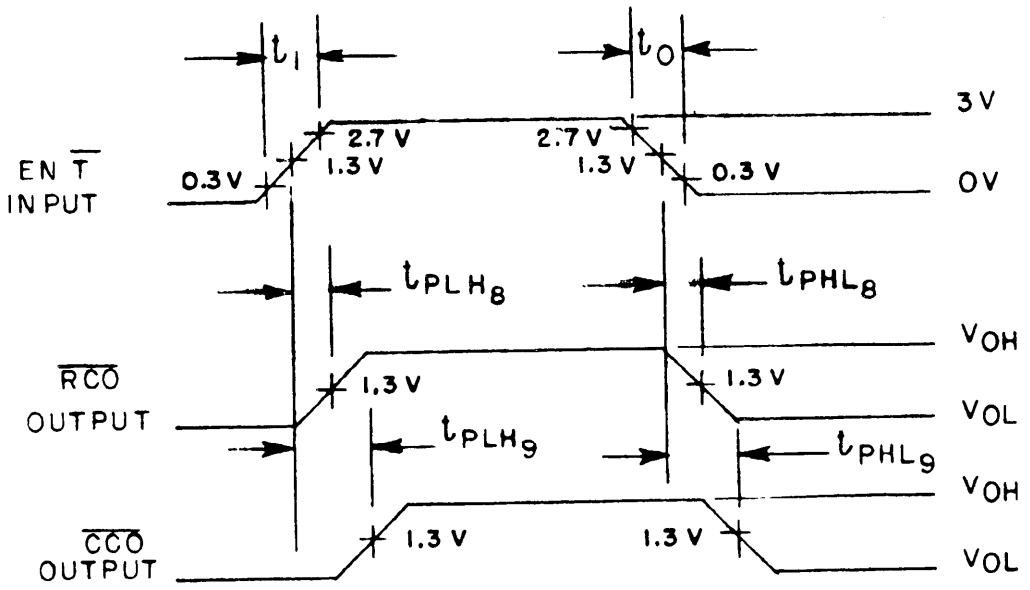
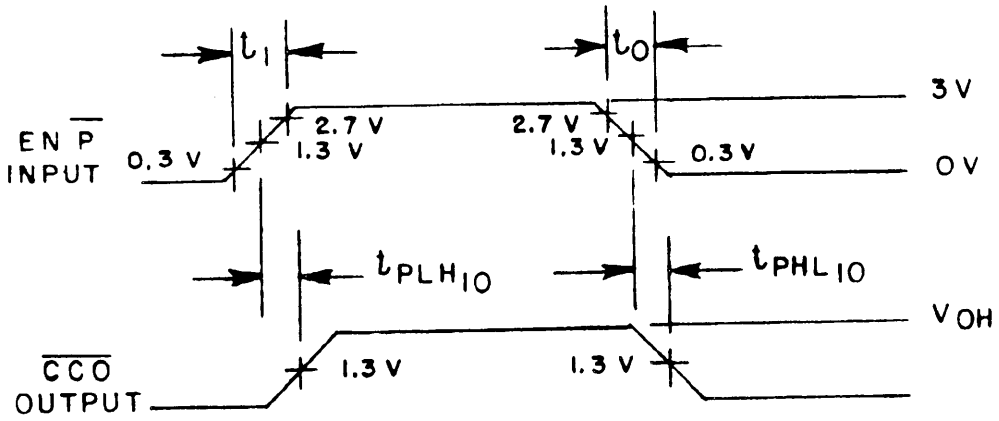


FIGURE 6. Switching time test circuit and waveforms for device type 05 - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:34Z
Check the source to verify that this is the current version before use.



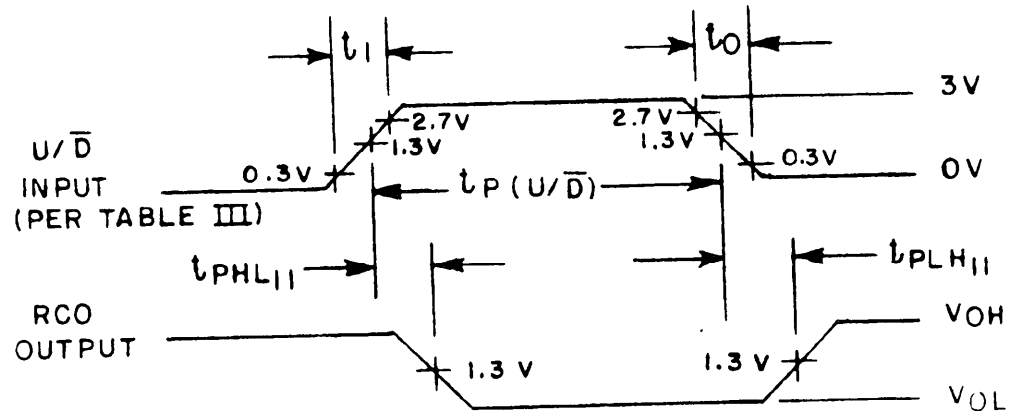
$\overline{ENT}$ TO $\overline{RCO}$/$\overline{CCO}$



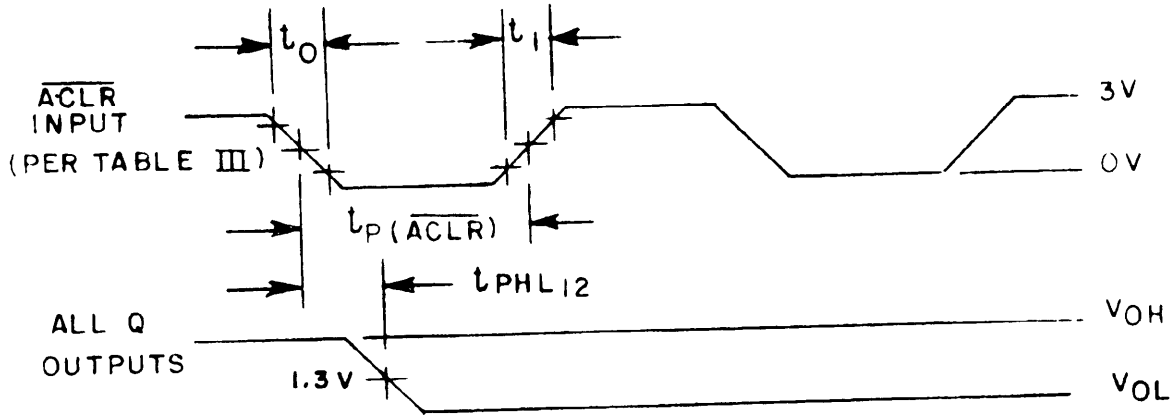
$\overline{ENP}$ TO $\overline{CCO}$

FIGURE 6. Switching time test circuit and waveforms for device type 05 - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:34Z
Check the source to verify that this is the current version before use.



U/$\overline{D}$ TO RCO

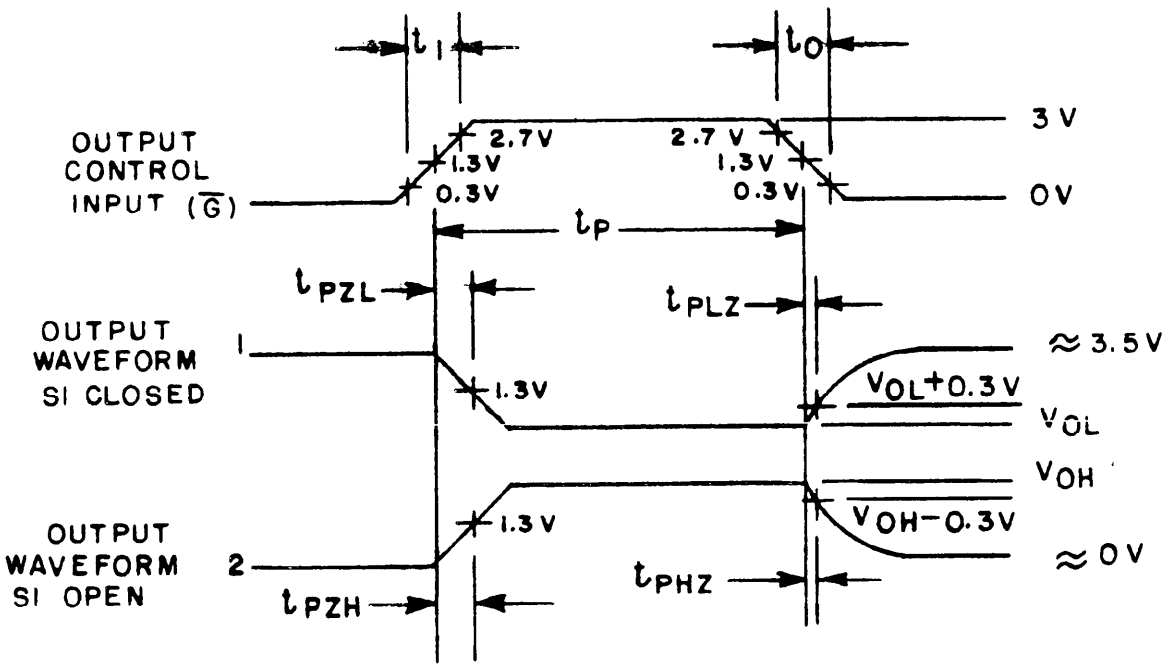


$\overline{ACLR}$

FIGURE 6. Switching time test circuit and waveforms for device type 05 - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:34Z
Check the source to verify that this is the current version before use.

Device type 05



TRI-STATE SWITCHING

FIGURE 6. Switching time test circuit and waveforms for device type 05 - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:34Z
Check the source to verify that this is the current version before use.

TABLE III. Group A inspection for device type 01.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Cases E,F Case 1/ 2 Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Test limits Min	Test limits Max	Unit
				2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20				
				$\overline{CLR}$	CLK	A	B	C	D	ENP	GND	$\overline{LOAD}$	ENT	QD	QC	QB	QA	RCO	V_{CC}				
1 T_C = +25°C	V_{OL1}	3007	1	4.5 V	2/			0.8 V	0.8 V		GND	GND		4 mA					4.5 V	QD		0.4	V
		"	2	"	"						"	"			4 mA				"	QC		"	"
		"	3	"	"		0.8 V				"	"				4 mA			"	QB		"	"
		"	4	"	"	0.8 V					"	"					4 mA		"	QA		"	"
		"	5	"	"						"	"	0.8 V					4 mA	"	RCO		"	"
	V_{OH1}	3006	6	"	"			2.0 V	2.0 V		"	"		-400 µA					"	QD	2.5		"
		"	7	"	"						"	"			-400 µA				"	QC	"		"
		"	8	"	"		2.0 V				"	"				-400 µA			"	QB	"		"
		"	9	"	"	2.0 V					"	"					-400 µA		"	QA	"		"
		"	10	"	"	2.0 V	0.8 V	0.8 V	2.0 V		"	"	2.0 V					-400 µA	"	RCO	"		"
	V_{IC}		11	-18 mA							"								"	$\overline{CLR}$		-1.5	"
			12		-18 mA						"								"	CLK		"	"
			13			-18 mA					"								"	A		"	"
			14				-18 mA				"								"	B		"	"
			15					-18 mA			"								"	C		"	"
			16						-18 mA		"								"	D		"	"
			17							18 mA	"								"	ENP		"	"
			18								"	-18 mA							"	$\overline{LOAD}$		"	"
			19								"		-18 mA						"	ENT		"	"
	I_{IL}	3009	20	0.4 V							"								5.5 V	$\overline{CLR}$	4/	4/	4/
		"	21		0.4 V						"								"	CLK	"	"	"
		"	22			0.4 V					"								"	A	"	"	"
		"	23				0.4 V				"								"	B	"	"	"
		"	24					0.4 V			"								"	C	"	"	"
		"	25						0.4 V		"								"	D	"	"	"
		"	26							0.4 V	"		4.5 V						"	ENP	"	"	"
		"	27							4.5 V	"		0.4 V						"	$\overline{LOAD}$	"	"	"
		"	28							4.5 V	"		4.5 V						"	ENT	"	"	"
	I_{IH1}	3010	29	2.7 V							"								"	$\overline{CLR}$		20	µA
		"	30		2.7 V						"								"	CLK		40	"
		"	31			2.7 V					"								"	A		20	"
		"	32				2.7 V				"								"	B		"	"
		"	33					2.7 V			"								"	C		"	"
		"	34						2.7 V		"								"	D		"	"
		"	35							GND	"		2.7 V						"	$\overline{LOAD}$		40*	"
		"	36							2.7 V	"		GND						"	ENP		20	"
		"	37							GND	"		GND						"	ENT		40	"
	I_{IH2}	"	38	7.0 V							"								"	$\overline{CLR}$		100	"
		"	39		7.0 V						"								"	CLK		200	"
		"	40			7.0 V					"								"	A		100	"
		"	41				7.0 V				"								"	B		"	"
		"	42					7.0 V			"								"	C		"	"
		"	43						7.0 V		"								"	D		"	"
		"	44							GND	"		7.0 V						"	$\overline{LOAD}$		200	"
		"	45							7.0 V	"		GND						"	ENP		100	"
		"	46							GND	"		GND						"	ENT		200	"
	I_{O} 3/	3011	47	4.5 V	2/			4.5 V	4.5 V		"			2.25 V					"	QD	-30	-112	mA
		"	48	"	"						"				2.25 V				"	QC	"	"	"
		"	49	"	"		4.5 V				"					2.25 V			"	QB	"	"	"
		"	50	"	"	4.5 V					"						2.25 V		"	QA	"	"	"
		"	51	"	"	4.5 V	GND	GND	4.5 V		"		GND					2.25 V	"	RCO	-15	-70	"
	I_{CC}	3005	52	GND	5.5 V	GND	GND	GND	GND	GND	"		GND						"	V_{CC}		21	"
		3005	53	GND	GND	GND	GND	GND	GND	GND	"		GND						"	V_{CC}		21	"
2	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = +125°C and V_{IC} tests are omitted.																						
3	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = -55°C and V_{IC} tests are omitted.																						

See footnotes at end of device type 01.

Source: https://assist.dla.mil -- Downloaded: 20260-21T14:34Z
Check the source to verify that this is the current version before use.

TABLE III. Group A inspection for device type 01 - Continued.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Cases E,F	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Test limits		
			Case 1/ 2	2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20		Min	Max	Unit
			Test no.	$\overline{CLR}$	CLK	A	B	C	D	ENP	GND	$\overline{LOAD}$	ENT	QD	QC	QB	QA	RCO	V_{CC}				
7 T_C = +25°C	Truth table tests 5/	3014	54	B 6/	A 6/	A	A	A	A	A	GND	A	A	L	L	L	L	L	4.5 V	See 7/			
			55	A	A	B	B	B	B	"	"	"	B	"	"	"	"	"	"	"			
			56	"	B	A	A	A	A	"	"	"	A	"	"	"	"	"	"	"			
			57	"	A	A	A	A	A	"	"	"	A	H	"	"	"	"	"	"			
			58	"	A	B	B	B	B	"	"	"	B	"	"	"	"	"	"	"			
			59	"	B	A	A	A	A	"	"	"	A	"	"	"	"	"	"	"			
			60	"	A	A	A	A	A	"	"	"	A	"	"	H	L	"	"	"			
			61	"	A	B	B	B	B	"	"	"	B	"	"	"	"	"	"	"			
			62	"	B	A	A	A	A	"	"	"	A	"	"	"	"	"	"	"			
			63	"	A	A	A	A	A	"	"	"	A	"	"	"	H	"	"	"			
			64	"	A	B	B	B	B	"	"	"	B	"	"	"	"	"	"	"			
			65	"	B	A	A	A	A	"	"	"	A	"	"	"	"	"	"	"			
			66	"	A	A	A	A	A	"	"	"	A	"	H	L	L	"	"	"			
			67	"	A	B	B	B	B	"	"	"	B	"	"	"	"	"	"	"			
			68	"	B	A	A	A	A	"	"	"	A	"	"	"	"	"	"	"			
			69	"	A	A	A	A	A	"	"	"	A	"	"	"	H	"	"	"			
			70	"	A	B	B	B	B	"	"	"	B	"	"	"	"	"	"	"			
			71	"	B	A	A	A	A	"	"	"	A	"	"	"	"	"	"	"			
			72	"	A	A	A	A	A	"	"	"	A	"	"	"	L	"	"	"			
			73	"	A	B	B	B	B	"	"	"	B	"	"	"	"	"	"	"			
			74	"	B	A	A	A	A	"	"	"	A	"	"	"	"	"	"	"			
			75	"	A	A	A	A	A	"	"	"	A	"	"	"	H	"	"	"			
			76	"	A	B	B	B	B	"	"	"	B	"	"	"	"	"	"	"			
			77	"	B	A	A	A	A	"	"	"	A	"	"	"	"	"	"	"			
			78	"	A	A	A	A	A	"	"	"	A	"	H	L	L	"	"	"			
			79	"	A	B	B	B	B	"	"	"	B	"	"	"	"	"	"	"			
			80	"	B	A	A	A	A	"	"	"	A	"	"	"	"	"	"	"			
			81	"	A	A	A	A	A	"	"	"	A	"	"	"	H	"	"	"			
			82	"	A	B	B	B	B	"	"	"	B	"	"	"	"	"	"	"			
			83	"	B	A	A	A	A	"	"	"	A	"	"	"	"	"	"	"			
			84	"	A	"	"	"	"	"	"	"	"	"	L	"	"	"	"	"			
			85	B	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"			
			86	A	"	"	"	"	"	B	"	"	"	"	"	"	"	"	"	"			
			87	"	B	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"			
			88	"	A	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"			
			89	"	"	B	B	B	B	A	"	"	B	"	"	"	"	"	"	"			
			90	"	"	A	A	A	A	"	"	"	"	"	"	"	"	"	"	"			
			91	"	B	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"			
			92	"	A	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"			
			93	"	A	"	B	B	B	"	"	"	B	"	"	"	"	"	"	"			
			94	"	B	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"			
			95	"	A	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"			
			96	"	A	"	A	A	"	"	"	"	"	"	"	"	"	"	"	"			
			97	"	B	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"			
			98	"	A	"	"	"	"	"	"	"	"	"	H	"	H	"	"	"			
			99	"	A	"	"	"	"	"	"	"	A	"	"	"	"	"	"	"			
			100	"	B	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"			
			101	"	A	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"			
			102	"	A	"	B	B	A	"	"	"	B	"	"	"	"	"	"	"			
			103	"	B	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"			
			104	"	A	"	"	"	"	"	"	"	"	H	L	L	"	"	"	"			
			105	"	A	"	"	"	"	"	"	"	A	"	"	"	"	"	"	"			
			106	"	B	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"			
			107	"	A	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"			

See footnotes at end of device type 01.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:34Z
Check the source to verify that this is the current version before use.

TABLE III. Group A inspection for device type 01 - Continued.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Cases E,F Case 1/ 2 Test no.	1 2 CLR	2 3 CLK	3 4 A	4 5 B	5 7 C	6 8 D	7 9 ENP	8 10 GND	9 12 LOAD	10 13 ENT	11 14 QD	12 15 QC	13 17 QB	14 18 QA	15 19 RCO	16 20 V_{CC}	Measured terminal	Test limits Min	Test limits Max	Unit
7 T_C = +25°C	Truth table tests 5/	3014	108	B	A	A	A	A	A	A	GND	B	A	L	L	L	L	L	4.5 V	See 7/			
		"	109	A	A	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"			
		"	110	"	B	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"			
		"	111	"	A	"	"	"	"	"	"	"	"	H	H	H	H	H	"	"			
		"	112	B	A	"	"	"	"	"	"	A	"	L	L	L	L	L	"	"			
		"	113	A	A	"	B	B	B	B	"	B	B	"	"	"	"	"	"	"			
		"	114	"	B	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"			
		"	115	"	A	"	"	"	"	"	"	"	"	"	"	"	H	"	"	"			
		"	116	"	A	B	"	"	A	A	"	"	A	"	"	"	"	"	"	"			
		"	117	"	B	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"			
		"	118	"	A	"	"	"	"	"	"	"	"	H	"	"	L	"	"	"			
		"	119	"	A	"	A	"	B	B	"	"	"	"	"	"	"	"	"	"			
		"	120	"	B	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"			
		"	121	"	A	"	"	"	"	"	"	"	"	L	"	H	"	"	"	"			
		"	122	"	A	"	B	A	A	"	"	"	"	"	"	"	"	"	"	"			
		"	123	"	B	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"			
		"	124	"	A	"	"	"	"	"	"	"	"	H	H	L	"	"	"	"			
		"	125	"	A	"	A	"	"	"	"	"	B	"	"	"	"	"	"	"			
		"	126	"	B	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"			
		"	127	"	A	"	"	"	"	"	"	"	"	"	"	H	"	"	"	"			
		"	128	"	A	A	B	B	"	A	"	"	A	"	"	"	"	"	"	"			
		"	129	" 6/	B 6/	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"			
		"	130	"	A	"	"	"	"	"	"	"	"	"	L	L	H	H	"	"			
		"	131	"	A	"	A	A	B	B	"	"	B	"	"	"	L	L	"	"			
		"	132	"	B	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"			
		"	133	"	A	"	"	"	"	"	"	"	"	"	H	H	"	"	"	"			
		"	134	"	A	B	"	B	A	A	"	"	"	"	"	"	"	"	"	"			
		"	135	"	B	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"			
		"	136	"	A	"	"	"	"	"	"	"	"	H	L	"	L	"	"	"			
		"	137	"	A	A	"	"	B	"	"	"	A	"	"	"	"	"	"	"			
		"	138	"	B	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"			
		"	139	"	A	"	"	"	"	"	"	"	"	L	"	"	H	"	"	"			
		"	140	"	A	B	B	A	"	"	"	"	"	"	"	"	"	"	"	"			
		"	141	"	B	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"			
		"	142	"	A	"	"	"	"	"	"	"	"	"	H	L	L	"	"	"			
		"	143	"	A	A	"	"	"	"	"	"	"	"	"	"	"	"	"	"			
		"	144	"	B	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"			
		"	145	"	A	"	"	"	"	"	"	"	"	"	"	"	H	"	"	"			
		"	146	"	A	B	A	"	"	"	"	"	"	"	"	"	"	"	"	"			
		"	147	"	B	"	"	"	"	"	"	"	"	"	"	H	"	"	"	"			
		"	148	"	A	"	"	"	"	"	"	"	"	"	"	"	L	"	"	"			
8	Same tests, terminal conditions, and limits as for subgroup 7 except T_C = +125°C and -55°C.																						
9 T_C = +25°C	f_{MAX} 8/	3003	149	4.5 V	IN					4.5 V	GND	4.5 V	4.5 V				OUT		5.0 V	QA	30		MHz
	t_{PLH1}	Fig. 6	150	"	"					"	"	"	"				OUT		"	CLK to QA	4	15	ns
		"	151	"	"					"	"	"	"			OUT			"	CLK to QB	"	"	"
		"	152	"	"					"	"	"	"		OUT				"	CLK to QC	"	"	"
		"	153	"	"					"	"	"	"	OUT					"	CLK to QD	"	"	"
	t_{PHL1}	"	154	"	"					"	"	"	"				OUT		"	CLK to QA	6	17	"
		"	155	"	"					"	"	"	"			OUT			"	CLK to QB	"	"	"
		"	156	"	"					"	"	"	"		OUT				"	CLK to QC	"	"	"
		"	157	"	"					"	"	"	"	OUT					"	CLK to QD	"	"	"

See footnotes at end of device type 01.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:34Z
Check the source to verify that this is the current version before use.

TABLE III. Group A inspection for device type 01 - Continued.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Cases E,F Case 1/ 2	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Test limits Min	Max	Unit
				2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20				
			Test no.	CLR	CLK	A	B	C	D	ENP	GND	LOAD	ENT	QD	QC	QB	QA	RCO	V_{CC}				
T_C = +25°C	t_{PLH2}	3003 Fig. 6	158	4.5 V	IN					4.5 V	GND	4.5 V	4.5 V					OUT	5.0 V	CLK to RCO	8	26	ns
	t_{PHL2}	"	159	"	"					"	"	"	"					OUT	"	CLK to RCO	7	23	"
	t_{PLH8}	"	160	"						"	"	"	IN					OUT	"	ENT to RCO	5	17	"
	t_{PHL8}	"	161	"						"	"	"	IN					OUT	"	ENT to RCO	4	13	"
	t_{PHL12}	" " " "	162 163 164 165	IN " " "	2/ " " "	4.5 V	4.5 V	4.5 V	4.5 V		" " " "	GND " " "		OUT	OUT	OUT	OUT		" " " "	CLR to QA CLR to QB CLR to QC CLR to QD	8 " " "	24 " " "	" " " "
	t_{PHL13}	"	166	"	"	4.5 V	GND	GND	4.5 V		"	"						OUT	"	CLR to RCO	11	28	"
10	f_{MAX} 8/	3003 Fig. 6	167	Same terminal conditions as subgroup 9, except T_C = +125°C and limits are as shown.																	25		MHz
	t_{PLH1}	"	168																		4	18	ns
	t_{PHL1}	"	169																		6	20	"
	t_{PLH2}	"	170																		8	30	"
	t_{PHL2}	"	171																		7	25	"
	t_{PLH8}	"	172																		5	20	"
	t_{PHL8}	"	173																		4	16	"
	t_{PHL12}	"	174																		8	28	"
	t_{PHL13}	"	175																		11	31	"
11	Same tests, terminal conditions, and limits as for subgroup 10, except T_C = -55°C.																						

1/ Case 2 pins not referenced are NC.

2/ Apply one clock pulse prior to test as follows: 2.5 V min/5.5 V max, 0 V

3/ Method 3011 shall be used, except the output voltage shall be as specified herein, and the output current shall be operating rather that short circuit current. The output conditions have been chosen to produce a current that closely approximates one-half of the true short circuit output current I_{OS}.

4/ I_{IL} limits in µA are as follows:

Test	Circuit A	Circuit B	Circuit C
I_{IL}	0/-200	0/-100	0/-200

5/ Only a summary of attributes data is required.

6/ A = 3.0 V min; B = 0.0 V or GND.

7/ H >1.5 V; L <1.5 V.

8/ The f_{MAX} minimum limit specified is the frequency of the input pulse. The output frequency shall be one-half of the input frequency.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:34Z
Check the source to verify that this is the current version before use.

TABLE III. Group A inspection for device type 02.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Cases E,F Case 1/ 2 Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Test limits Min	Test limits Max	Unit
					CLR	CLK	A	B	C	D	ENP	GND	LOAD	ENT	QD	QC	QB	QA	RCO	V_{CC}			
1 T_C = +25°C	V_{OL1}	3007	1	4.5 V	2/				0.8 V	0.8 V		GND	GND		4 mA					4.5 V	QD		0.4	V
			2	"	"							"	"			4 mA				"	QC			"
			3	"	"			0.8 V				"	"		4 mA					"	QB			"
			4	"	"	0.8 V						"	"				4 mA			"	QA			"
			5	"	"							"	"	0.8 V				4 mA		"	RCO			"
	V_{OH1}	3006	6	"	"				2.0 V			"	"	-400 µA					"	QD	2.5		"
			7	"	"					2.0 V		"	"		-400 µA					"	QC	"		"
			8	"	"		2.0 V					"	"			-400 µA				"	QB	"		"
			9	"	"	2.0 V						"	"				-400 µA			"	QA	"		"
			10	"	"	2.0 V	0.8 V		2.0 V			"	"	2.0 V				-400 µA		"	RCO	"		"
	V_{IC}		11	-18 mA	-18 mA															"	$\overline{CLR}$		-1.5	V
			12			-18 mA														"	CLK			"
			13				18 mA													"	A			"
			14					-18 mA												"	B			"
			15						-18 mA											"	C			"
			16							18 mA										"	D			"
			17										-18 mA							"	ENP			"
			18											-18 mA						"	$\overline{LOAD}$			"
			19																	"	ENT			"
	I_{IL}	3009	20	0.4 V															5.5 V	$\overline{CLR}$	4/	4/	4/
			21		0.4 V															"	CLK	"	"	"
			22			0.4 V														"	A	"	"	"
			23				0.4 V													"	B	"	"	"
			24					0.4 V												"	C	"	"	"
			25						0.4 V											"	D	"	"	"
			26							0.4 V		4.5 V	4.5 V							"	ENP	"	"	"
			27							4.5 V		0.4 V	4.5 V							"	$\overline{LOAD}$	"	"	"
			28							4.5 V		4.5 V	0.4 V							"	ENT	"	"	"
	I_{IH1}	3010	29	2.7 V																"	$\overline{CLR}$		20	µA
			30		2.7 V															"	CLK		"	"
			31			2.7 V														"	A		"	"
			32				2.7 V													"	B		"	"
			33					2.7 V												"	C		"	"
			34						2.7 V											"	D		"	"
			35							GND		2.7 V	GND							"	$\overline{LOAD}$		"	"
			36							2.7 V		GND	GND							"	ENP		80	"
			37							GND		GND	2.7 V							"	ENT		40	"
	I_{IH2}	"	38	7.0 V																"	$\overline{CLR}$		100	"
			39		7.0 V															"	CLK		"	"
			40			7.0 V														"	A		"	"
			41				7.0 V													"	B		"	"
			42					7.0 V												"	C		"	"
			43						7.0 V											"	D		"	"
			44							GND		7.0 V	GND							"	$\overline{LOAD}$		"	"
			45							7.0 V		GND	GND							"	ENP		200	"
			46							GND		GND	7.0 V							"	ENT		200	"
	I_O 3/	3011	47	4.5 V	2/				4.5 V					2.25 V						"	QD	-30	-112	mA
			48	"	"			4.5 V							2.25 V					"	QC	"	"	"
			49	"	"											2.25 V				"	QB	"	"	"
			50	"	"	4.5 V											2.25 V			"	QA	"	"	"
			51	"	"	4.5 V	GND	GND	4.5 V			GND	4.5 V					2.25 V		"	RCO	-15	-70	"
	I_{CC}	3005	52	GND	5.5 V	GND	GND	GND	GND	GND	"	"	GND							"	V_{CC}		21	"
		3005	53	GND	GND	GND	GND	GND	GND	GND	"	"	GND							"	V_{CC}		21	"
2	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = +125°C and V_{IC} tests are omitted.																						
3	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = -55°C and V_{IC} tests are omitted.																						

See footnotes at end of device type 02.

Source: https://assist.dla.mil -- Downloaded: 2026-02-21T14:34Z
Check the source to verify that this is the current version before use.

TABLE III. Group A inspection for device type 02 - Continued.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Cases E,F Case 1/ 2 Test no.	1 CLR	2 CLK	3 A	4 B	5 C	6 D	7 ENP	8 GND	9 LOAD	10 ENT	11 QD	12 QC	13 QB	14 QA	15 RCO	16 V_{CC}	Measured terminal	Test limits Min	Test limits Max	Unit
			Cases E,F Case 1/ 2 Test no.	2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20				
				CLR	CLK	A	B	C	D	ENP	GND	LOAD	ENT	QD	QC	QB	QA	RCO	V_{CC}				
7 T_C = +25°C	Truth table tests 5/	3014	54	B 6/	B	B	B	B	B	A 6/	GND	B	A	X	X	X	X	X	4.5 V	See 7/			
		"	55	B	A	"	"	"	"	"	"	"	"	L	L	L	L	L	"	"			
		"	56	A	B	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"			
		"	57	A	A	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"			
		"	58	B	A	A	A	A	A	"	"	A	"	"	"	"	"	"	"	"			
		"	59	"	B	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"			
		"	60	"	A	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"			
		"	61	A	A	B	B	B	B	"	"	"	B	"	"	"	"	"	"	"			
		"	62	"	B	A	A	A	A	"	"	"	A	"	"	"	"	"	"	"			
		"	63	"	A	A	A	A	A	"	"	"	A	"	"	"	H	"	"	"			
		"	64	"	A	B	B	B	B	"	"	"	B	"	"	"	"	"	"	"			
		"	65	"	B	A	A	A	A	"	"	"	A	"	"	"	"	"	"	"			
		"	66	"	A	A	A	A	A	"	"	"	A	"	"	"	L	"	"	"			
		"	67	"	A	B	B	B	B	"	"	"	B	"	"	"	"	"	"	"			
		"	68	"	B	A	A	A	A	"	"	"	A	"	"	"	"	"	"	"			
		"	69	"	A	A	A	A	A	"	"	"	A	"	"	"	H	"	"	"			
		"	70	"	A	B	B	B	B	"	"	"	B	"	"	"	"	"	"	"			
		"	71	"	B	A	A	A	A	"	"	"	A	"	"	"	"	"	"	"			
		"	72	"	A	A	A	A	A	"	"	"	A	"	H	L	L	"	"	"			
		"	73	"	A	B	B	B	B	"	"	"	B	"	"	"	"	"	"	"			
		"	74	"	B	A	A	A	A	"	"	"	A	"	"	"	"	"	"	"			
		"	75	"	A	A	A	A	A	"	"	"	A	"	"	"	H	"	"	"			
		"	76	"	A	B	B	B	B	"	"	"	B	"	"	"	"	"	"	"			
		"	77	"	B	A	A	A	A	"	"	"	A	"	"	"	"	"	"	"			
		"	78	"	A	A	A	A	A	"	"	"	A	"	"	H	"	L	"	"			
		"	79	"	A	B	B	B	B	"	"	"	B	"	"	"	"	"	"	"			
		"	80	"	B	A	A	A	A	"	"	"	A	"	"	"	"	"	"	"			
		"	81	"	A	A	A	A	A	"	"	"	A	"	"	"	"	H	"	"			
		"	82	"	A	B	B	B	B	"	"	"	B	"	"	"	"	"	"	"			
		"	83	"	B	A	A	A	A	"	"	"	A	"	"	"	"	"	"	"			
		"	84	"	A	A	A	A	A	"	"	"	A	H	L	L	L	L	"	"			
		"	85	"	A	B	B	B	B	"	"	"	B	"	"	"	"	"	"	"			
		"	86	"	B	B	B	B	B	"	"	"	B	"	"	"	"	"	"	"			
		"	87	"	B	A	A	A	A	"	"	"	A	"	"	"	"	"	"	"			
		"	88	"	A	A	A	A	A	"	"	"	A	"	"	"	"	H	H	"			
		"	89	"	A	B	B	B	B	"	"	"	B	"	"	"	"	"	L	"			
		"	90	"	B	A	A	A	A	"	"	"	A	"	"	"	"	"	H	"			
		"	91	"	A	"	"	"	"	"	"	"	"	"	"	"	"	"	L	"			
		"	92	"	A	"	"	"	"	B	"	"	"	"	"	"	"	"	"	"			
		"	93	"	B	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"			
		"	94	"	A	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"			
		"	95	"	B	"	"	"	"	A	"	"	B	"	"	"	"	"	"	"			
		"	96	"	A	"	B	B	B	"	"	"	"	"	"	"	"	"	"	"			
		"	97	"	B	"	"	"	"	"	"	B	"	"	"	"	"	"	"	"			
		"	98	"	A	"	"	"	"	"	"	"	"	"	"	"	H	"	"	"			
		"	99	"	A	"	A	A	A	"	"	"	"	"	"	"	"	"	"	"			
		"	100	"	B	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"			
		"	101	"	A	"	"	"	"	"	"	"	"	"	H	H	"	"	"	"			
		"	102	"	A	"	"	"	"	"	"	A	"	"	"	"	"	"	"	"			
		"	103	"	B	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"			
		"	104	"	A	A	"	"	"	"	"	"	"	"	"	"	"	"	"	"			
		"	105	"	B	"	B	B	"	"	"	B	"	"	"	"	"	"	"	"			
		"	106	"	A	"	"	"	"	"	"	B	"	H	L	L	"	"	"	"			
		"	107	"	A	"	"	"	"	"	"	A	"	"	"	"	"	"	"	"			
		"	108	"	B	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"			
		"	109	"	A	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"			
		"	110	B	A	"	A	A	"	"	"	"	A	"	"	"	"	H	L	"			
		"	111	B	A	"	"	"	"	"	"	"	"	L	"	"	L	"	"	"			
		"	112	A	A	"	"	"	"	"	"	B	"	"	"	"	"	"	"	"			
		"	113	"	B	"	B	B	"	"	"	"	"	"	"	"	"	"	"	"			
		"	114	"	A	"	"	"	"	"	"	"	"	H	"	"	H	H	"	"			
		"	115	"	B	"	"	"	"	"	"	"	"	H	"	"	H	H	"	"			

See footnotes at end of device type 02.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:34Z
Check the source to verify that this is the current version before use.

TABLE III. Group A inspection for device type 02 - Continued.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Cases E,F Case 1/ 2 Test no.	1 2 CLK	2 3 CLK	3 4 A	4 5 B	5 7 C	6 8 D	7 9 ENP	8 10 GND	9 12 LOAD	10 13 ENT	11 14 QD	12 15 QC	13 17 QB	14 18 QA	15 19 RCO	16 20 V_{CC}	Measured terminal	Test limits Min	Test limits Max	Unit
7 T_C = +25°C	Truth table tests 5/	3014	116	A	A	A	B	B	B	A	GND	B	A	H	L	L	H	H	4.5 V	See 7/			
		"	117	B	B	"	"	"	"	"	"	"	"	H	"	"	H	H	"	"			
		"	118	B	A	"	"	"	"	"	"	"	"	L	"	"	L	L	"	"			
		"	119	A	A	B	"	"	"	"	"	"	B	"	"	"	"	"	"	"			
		"	120	"	B	"	A	"	"	"	"	"	"	"	"	"	"	"	"	"			
		"	121	"	A	"	"	"	"	"	"	"	"	"	"	H	"	"	"	"			
		"	122	"	A	"	B	"	A	"	"	"	"	"	"	"	"	"	"	"			
		"	123	"	B	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"			
		"	124	"	A	"	"	"	"	"	"	"	"	H	"	L	"	"	"	"			
		"	125	"	A	"	"	"	B	"	"	"	A	"	"	"	"	"	"	"			
		"	126	"	B	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"			
		"	127	"	A	"	"	"	"	"	"	"	"	L	"	"	"	"	"	"			
		"	128	"	A	"	A	A	A	"	"	"	"	"	"	"	"	"	"	"			
		"	129	"	B	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"			
		"	130	"	A	"	"	"	"	"	"	"	"	H	H	H	"	"	"	"			
		"	131	"	A	A	"	B	B	"	"	"	"	"	"	"	"	"	"	"			
		"	132	"	B	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"			
		"	133	"	A	"	"	"	"	"	"	"	"	L	L	"	H	"	"	"			
		"	134	"	A	B	B	A	"	"	"	"	"	"	"	"	"	"	"	"			
		"	135	"	B	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"			
		"	136	"	A	"	"	"	"	"	"	"	"	"	H	L	L	"	"	"			
		"	137	"	A	A	"	"	"	"	"	"	"	"	"	"	"	"	"	"			
		"	138	"	B	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"			
		"	139	"	A	"	"	"	"	"	"	"	"	"	"	"	H	"	"	"			
		"	140	"	A	B	A	"	"	"	"	"	"	"	"	"	"	"	"	"			
		"	141	"	B	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"			
		"	142	"	A	"	"	"	"	"	"	"	"	"	"	H	L	"	"	"			
		"	143	"	A	"	B	B	A	"	"	"	"	"	"	"	"	"	"	"			
		"	144	"	B	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"			
		"	145	"	A	"	"	"	"	"	"	"	"	H	L	L	"	"	"	"			
		"	146	"	A	A	"	"	"	"	"	"	"	"	"	"	"	"	"	"			
		"	147	"	B	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"			
		"	148	"	A	"	"	"	"	"	"	"	"	"	"	"	H	H	"	"			
		"	149	"	A	"	A	A	"	"	"	A	"	"	"	"	"	"	"	"			
		"	150	"	B	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"			
		"	151	"	A	"	"	"	"	"	"	"	"	"	"	"	L	L	"	"			
8	Same tests, terminal conditions, and limits as for subgroup 7, except T_C = +125°C and -55°C.																						
9 T_C = +25°C	f_{MAX} 8/	3003 Fig. 6	152	4.5 V	IN					4.5 V	GND	4.5 V	4.5 V				OUT		5.0 V	QA	30		MHz
	t_{PLH1}	"	153	"	"					"	"	"	"				OUT		"	CLK to QA	4	15	ns
		"	154	"	"					"	"	"	"			OUT			"	CLK to QB	"	"	"
		"	155	"	"					"	"	"	"		OUT				"	CLK to QC	"	"	"
		"	156	"	"					"	"	"	"	OUT					"	CLK to QD	"	"	"
	t_{PHL1}	"	157	"	"					"	"	"	"				OUT		"	CLK to QA	6	17	"
		"	158	"	"					"	"	"	"			OUT			"	CLK to QB	"	"	"
		"	159	"	"					"	"	"	"		OUT				"	CLK to QC	"	"	"
		"	160	"	"					"	"	"	"	OUT					"	CLK to QD	"	"	"

See footnotes at end of device type 02.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:34Z
Check the source to verify that this is the current version before use.

TABLE III. Group A inspection for device type 02 - Continued.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Cases E,F Case 1/ 2 Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Test limits Min	Test limits Max	Unit
				2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20				
				$\overline{CLR}$	CLK	A	B	C	D	ENP	GND	$\overline{LOAD}$	ENT	QD	QC	QB	QA	RCO	V_{CC}				
9 T_C = +25°C	t_{PLH2}	3003 Fig. 6	161	4.5 V	IN					4.5 V	GND	4.5 V	4.5 V					OUT	5.0 V	CLK to RCO	8	26	ns
	t_{PHL2}	"	162	"	"					"	"	"	"					OUT	"	CLK to RCO	7	23	"
	t_{PLH8}	"	163	"						"	"	"	IN					OUT	"	ENT to RCO	5	17	"
	t_{PHL8}	"	164	"						"	"	"	IN					OUT	"	ENT to RCO	4	13	"
10	f_{MAX} 8/	3003 Fig. 6	165	Same terminal conditions as subgroup 9, except T_C = +125°C and limits are as shown.																	25		MHz
	t_{PLH1}	"	166																		4	18	ns
	t_{PHL1}	"	167																		6	20	"
	t_{PLH2}	"	168																		8	30	"
	t_{PHL2}	"	169																		7	25	"
	t_{PLH8}	"	170																		5	20	"
	t_{PHL8}	"	171																		4	16	"
11	Same tests, terminal conditions, and limits as for subgroup 10, except T_C = -55°C.																						

1/ Case 2 pins not referenced are NC.

2/ Apply one clock pulse prior to test as follows: ---2.5 V min/5.5 V max; 0 V

3/ Method 3011 shall be used, except the output voltage shall be as specified herein, and the output current shall be operating rather that short circuit current. The output conditions have been chosen to produce a current that closely approximates one-half of the true short circuit output current I_{OS}.

4/ I_{IL} limits in µA are as follows:

Test	Circuit A	Circuit B	Circuit C
I_{IL}	0/-200	0/-100	0/-200

5/ Only a summary of attributes data is required.

6/ A = 3.0 V min; B = 0.0 V or GND.

7/ H >1.5 V; L <1.5 V.

8/ The f_{MAX} minimum limit specified is the frequency of the input pulse. The output frequency shall be one-half of the input frequency.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:34Z
Check the source to verify that this is the current version before use.

TABLE III. Group A inspection for device type 03.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Cases E,F Case 1/ 2 Test no.	1 U/D	2 CLK	3 A	4 B	5 C	6 D	7 ENP	8 GND	9 LOAD	10 ENT	11 QD	12 QC	13 QB	14 QA	15 RCO	16 V_{CC}	Measured terminal	Test limits Min	Test limits Max	Unit
			Cases E,F Case 1/ 2 Test no.	2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20				
1 T_C = +25°C	V_{OH1}	3006	1	4.5 V	2/						GND	0.8 V	0.8 V	-400 µA					4.5 V	QD	2.5		V
		"	2	"	"						"	"	"		-400 µA				"	QC	"		"
		"	3	"	"		2.0 V				"	"	"						"	QB	"		"
		"	4	"	"	2.0 V					"	"	"			-400 µA			"	QA	"		"
		"	5	0.8 V	"			2.0 V	2.0 V	0.8 V	"	"	"					-400 µA	"	RCO	"		"
	V_{OL1}	3007	6	4.5 V	"					4.5 V	"	"	0.5 V	4 mA					"	QD		0.4	"
		"	7	"	"			0.8 V		"	"	"	"		4 mA				"	QC		"	"
		"	8	"	"		0.8 V		0.8 V	"	"	"	"			4 mA			"	QB		"	"
		"	9	"	"	0.8 V				"	"	"	"				4 mA		"	QA		"	"
		"	10	0.8 V	"					0.8 V	"	"	0.8 V					4 mA	"	RCO		"	"
	V_{IC}		11	-18 mA	-18 mA						"								"	U/D		-1.5	"
			12			-18 mA					"								"	CP		"	"
			13				-18 mA				"								"	A		"	"
			14					-18 mA			"								"	B		"	"
			15						-18 mA		"								"	C		"	"
			16							-18 mA	"								"	D		"	"
			17								"	-18 mA							"	ENP		"	"
			18								"		-18 mA						"	LOAD		"	"
			19								"								"	ENT		"	"
	I_{IL}	3009	20	0.4 V							"								5.5 V	U/D	4/	4/	4/
		"	21		0.4 V						"								"	CP	"	"	"
		"	22			0.4 V					"								"	A	"	"	"
		"	23				0.4 V				"								"	B	"	"	"
		"	24					0.4 V			"								"	C	"	"	"
		"	25						0.4 V		"								"	D	"	"	"
		"	26							0.4 V	"								"	ENP	"	"	"
		"	27								"	0.4 V							"	LOAD	"	"	"
		"	28								"		0.4 V						"	ENT	"	"	"
	I_{IH1}	3010	29	2.7 V							"								"	U/D		20	µA
		"	30		2.7 V						"								"	CLK		"	"
		"	31			2.7 V					"								"	A		"	"
		"	32				2.7 V				"								"	B		"	"
		"	33					2.7 V			"								"	C		"	"
		"	34						2.7 V		"								"	D		"	"
		"	35							2.7 V	"								"	ENP		"	"
		"	36								"	2.7 V							"	LOAD		"	"
		"	37								"		2.7 V						"	ENT		"	"
	I_{IH2}	"	38	7.0 V							"								"	U/D		100	"
		"	39		7.0 V						"								"	CLK		"	"
		"	40			7.0 V					"								"	A		"	"
		"	41				7.0 V				"								"	B		"	"
		"	42					7.0 V			"								"	C		"	"
		"	43						7.0 V		"								"	D		"	"
		"	44							7.0 V	"								"	ENP		"	"
		"	45								"	7.0 V							"	LOAD		"	"
		"	46								"		7.0 V						"	ENT		"	"

See footnotes at end of device type 03.

Source: https://assist.dla.mil -- Downloaded: 2026-02-21T14:34Z
Check the source to verify that this is the current version before use.

TABLE III. Group A inspection for device type 03 - Continued.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Cases E,F Case 1/ 2 Test no.	1 U/D	2 CLK	3 A	4 B	5 C	6 D	7 ENP	8 GND	9 LOAD	10 ENT	11 QD	12 QC	13 QB	14 QA	15 RCO	16 V_{CC}	Measured terminal	Test limits Min	Test limits Max	Unit
			Case 1/ 3 Test no.	2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20				
1 T_C = +25°C	I_O 2/	3011	47	5.5 V	2/	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	GND	GND	5.5 V	2.25 V	2.25 V				5.5 V	QD	-30	-112	mA
		"	48	"	"	"	"	"	"	"	"	"	"			2.25 V			"	QC	"	"	"
		"	49	"	"	"	"	"	"	"	"	"	"				2.25 V		"	QB	"	"	"
		"	50	"	"	"	"	"	"	"	"	"	"					2.25 V	"	QA	"	"	"
		"	51	"	"	"	"	"	"	"	"	"	"						"	RCO	-15	-70	"
	I_{CC}	3005	52	GND	2/	GND	GND	GND	GND	GND	"	5.5 V	GND						"	V_{CC}		25	"
2	Same tests, terminal conditions, and limits, except T_C = +125°C and V_{IC} tests are omitted.																						
3	Same tests, terminal conditions, and limits, except T_C = -55°C and V_{IC} tests are omitted.																						
7 T_C = +25°C	Truth table tests 5/	3014	53	A 6/	B 6/	B	B	B	B	B	GND	B	B	X	X	X	X	X	4.5 V	See 7/			
		"	54	"	A	"	"	"	"	"	"	"	A	L 6/	L	L	L	H 6/	"	"			
		"	55	"	B	"	"	"	"	"	"	"	"	"	"	H	H	"	"	"			
		"	56	"	A	"	"	"	"	"	"	"	"	"	H	"	L	"	"	"			
		"	57	"	B	"	"	"	"	"	"	"	"	"	"	L	H	"	"	"			
		"	58	"	A	"	"	"	"	"	"	"	"	"	"	H	L	"	"	"			
		"	59	"	B	"	"	"	"	"	"	"	"	"	"	L	H	"	"	"			
		"	60	"	A	"	"	"	"	"	"	"	"	"	"	H	H	"	"	"			
		"	61	"	B	"	"	"	"	"	"	"	"	"	"	L	L	"	"	"			
		"	62	"	A	"	"	"	"	"	"	"	"	"	"	H	L	"	"	"			
		"	63	"	B	"	"	"	"	"	"	"	"	"	"	L	H	"	"	"			
		"	64	"	A	"	"	"	"	"	"	"	"	"	"	H	H	"	"	"			
		"	65	"	B	"	"	"	"	"	"	"	"	"	"	L	L	"	"	"			
		"	66	"	A	"	"	"	"	"	"	"	"	"	"	H	L	"	"	"			
		"	67	"	B	"	"	"	"	"	"	"	"	"	"	L	H	"	"	"			
		"	68	"	A	"	"	"	"	"	"	"	"	"	"	H	H	"	"	"			
		"	69	"	B	"	"	"	"	"	"	"	"	"	"	L	L	"	"	"			
		"	70	"	A	"	"	"	"	"	"	"	"	"	"	H	L	"	"	"			
		"	71	"	B	"	"	"	"	"	"	"	"	"	"	L	H	"	"	"			
		"	72	"	A	"	"	"	"	"	"	"	"	"	"	H	H	"	"	"			
		"	73	"	B	"	"	"	"	A	"	"	"	"	"	L	L	"	"	"			
		"	74	"	A	"	"	"	"	B	"	"	"	"	"	H	L	"	"	"			
		"	75	"	B	"	"	"	"	"	"	"	"	"	"	L	H	"	"	"			
		"	76	"	A	"	"	"	"	"	"	"	"	"	"	H	H	"	"	"			
		"	77	"	B	"	"	"	"	"	"	"	"	"	"	L	L	"	"	"			
		"	78	"	A	"	"	"	"	"	"	"	"	"	"	H	L	"	"	"			
		"	79	"	B	"	"	"	"	"	"	"	"	"	"	L	H	"	"	"			
		"	80	"	A	"	"	"	A	"	"	"	"	"	"	H	H	"	"	"			
		"	81	"	B	"	"	"	"	"	"	B	"	"	"	L	L	"	"	"			
		"	82	"	A	"	"	"	"	"	"	A	"	"	"	H	L	"	"	"			
		"	83	"	B	"	"	"	"	"	"	"	"	"	"	L	H	"	"	"			
		"	84	"	A	"	"	"	"	"	"	"	"	"	"	H	H	"	"	"			
		"	85	"	B	"	"	"	"	"	"	"	"	"	"	L	L	"	"	"			
		"	86	"	A	"	"	"	"	"	"	"	"	"	"	H	L	"	"	"			
		"	87	"	B	"	"	"	"	"	"	"	"	"	"	L	H	"	"	"			
		"	88	"	A	"	"	"	"	"	"	"	"	"	"	H	H	"	"	"			
		"	89	"	B	"	"	"	"	"	"	"	"	"	"	L	L	"	"	"			
		"	90	"	A	"	"	"	"	"	"	"	"	"	"	H	L	"	"	"			
		"	91	"	B	"	"	"	"	"	"	"	"	"	"	L	H	"	"	"			
		"	92	"	A	"	"	"	"	"	"	"	"	"	"	H	H	"	"	"			
		"	93	"	B	"	"	"	"	"	"	"	"	"	"	L	L	"	"	"			
		"	94	"	B	"	"	"	"	"	"	"	"	"	"	H	L	"	"	"			

See footnotes at end of device type 03.

Source: https://assist.dla.mil -- Downloaded: 2026-02-21T14:34Z
Check the source to verify that this is the current version before use.

TABLE III. Group A inspection for device type 03 - Continued.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Cases E,F Case 1/ 2 Test no.	1 U/$\overline{D}$	2 CLK	3 A	4 B	5 C	6 D	7 $\overline{ENP}$	8 GND	9 $\overline{LOAD}$	10 $\overline{ENT}$	11 QD	12 QC	13 QB	14 QA	15 $\overline{RCO}$	16 V_{CC}	Measured terminal	Test limits Min	Test limits Max	Unit
			Case 1/ 2	2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20				
7 T_C = +25°C	Truth table test 5/	3014	95	B	A	A	B	B	A	B	GND	A	B	L	H	H	L	L	4.5 V	See 7/			
	"	"	96	"	B	" 6/	"	"	"	"	"	"	"	" 6/	" 6/	H	L	H	"				
	"	"	97	"	A	"	"	"	"	"	"	"	"	"	"	L	H	"	"				
	"	"	98	"	B	"	"	"	"	"	"	"	"	"	"	"	H	"	"				
	"	"	99	"	A	"	"	"	"	"	"	"	"	"	"	"	L	L	"				
	"	"	100	"	B	"	"	"	"	A	"	"	"	"	"	"	"	"	"				
	"	"	101	"	A	"	"	"	"	A	"	"	"	"	"	"	"	"	"				
	"	"	102	"	B	"	"	"	"	B	"	"	"	"	"	"	"	"	"				
	"	"	103	"	B	"	"	"	"	"	"	"	A	"	"	"	"	H	"				
	"	"	104	"	A	"	"	"	"	"	"	"	A	"	"	"	"	H	"				
	"	"	105	"	B	"	"	"	"	"	"	"	B	"	"	"	"	L	"				
	"	"	106	"	B	B	A	A	"	"	"	B	"	"	"	"	L	"				
	"	"	107	"	A	B	A	A	"	"	"	B	"	"	"	H	H	H	"				
8	Same tests, terminal conditions, and limits as for subgroup 7, except T_C = +125°C and -55°C.																						
9 T_C = +25°C	f_{MAX} 8/	See Fig. 7	108	5.0 V	IN						GND	5.0 V					OUT		5.0 V	CP to Qn	30		MHz
			109	GND	"						"	5.0 V					OUT		"	CP to Qn	30		MHz
	t_{PLH1}	3003 Fig. 7	110	5.0 V	"	4.5 V				GND	"	GND	GND				OUT		"	CLK to QA	5	16	ns
			111	"	"		4.5 V			"	"	"	"			OUT			"	CLK to QB	"	"	"
			112	"	"			4.5 V		"	"	"	"		OUT				"	CLK to QC	"	"	"
			113	"	"				4.5 V	"	"	"	"	OUT					"	CLK to QD	"	"	"
	t_{PHL1}	"	114	"	"	GND				"	"	"	"				OUT		"	CLK to QA	"	"	"
		"	115	"	"		GND			"	"	"	"			OUT			"	CLK to QB	"	"	"
		"	116	"	"			GND		"	"	"	"		OUT				"	CLK to QC	"	"	"
		"	117	"	"				GND	"	"	"	"	OUT					"	CLK to QD	"	"	"
	t_{PLH2}	"	118	"	"	5.0 V	GND	GND	5.0 V	"	"	8/	"					OUT	"	CLK to $\overline{RCO}$	10	28	"
	t_{PHL2}	"	119	"	"	"	"	"	"	"	"	8/	"					"	"	CLK to $\overline{RCO}$	6	18	"
	t_{PLH8}	"	120	"	2/	"	"	"	"	"	"	GND	IN					"	"	$\overline{ENT}$ to $\overline{RCO}$	5	16	"
	t_{PHL8}	"	121	"	"	"	"	"	"		"	"	IN					"	"	$\overline{ENT}$ to $\overline{RCO}$	3	13	"
	t_{PLH11}	"	122	IN	"	GND	"	"	GND	GND	"	"	GND					"	"	U/$\overline{D}$ to $\overline{RCO}$	5	23	"
	t_{PHL11}	"	123	IN	"	GND	"	"	GND	GND	"	"	GND					"	"	U/$\overline{D}$ to $\overline{RCO}$	5	19	"

See footnotes at end of device type 03.

Source: https://assist.dla.mil -- Downloaded: 2026-05-21T14:34Z
Check the source to verify that this is the current version before use.

TABLE III. Group A inspection for device type 03 - Continued.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Cases E,F Case 1/ 2 Test no.	1 2 U/D	2 3 CLK	3 4 A	4 5 B	5 7 C	6 8 D	7 9 ENP	8 10 GND	9 12 LOAD	10 13 ENT	11 14 QD	12 15 QC	13 17 QB	14 18 QA	15 19 RCO	16 20 V_{CC}	Measured terminal	Test limits Min	Max	Unit
10 T_C = +125°C	f_{MAX} 8/																				25		MHz
	t_{PLH1}																				5	19	ns
	t_{PHL1}																				5	20	"
	t_{PLH2}																				10	32	"
	t_{PHL2}	Same tests and terminal conditions as for subgroup 9, and limits are as shown.																			6	22	"
	t_{PLH}																				5	19	"
	t_{PHL}																				3	16	"
	t_{PLH11}																				5	25	"
	t_{PHL11}																				5	23	"
11	Same tests, terminal conditions, and limits as for subgroup 9, except T_C = -55°C.																						

1/ Case 2 pins not referenced are NC.

2/ Apply one clock pulse prior to test as follows: ---2.5 V min/5.5 V max, 0 V

3/ Method 3011 shall be used, except the output voltage shall be as specified herein, and the output current shall be operating rather that short circuit current. The output conditions have been chosen to produce a current that closely approximates one-half of the true short circuit output current I_{OS}.

4/ I_{IL} limits in µA are as follows:

Test	Circuit A	Circuit B	Circuit C
I_{IL}	0/-200	0/-100	0/-200

5/ Only a summary of attributes data is required.

6/ A = 3.0 V min; B = 0.0 V or GND.

7/ H >1.5 V; L <1.5 V.

8/ The f_{MAX} minimum limit specified is the frequency of the input pulse. The output frequency shall be one-half of the input frequency.

Source: https://assist.dla.mil -- Downloaded: 2026-02-21T14:34Z
Check the source to verify that this is the current version before use.

TABLE III. Group A inspection for device type 04.
Terminal conditions (pins not designated may be high ≥2.0 V, or low ≤0.8 V, or open).

Subgroup	Symbol	MIL-STD-883	Cases R,S,X Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	Measured terminal	Limits Min	Limits Max	Unit
				$\overline{ALOAD}$	CLK	A	B	C	D	ENP	$\overline{ACLR}$	$\overline{SCLR}$	GND	$\overline{SLOAD}$	ENT	QD	QC	QB	QA	G	CCO	RCO	V_{CC}				
1 T_C = +25°C	V_{OH1}	3006	1	2.0 V	1/	2.0 V	0.8 V	2.0 V	2.0 V	2.0 V	2.0 V	2.0 V	GND	0.8 V	2.0 V					GND		-400 µA	4.5 V	RCO	2.5		V
		"	2	2.0 V	"	"	0.8 V	2.0 V	2.0 V	2.0 V	"	"	"	"	2.0 V					"	-400 µA		"	CCO	2.5		"
	V_{OH2}	"	3	0.8 V	"	"				0.8 V	"	"	"	"	0.8 V			-1 mA	-1 mA	"			"	QA	2.4		"
		"	4	"	"		2.0 V			"	"	"	"	"	"		-1 mA			"			"	QB	"		"
		"	5	"	"			2.0 V		"	"	"	"	"	"	-1 mA				"			"	QC	"		"
		"	6	"	"				2.0 V	"	"	"	"	"	"					"			"	QD	"		"
	V_{OL1}	3007	7	2.0 V	"	2.0 V	0.8 V	0.8 V	2.0 V	"	"	"	"	"	"					"		4 mA	"	RCO		0.4	"
		"	8	2.0 V	"	2.0 V	0.8 V	0.8 V	2.0 V	"	"	"	"	"	"					"	4 mA		"	CCO		"	"
	V_{OL2}	"	9	0.8 V	"	0.8 V				"	"	"	"	"	"					"			"	QA		"	"
		"	10	"	"		0.8 V			"	"	"	"	"	"			12 mA	12 mA	"			"	QB		"	"
		"	11	"	"			0.8 V		"	"	"	"	"	"		12 mA			"			"	QC		"	"
		"	12	"	"				0.8 V	"	"	"	"	"	"	12 mA				"			"	QD		"	"
	V_{IC}		13	-18 mA																			"				
			14		-18 mA																			$\overline{ALOAD}$		-1.5	"
			15			-18 mA																		CLK		"	"
			16				-18 mA																	A		"	"
			17					-18 mA																B		"	"
			18						-18 mA															C		"	"
			19							-18 mA														D		"	"
			20								-18 mA													ENP		"	"
			21									-18 mA												$\overline{ACLR}$		"	"
			22																					$\overline{SCLR}$		"	"
			23											-18 mA										$\overline{SLOAD}$		"	"
			24												-18 mA					-18 mA				ENT		"	"
																								G		"	"
	I_{IL}	3009	25	0.4 V																			5.5 V	$\overline{ALOAD}$	2/	2/	2/
		"	26		0.4 V																		"	CLK	"	"	"
		"	27			0.4 V																	"	A	"	"	"
		"	28				0.4 V																"	B	"	"	"
		"	29					0.4 V															"	C	"	"	"
		"	30						0.4 V														"	D	"	"	"
		"	31							0.4 V													"	ENP	"	"	"
		"	32								0.4 V												"	$\overline{ACLR}$	"	"	"
		"	33									0.4 V											"	$\overline{SCLR}$	"	"	"
		"	34											0.4 V									"	$\overline{SLOAD}$	"	"	"
		"	35												0.4 V								"	ENT	"	"	"
		"	36																	0.4 V			"	G	"	"	"
	I_{IH1}	3010	37							2.7 V													"	ENP		40	"
		"	38												2.7 V								"	ENT		40	"
		"	39	2.7 V																			"	$\overline{ALOAD}$		20	"
		"	40		2.7 V																		"	CLK		"	"
		"	41			2.7 V																	"	A		"	"
		"	42				2.7 V																"	B		"	"
		"	43					2.7 V															"	C		"	"
		"	44						2.7 V														"	D		"	"
		"	45								2.7 V												"	$\overline{ACLR}$		"	"
		"	46									2.7 V											"	$\overline{SCLR}$		"	"
		"	47											2.7 V									"	$\overline{SLOAD}$		"	"
		"	48																	2.7 V			"	G		"	"

See footnotes at end of device type 04.

MIL-M-38510/381(USAF)

Source: https://assist.dla.mil -- Downloaded: 2026-02-21T14:34Z
Check the source to verify that this is the current version before use.

TABLE III. Group A inspection for device type 04 - Continued.
Terminal conditions (pins not designated may be high ≥2.0 V, or low ≤0.8 V, or open).

Subgroup	Symbol	MIL-STD-883 Test no.	Cases R,S,2	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	Measured terminal	Limits		Unit
				ALOAD	CLK	A	B	C	D	ENP	ACLR	SCLR	GND	SLOAD	ENT	QD	QC	QB	QA	$\overline{G}$	CCO	RCO	V_{CC}		Min	Max	
1 T_C = +25°C	I_{IH2}	3010	49							7.0 V			GND		7.0 V								5.5 V	ENP		200	µA
		"	50										"										"	ENT		200	"
		"	51	7.0 V									"										"	ALOAD		100	"
		"	52		7.0 V								"										"	CLK	"	"	"
		"	53			7.0 V							"										"	A	"	"	"
		"	54				7.0 V						"										"	B	"	"	"
		"	55					7.0 V					"										"	C	"	"	"
		"	56						7.0 V				"										"	D	"	"	"
		"	57								7.0 V		"										"	$\overline{ACLR}$	"	"	"
		"	58									7.0 V	"										"	$\overline{SCLR}$	"	"	"
		"	59										"	7.0 V									"	$\overline{SLOAD}$	"	"	"
		"	60										"							7.0 V				"	$\overline{G}$	"	"	"
	I_O 3/	3011	61	5.5 V	1/	5.5 V	GND	GND	5.5 V	5.5 V	5.5 V	5.5 V	"	GND	5.5 V					GND		2.25 V	"	RCO	-15	-70	mA
		"	62	5.5 V	"	"	GND	GND	5.5 V	5.5 V	"	"	"	"	5.5 V					"	2.25 V		"	CCO	-15	-70	"
		"	63		"						"	"	"	"	GND				2.25 V	"			"	QA	-30	-112	"
		"	64		"		5.5 V				"	"	"	"	"			2.25 V		"			"	QB	"	"	"
		"	65		"			5.5 V			"	"	"	"	"	2.25 V	2.25 V						"	QC	"	"	"
		"	66		"				5.5 V		"	"	"	"	"								"	QD	"	"	"
	I_{OZL}		67	2.0 V	"	2.0 V	2.0 V	2.0 V	2.0 V	2.0 V	2.0 V	2.0 V	"	0.8 V	2.0 V					2.0 V				QA		-20	µA
			68	"	"	"	"	"	"	"	"	"	"	"	"			0.4 V	0.4 V	"				QB		"	"
			69	"	"	"	"	"	"	"	"	"	"	"	"		0.4 V							QC		"	"
			70	"	"	"	"	"	"	"	"	"	"	"	"	0.4 V								QD		"	"
	I_{OZH}		71	"	"	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	"	"	"	"	0.8 V					"				QA		20	"
			72	"	"	"	"	"	"	"	"	"	"	"	"			2.7 V	2.7 V	"				QB		"	"
			73	"	"	"	"	"	"	"	"	"	"	"	"		2.7 V							QC		"	"
			74	"	"	"	"	"	"	"	"	"	"	"	"	2.7 V								QD		"	"
	I_{CCH}	3005	75	4.5 V	"	4.5 V	4.5 V	4.5 V	4.5 V	4.5 V	4.5 V	4.5 V	"	GND	4.5 V					GND			"	V_{CC}		27	mA
	I_{CCL}	"	76	GND	"	GND	GND	GND	GND	GND	GND	GND	"	"	GND					GND			"	V_{CC}		33	"
	I_{CCZ}	"	77	GND	"	GND	GND	GND	GND	GND	4.5 V	4.5 V	"	"	GND					4.5 V			"	V_{CC}		36	"
2	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = +125°C and V_{IC} tests are omitted.																										
3	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = -55°C and V_{IC} tests are omitted.																										
7 T_C = +25°C	Truth table tests 4/	3015 Fig.8	78	A 5/	A	A	A	A	A	A	B 5/	A	GND	A	A	L 5/	L	L	L	L H 5/	B	L	L	4.5 V	See 6/		
		"	79	B	B	B	B	B	B	B	A	A	"	B	B	"	"	"	"	H	"	"	"		"		
		"	80	A	A	A	A	A	A	A	"	A	"	A	A	"	"	H	"	L	"	"	"		"		
		"	81	"	B	B	B	B	B	B	"	B	"	"	B	"	"	"	"	H	"	"	"		"		
		"	82	"	A	A	A	A	A	A	"	A	"	"	A	"	"	"	"	L	"	"	"		"		
		"	83	"	B	B	B	B	B	B	"	B	"	"	B	"	"	"	"	H	"	"	"		"		
		"	84	"	A	A	A	A	A	A	"	A	"	"	A	"	"	"	"	L	"	"	"		"		
		"	85	"	B	B	B	B	B	B	"	B	"	"	B	"	"	"	"	H	"	"	"		"		
		"	86	"	A	A	A	A	A	A	"	A	"	"	A	"	"	"	"	L	"	"	"		"		
		"	87	"	B	B	B	B	B	B	"	B	"	"	B	"	"	"	"	H	"	"	"		"		
		"	88	"	A	A	A	A	A	A	"	A	"	"	A	"	"	"	"	L	"	"	"		"		
		"	89	"	B	B	B	B	B	B	"	B	"	"	B	"	"	"	"	H	"	"	"		"		
		"	90	"	A	A	A	A	A	A	"	A	"	"	A	"	"	"	"	L	"	"	"		"		
		"	91	"	A	A	A	A	A	A	"	A	"	"	A	"	"	"	"	H	"	"	"		"		

See footnotes at end of device type 04.

Source: https://assist.dla.mil -- Downloaded: 2026-02-21T14:34Z
Check the source to verify that this is the current version before use.

Specified herein.

TABLE III. Group A inspection for device type 04 - Continued.
Terminal conditions (pins not designated may be high ≥2.0 V, or low ≤0.8 V, or open).

Subgroup	Symbol	MIL-STD-883	Cases R,S,2 Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	Measured terminal	Limits Min	Limits Max	Unit
				$\overline{ALOAD}$	CLK	A	B	C	D	ENP	$\overline{ACLR}$	$\overline{SCLR}$	GND	$\overline{SLOAD}$	ENT	QD	QC	QB	QA	$\overline{TC}$	CCO	RCO	V_{CC}				
7 T_C = +25°C	Truth table tests 4/	3015 Fig.8	92	A 5/	B	B 5/	B	B	B	A	A	A	GND	A	A	L	H	H	H	B	L	L	4.5 V	See 6/			
			93	"	A	"	"	"	"	"	"	"	"	"	"	H	L	L	L	"	"	"	"	"			
			94	"	B	"	"	"	"	"	"	"	"	"	"	"	"	"	L	"	"	"	"	"			
			95	"	A	"	"	"	"	"	"	"	"	"	"	"	"	"	H	"	"	H	"	"			
			96	"	B	"	"	"	"	"	"	"	"	"	"	"	"	"	H	"	H	"	"	"			
			97	"	A	"	"	"	"	"	"	"	"	"	"	"	"	"	L	"	L	"	"	"			
			98	"	B	A	"	A	"	"	"	"	"	B	"	"	"	"	L	"	"	"	"	"			
			99	"	A	A	"	A	"	"	"	"	"	B	"	H	"	"	H	"	"	"	"	"			
			100	"	B	B	"	B	"	"	"	"	"	A	"	"	"	"	H	"	"	"	"	"			
			101	"	A	"	"	"	"	"	"	"	"	"	"	"	"	H	L	"	"	"	"	"			
			102	"	B	"	"	"	"	B	"	"	"	"	"	"	"	"	"	"	"	"	"	"			
			103	"	A	A	"	"	A	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"			
			104	"	B	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"			
			105	"	A	"	"	"	"	"	"	"	"	"	B	"	"	"	"	"	"	"	"	"			
			106	"	B	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"			
			107	"	A	"	"	"	"	A	"	"	"	"	"	"	"	"	"	"	"	"	"	"			
			108	"	B	"	"	"	"	"	"	"	"	"	A	"	"	"	"	"	"	"	"	"			
			109	"	A	"	A	A	"	"	"	"	"	"	"	"	"	"	H	"	"	"	"	"			
			110	"	B	"	"	"	"	"	"	"	"	"	"	"	"	"	H	"	"	"	"	"			
			111	"	A	"	"	"	"	"	"	"	"	"	H	"	L	"	L	"	"	"	"	"			
			112	"	B	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"			
			113	"	A	"	"	"	"	"	"	"	"	"	"	"	"	"	H	"	H	"	"	"			
			114	"	B	"	"	"	"	B	"	"	"	"	B	"	"	"	"	"	"	H	"	"			
			115	"	A	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	L	"	"			
			116	"	B	"	"	"	"	"	"	B	"	"	"	"	"	"	"	"	"	H	"	"			
			117	"	A	"	"	"	"	A	"	"	"	"	A	L	"	"	L	"	L	L	"	"			
			118	"	B	"	"	"	"	A	"	"	"	"	A	L	"	"	L	"	L	L	"	"			
8	Same tests, terminal conditions, and limits as for subgroup 7, except T_C = +125°C and -55°C.																										
9 T_C = +25°C	f_{MAX} 8/	3003 Fig. 8	119	4.5 V	IN					4.5 V	4.5 V	4.5 V	GND	4.5 V	4.5 V				OUT	GND			5.0 V	Q outputs	20		MHz
	t_{PLH1}	"	120	7/	IN	GND				"	"	"	"	"	"				OUT	"			"	CLK to QA	4	12	ns
		"	121	"	"		GND			"	"	"	"	"	"			OUT		"			"	CLK to QB	"	"	"
		"	122	"	"			GND		"	"	"	"	"	"		OUT			"			"	CLK to QC	"	"	"
		"	123	"	"				GND	"	"	"	"	"	"	OUT				"			"	CLK to QD	"	"	"
	t_{PHL1}	"	124	"	"	4.5 V				"	"	"	"	"	"				OUT	"			"	CLK to QA	5	18	"
		"	125	"	"		4.5 V			"	"	"	"	"	"			OUT		"			"	CLK to QB	"	"	"
		"	126	"	"			4.5 V		"	"	"	"	"	"		OUT			"			"	CLK to QC	"	"	"
		"	127	"	"				4.5 V	"	"	"	"	"	"	OUT				"			"	CLK to QD	"	"	"
	t_{PLH2}	"	128	"	"	GND	GND	GND	4.5 V	"	"	"	"	"	"					"		OUT	"	CLK to RCO	9	29	"
	t_{PHL2}	"	129	"	"	4.5 V	"	"	"	"	"	"	"	"	"					"		OUT	"	CLK to RCO	8	24	"
	t_{PLH3}	"	130	"	"	"	"	"	"	"	"	"	"	"	"					"	OUT		"	CLK to CCO	8	26	"
	t_{PHL3}	"	131	"	"	"	"	"	"	"	"	"	"	"	"					"	OUT		"	CLK to CCO	5	16	"
	t_{PLH4}	"	132	IN		4.5 V				"	"	"	"	"	"				OUT	"			"	$\overline{ALOAD}$ to QA	10	35	"
		"	133	"			4.5 V			"	"	"	"	"	"			OUT		"			"	$\overline{ALOAD}$ to QB	"	"	"
		"	134	"				4.5 V		"	"	"	"	"	"		OUT			"			"	$\overline{ALOAD}$ to QC	"	"	"
		"	135	"					4.5 V	"	"	"	"	"	"	OUT				"			"	$\overline{ALOAD}$ to QD	"	"	"

See footnotes at end of table.

Source: https://assist.dla.mil -- Downloaded: 2026-02-10T14:34Z
Check the source to verify that this is the current version before use.

TABLE III. Group A inspection for device type 04 - Continued.
Terminal conditions (pins not designated may be high ≥2.0 V, or low ≤0.8 V, or open).

Subgroup	Symbol	MIL-STD-883	Cases R,S,2 Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	Measured terminal	Limits		Unit
				$\overline{ALOAD}$	CLK	A	B	C	D	ENP	$\overline{ACLR}$	$\overline{SCLR}$	GND	$\overline{SLOAD}$	ENT	QD	QC	QB	QA	$\overline{G}$	CCO	RCO	V_{CC}		Min	Max	
9 T_C = +25°C	t_{PHL4}	3003 Fig. 8	136 137 138 139	IN " " "		GND	GND	GND	GND	4.5 V " " "	4.5 V " " "	4.5 V " " "	GND " " "	4.5 V " " "	4.5 V " " "	OUT	OUT	OUT	OUT	GND " " "			5.0 V " " "	$\overline{ALOAD}$ to QA $\overline{ALOAD}$ to QB $\overline{ALOAD}$ to QC $\overline{ALOAD}$ to QD	7 " " "	23 " " "	ns " " "
	t_{PLH5}	"	140	"		4.5 V	GND	GND	4.5 V	"	"	"	"	"	"					"		OUT	"	$\overline{ALOAD}$ to RCO	15	42	"
	t_{PHL5}	"	141	"		GND	"	"	GND	"	"	"	"	"	"					"		OUT	"	$\overline{ALOAD}$ to RCO	12	30	"
	t_{PLH6}	"	142	"		4.5 V	"	"	4.5 V	"	"	"	"	"	"					"	OUT		"	$\overline{ALOAD}$ to CCO	25	55	"
	t_{PHL6}	"	143	"		GND	"	"	GND	"	"	"	"	"	"					"	OUT		"	$\overline{ALOAD}$ to CCO	12	33	"
	t_{PLH7}	" " " "	144 145 146 147	GND " " "		IN	IN	IN	IN	" " " "	" " " "	" " " "	" " " "	" " " "	" " " "	OUT	OUT	OUT	OUT	" " " "			" " " "	A to QA B to QB C to QC D to QD	8 " " "	30 " " "	" " " "
	t_{PHL7}	" " " "	148 149 150 151	" " " "		IN	IN	IN	IN	" " " "	" " " "	" " " "	" " " "	" " " "	" " " "	OUT	OUT	OUT	OUT	" " " "			" " " "	A to QA B to QB C to QC D to QD	7 " " "	22 " " "	" " " "
	t_{PLH8}	"	152	7/		4.5 V	GND	GND	4.5 V	"	"	"	"	"	IN					"		OUT	"	ENT to RCO	5	16	"
	t_{PHL8}	"	153	"		"	"	"	"	"	"	"	"	"	"					"		OUT	"	ENT to RCO	4	14	"
	t_{PLH9}	"	154	"		"	"	"	"	"	"	"	"	"	"					"	OUT		"	ENT to CCO	12	32	"
	t_{PHL9}	"	155	"		"	"	"	"	"	"	"	"	"	"					"	"		"	ENT to CCO	4	12	"
	t_{PLH10}	"	156	"		"	"	"	"	IN	"	"	"	"	4.5 V					"	"		"	ENP to CCO	5	18	"
	t_{PHL10}	"	157	"		"	"	"	"	IN	"	"	"	"	"					"	"		"	ENP to CCO	4	12	"
	t_{PHL12}	" " " "	158 159 160 161	" " " "		"	4.5 V	4.5 V	4.5 V	4.5 V " " "	IN " " "	" " " "	" " " "	" " " "	" " " "	OUT	OUT	OUT	OUT	" " " "			" " " "	$\overline{ACLR}$ to QA $\overline{ACLR}$ to QB $\overline{ACLR}$ to QC $\overline{ACLR}$ to QD	7 " " "	22 " " "	" " " "
	t_{PZH}	" " " "	162 163 164 165	" " " "		4.5 V	4.5 V	4.5 V	4.5 V	" " " "	" " " "	" " " "	" " " "	" " " "	" " " "	OUT	OUT	OUT	OUT	IN " " "			" " " "	G to QA G to QB G to QC G to QD	5 " " "	19 " " "	" " " "
	t_{PZL}	" " " "	166 167 168 169	" " " "		GND	GND	GND	GND	" " " "	" " " "	" " " "	" " " "	" " " "	" " " "	OUT	OUT	OUT	OUT	" " " "			" " " "	G to QA G to QB G to QC G to QD	8 " " "	23 " " "	" " " "

Source: https://assist.dla.mil -- Downloaded: 2026-02-21T14:34Z
Check the source to verify that this is the current version before use.

TABLE III. Group A inspection for device type 04 - Continued.
Terminal conditions (pins not designated may be high ≥2.0 V, or low ≤0.8 V, or open).

Subgroup	Symbol	MIL-STD-883 Test no.	Cases R,S,Z	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	Measured terminal	Limits		Unit
				$\overline{ALOAD}$	CLK	A	B	C	D	ENP	$\overline{ACLR}$	$\overline{SCLR}$	GND	$\overline{SLOAD}$	ENT	QD	QC	QB	QA	$\overline{G}$	CCO	RCO	V_{CC}		Min	Max	
9 T_C = +25°C	t_{PHZ}	3003 Fig. 8	170	7/		4.5 V	4.5 V			4.5 V	4.5 V	4.5 V	GND	4.5 V	4.5 V					IN			5.0 V	G to QA	2	10	ns
		"	171	"		"	"	4.5 V		"	"	"	"	"	"					"			"	G to QB	"	"	"
		"	172	"		"	"	"		"	"	"	"	"	"		OUT			"			"	G to QC	"	"	"
		"	173	"		"	"	"	4.5 V	"	"	"	"	"	"	OUT				"			"	G to QD	"	"	"
	t_{PLZ}	"	174	"		GND	GND			"	"	"	"	"	"					"			"	G to QA	4	15	"
		"	175	"		"	"	GND		"	"	"	"	"	"		OUT	OUT	OUT	"			"	G to QB	"	"	"
		"	176	"		"	"	"		"	"	"	"	"	"		"			"			"	G to QC	"	"	"
		"	177	"		"	"	"	GND	"	"	"	"	"	"	OUT	"			"			"	G to QD	"	"	"
10 T_C = +125°C	f_{MAX} 8/	3003 Fig. 8																							18		MHz
	t_{PLH1}	"																							4	15	ns
	t_{PHL1}	"																							5	21	"
	t_{PLH2}	"																							9	35	"
	t_{PHL2}	"																							8	29	"
	t_{PLH3}	"																							8	31	"
	t_{PHL3}	"																							5	20	"
	t_{PLH4}	"	Same tests and terminal conditions as subgroup 9, except T_C = +125°C.																						10	38	"
	t_{PHL4}	"																							7	27	"
	t_{PLH5}	"																							15	55	"
	t_{PHL5}	"																							12	35	"
	t_{PLH6}	"																							25	65	"
	t_{PHL6}	"																							12	42	"
	t_{PLH7}	"																							8	35	"
	t_{PHL7}	"																							7	27	"
	t_{PLH8}	"																							5	20	"
	t_{PHL8}	"																							4	18	"

See footnotes at end of device type 04.

MIL-M-38510/381(USAF)

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:34Z
Check the source to verify that this is the current version before use.

TABLE III. Group A inspection for device type 04 - Continued.
Terminal conditions (pins not designated may be high ≥2.0 V, or low ≤0.8 V, or open).

Subgroup	Symbol	MIL-STD-883	Cases R,S,2 Test no.	1 ALOAD	2 CLK	3 A	4 B	5 C	6 D	7 ENP	8 ACLR	9 SCLR	10 GND	11 SLOAD	12 ENT	13 QD	14 QC	15 QB	16 QA	17 $\overline{G}$	18 CCO	19 RCO	20 V_{CC}	Measured terminal	Limits Min	Limits Max	Unit
10 T_C = +125°C	t_{PLH9}	3003 Fig. 8																							12	35	ns
	t_{PHL9}	"																							4	15	"
	t_{PLH10}	"																							5	22	"
	t_{PHL10}	"																							4	14	"
	t_{PHL12}	"	Same tests and terminal conditions as subgroup 9 except T_C = +125°C.																						7	28	"
	t_{PZH}	"																							5	24	"
	t_{PZL}	"																							8	28	"
	t_{PHZ}	"																							2	15	"
	t_{PLZ}	"																							4	20	"
11 T_C = -55°C	Same tests, terminal conditions, and limits as subgroup 10 except T_C = -55°C.																										

1/ Apply one clock pulse prior to test as follows: ---2.5 V min/5.5 V max, 0 V

2/ I_{IL} limits in μA are as follows:

Test	Circuit A	Circuit B	Circuit C
I_{IL}	0/-200	0/-100	0/-200

3/ Method 3011 shall be used, except the output voltage shall be as specified herein, and the output current shall be operating rather that short circuit current. The output conditions have been chosen to produce a current that closely approximates one-half of the true short circuit output current I_{OS}.

4/ Only a summary of attributes data is required.

5/ A = 3.0 V min; B = 0.0 V or GND.

6/ H >1.5 V; L <1.5 V.

7/ Apply one pulse prior to test as follows: 2.5 V min/5.5 V max, ---0 V

8/ The f_{MAX} minimum limit specified is the frequency of the input pulse. The output frequency shall be one-half of the input frequency.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:34Z
Check the source to verify that this is the current version before use.

TABLE III. Group A inspection for device type 05.
Terminal conditions (pins not designated may be high ≥2.0 V, or low ≤0.8 V, or open).

Subgroup	Symbol	MIL-STD-883	Cases R,S,2 Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	Measured terminal	Limits Min	Limits Max	Unit
				U/D	CLK	A	B	C	D	ENP	ACLR	SCLR	GND	LOAD	ENT	QD	QC	QB	QA	G	CCO	RCO	V_{CC}				
1 T_C = +25°C	V_{OH1}	3006	1 2	2.0 V 2.0 V	1/	0.8 V 0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	2.0 V	2.0 V	GND	0.8 V	0.8 V					0.8 V	-600 µA	-400 µA	4.5 V	RCO CCO	2.5 2.5		V
	V_{OH2}		3 4 5 6			2.0 V 0.8 V	2.0 V 0.8 V	2.0 V 0.8 V	2.0 V							-1 mA	-1 mA	-1 mA	-1 mA					QA QB QC QD	2.4		
	V_{OL1}	3007	7 8			2.0 V 2.0 V															4 mA	4 mA		RCO CCO		0.4	
	V_{OL2}		9 10 11 12			0.8 V 2.0 V	2.0 V 0.8 V 2.0 V 2.0 V	2.0 V 2.0 V 0.8 V 2.0 V	2.0 V 2.0 V 0.8 V							12 mA	12 mA	12 mA	12 mA					QA QB QC QD			
	V_{IC}		13 14 15 16 17 18 19 20 21 22 23 24	-18 mA	-18 mA	-18 mA	-18 mA	-18 mA	-18 mA	-18 mA	-18 mA	-18 mA		-18 mA	-18 mA					-18 mA				U/D CLK A B C D ENP ACLR SCLR LOAD ENT G		-1.5	
	I_{IL}	3009	25 26 27 28 29 30 31 32 33 34 35 36	0.4 V	0.4 V	0.4 V	0.4 V	0.4 V	0.4 V	0.4 V	0.4 V	0.4 V		0.4 V	0.4 V					0.4 V			5.5 V	U/D CLK A B C D ENP ACLR SCLR LOAD ENT G	2/	2/	2/
	I_{IH1}	3010	37 38 39 40 41 42 43 44 45 46 47 48	2.7 V	2.7 V	2.7 V	2.7 V	2.7 V	2.7 V	2.7 V	2.7 V	2.7 V		2.7 V	2.7 V					2.7 V				U/D CLK A B C D ENP ACLR SCLR LOAD ENT G		20	µA

See footnotes at end of device type 05.

Source: https://assist.dla.mil -- Downloaded: 2026-02-21T14:34Z
Check the source to verify that this is the current version before use.

TABLE III. Group A inspection for device type 05 - Continued.
Terminal conditions (pins not designated may be high ≥2.0 V, or low ≤0.8 V, or open).

Subgroup	Symbol	MIL-STD-883	Cases R,S,Z Test no.	1 U/$\overline{D}$	2 CLK	3 A	4 B	5 C	6 D	7 $\overline{ENP}$	8 $\overline{ACLR}$	9 $\overline{SCLR}$	10 GND	11 $\overline{LOAD}$	12 $\overline{ENT}$	13 QD	14 QC	15 QB	16 QA	17 $\overline{G}$	18 $\overline{CCO}$	19 $\overline{RCO}$	20 V_{CC}	Measured terminal	Limits Min	Limits Max	Unit
1 T_C = +25°C	I_{IH2}	3010	49	7.0 V									GND										5.5 V	U/$\overline{D}$		100	µA
			50		7.0 V								"										"	CLK		"	"
			51			7.0 V							"										"	A		"	"
			52				7.0 V						"										"	B		"	"
			53					7.0 V					"										"	C		"	"
			54						7.0 V				"										"	D		"	"
			55							7.0 V			"										"	$\overline{ENP}$		"	"
			56								7.0 V		"										"	$\overline{ACLR}$		"	"
			57									7.0 V	"										"	$\overline{SCLR}$		"	"
			58										"	7.0 V									"	$\overline{LOAD}$		"	"
			59										"		7.0 V								"	$\overline{ENT}$		"	"
			60										"						7.0 V				"	G		"	"
	I_O 3/	3011	61	5.5 V	1/	5.5 V	GND	GND	5.5 V	5.5 V	5.5 V	5.5 V	"	GND	5.5 V					GND		2.25 V	"	$\overline{RCO}$	-15	-70	mA
			62	"	"	"	GND	GND	5.5 V	"	"	"	"	"	"					"			"	$\overline{CCO}$	-15	-70	"
			63	"	"	"	5.5 V			"	"	"	"	"	"				2.25 V	"	2.25 V			"	QA	-30	-112	"
			64	"	"	"		5.5 V		"	"	"	"	"	"			2.25 V		"			"	QB	"	"	"
			65	"	"	"			5.5 V	"	"	"	"	"	"	2.25 V	2.25 V			"			"	QC	"	"	"
			66	"	"	"				"	"	"	"	"	"					"			"	QD	"	"	"
	I_{OZL}		67	2.0 V	"	2.0 V				2.0 V	2.0 V	2.0 V	"	0.8 V	2.0 V				0.4 V	2.0 V			"	QA		-20	µA
			68	"	"	"	2.0 V			"	"	"	"	"	"			0.4 V		"			"	QB		"	"
			69	"	"	"		2.0 V		"	"	"	"	"	"		0.4 V			"			"	QC		"	"
			70	"	"	"			2.0 V	"	"	"	"	"	"	0.4 V				"			"	QD		"	"
	I_{OZH}		71	"	"	10.8 V				"	"	"	"	"	"					"			"	QA		20	"
			72	"	"	"	10.8 V			"	"	"	"	"	"				2.7 V	"			"	QB		"	"
			73	"	"	"		10.8 V		"	"	"	"	"	"		2.7 V	2.7 V		"			"	QC		"	"
			74	"	"	"			10.8 V	"	"	"	"	"	"	2.7 V				"			"	QD		"	"
	I_{CCH}	3005	75	4.5 V	"	4.5 V	4.5 V	4.5 V	4.5 V	4.5 V	4.5 V	4.5 V	"	GND	4.5 V					GND			"	V_{CC}		26	mA
	I_{CCL}	"	76	GND	"	GND	GND	GND	GND	GND	GND	GND	"	"	GND					GND			"	V_{CC}		32	"
	I_{CCZ}	"	77	GND	"	GND	GND	GND	GND	GND	4.5 V	4.5 V	"	"	GND					4.5 V			"	V_{CC}		32	"
2	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = +125°C and V_{IC} tests are omitted.																										
3	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = -55°C and V_{IC} tests are omitted.																										
7 T_C = +25°C	Truth table tests 4/	3014	78	A 5/	B 5/	A	A	A	A	B	B	A	GND	A	B	L 6/	L	L	L	B	H 6/	H	4.5 V	See 6/			
			79	"	B	"	B	B	B	"	A	"	"	B	"	"	"	"	H	"	"	"	"					
			80	"	A	"	"	A	A	"	B	"	"	A	"	"	"	H	L	"	"	"	"					
			81	"	B	B	"	B	B	"	A	"	"	B	"	"	"	L	H	"	"	"	"					
			82	"	A	A	"	A	A	"	B	"	"	A	"	"	"	H	L	"	"	"	"					
			83	"	B	B	"	B	B	"	A	"	"	B	"	"	"	L	H	"	"	"	"					
			84	"	A	A	"	A	A	"	B	"	"	A	"	"	"	H	L	"	"	"	"					
			85	"	B	B	"	B	B	"	A	"	"	B	"	"	"	L	H	"	"	"	"					
			86	"	A	A	"	A	A	"	B	"	"	A	"	"	"	H	L	"	"	"	"					
			87	"	B	B	"	B	B	"	A	"	"	B	"	"	"	L	H	"	"	"	"					
			88	"	A	A	"	A	A	"	B	"	"	A	"	"	"	H	L	"	"	"	"					
			89	"	B	B	"	B	B	"	A	"	"	B	"	"	"	L	H	"	"	"	"					
			90	"	A	A	"	A	A	"	B	"	"	A	"	"	"	H	L	"	"	"	"					
			91	"	B	B	"	B	B	"	A	"	"	B	"	"	"	L	H	"	"	"	"					

See footnotes at end of device type 05.

Source: https://assist.dla.mil -- Downloaded: 2026-02-21T14:34Z
Check the source to verify that this is the current version before use.

TABLE III. Group A inspection for device type 05 - Continued.
Terminal conditions (pins not designated may be High ≥2.0 V, or low ≤0.8 V, or open).

Subgroup	Symbol	MIL-STD-883	Cases R,S,2 Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	Measured terminal	Limits Min	Limits Max	Unit
				U/D	CLR	A	B	C	D	ENP	ACLR	SCLR	GND	LOAD	ENT	QD	QC	QB	QA	G	CCO	RCO	V_{CC}				
7 T_C = +25°C	Truth table tests	3014	92	A 5/	A	B 5/	B	B	B	B	A	A	GND	A	B	L 6/	H	H	H	B	H 6/	H	4.5 V	See 6/			
	"	"	93	"	B	"	"	"	"	A	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	
	"	"	94	"	A	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	
	"	"	95	"	B	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	
	"	"	96	"	A	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	
	"	"	97	"	B	"	"	"	"	"	"	"	"	"	"	A	"	"	"	"	"	"	"	"	"	"	"	
	"	"	98	"	A	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	
	"	"	99	"	B	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	
	"	"	100	"	A	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	
	"	"	101	"	B	"	"	"	"	"	"	"	"	"	"	B	"	"	"	"	"	"	"	"	"	"	"	
	"	"	102	"	A	"	"	"	"	"	"	"	"	"	"	"	H	L	L	L	"	"	"	"	"	"	"	
	"	"	103	"	B	"	"	"	"	"	"	"	"	"	"	"	"	"	"	H	"	"	"	"	"	"	"	
	"	"	104	"	A	"	"	"	"	"	"	"	"	"	"	"	"	"	"	H	"	"	L	"	"	"	"	
	"	"	105	"	B	"	"	"	"	"	"	"	"	"	"	"	"	"	"	H	"	L	L	"	"	"	"	
	"	"	106	"	A	"	"	"	"	"	"	"	"	"	"	"	L	"	"	L	"	H	H	"	"	"	"	
	"	"	107	"	B	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	
	"	"	108	"	A	"	A	A	A	"	"	"	"	B	"	"	"	"	"	"	"	"	"	"	"	"	"	
	"	"	109	"	B	"	"	"	"	"	"	"	"	B	"	"	"	H	H	H	"	"	"	"	"	"	"	
	"	"	110	B	A	A	"	"	"	"	"	"	"	A	"	"	"	"	"	H	"	"	"	"	"	"	"	
	"	"	111	"	B	"	"	"	"	"	"	"	"	"	"	"	"	"	"	L	"	"	"	"	"	"	"	
	"	"	112	"	A	"	"	"	"	"	"	"	"	"	"	"	"	"	"	L	"	"	"	"	"	"	"	
	"	"	113	"	B	"	"	"	"	"	"	"	"	"	"	"	"	"	"	H	"	"	"	"	"	"	"	
	"	"	114	"	A	"	"	"	"	"	"	"	"	"	"	"	"	"	"	H	"	"	"	"	"	"	"	
	"	"	115	"	B	"	"	"	"	"	"	"	"	"	"	"	"	"	"	L	"	"	"	"	"	"	"	
	"	"	116	"	A	"	"	"	"	"	"	"	"	"	"	"	"	L	H	H	"	"	"	"	"	"	"	
	"	"	117	"	B	"	"	"	"	"	"	"	"	"	"	"	"	"	"	L	"	"	"	"	"	"	"	
	"	"	118	"	A	"	"	"	"	"	"	"	"	"	"	"	"	"	"	L	"	"	"	"	"	"	"	
	"	"	119	"	B	"	"	"	"	"	"	"	"	"	"	"	"	"	"	H	"	"	"	"	"	"	"	
	"	"	120	"	A	"	"	"	"	"	"	"	"	"	"	"	"	"	"	H	"	"	"	"	"	"	"	
	"	"	121	"	B	"	"	"	"	"	"	"	"	"	"	"	"	"	"	L	"	"	"	"	"	"	"	
	"	"	122	"	A	"	"	"	"	"	"	"	"	"	"	"	"	"	"	H	"	"	"	"	"	"	"	
	"	"	123	"	B	"	"	"	"	"	"	"	"	"	"	"	"	"	"	L	"	"	L	"	"	"	"	
	"	"	124	"	A	"	"	"	"	"	"	"	"	"	"	"	"	"	"	L	"	"	L	"	"	"	"	
	"	"	125	"	B	"	"	"	"	"	"	"	"	"	"	"	H	"	"	H	"	"	H	"	"	"	"	
	"	"	126	"	A	"	"	"	"	"	"	"	"	"	"	"	"	"	"	H	"	"	"	"	"	"	"	
	"	"	127	"	B	"	"	"	"	"	"	"	"	"	"	"	"	"	"	L	"	"	"	"	"	"	"	
	"	"	128	"	A	"	"	"	"	"	"	"	"	"	"	"	"	"	"	L	"	"	"	"	"	"	"	
	"	"	129	"	B	"	"	"	"	"	"	"	"	"	"	"	L	H	H	H	"	"	"	"	"	"	"	
	"	"	130	"	A	"	"	"	"	"	"	"	"	"	"	"	"	H	H	H	"	"	"	"	"	"	"	
	"	"	131	"	B	"	"	"	"	"	"	"	"	B	"	"	"	L	L	L	"	"	"	"	"	"	"	
	"	"	132	"	B	"	"	"	"	"	"	"	"	"	"	"	"	"	"	L	"	"	"	"	"	"	"	
8	Same tests, terminal conditions, and limits as for subgroup 7, except T_C = +125°C and -55°C.																										
9 T_C = +25°C	f_{MAX}	3003 Fig. 9	133	4.5 V GND	IN					GND	4.5 V	4.5 V	GND	4.5 V	GND				OUT OUT	GND			5.0 V	() outputs () outputs	20		MHz
			134																								
	t_{PLH1}	"	135	"	"	4.5 V	4.5 V	4.5 V	4.5 V	"	"	"	"	GND	"	OUT	OUT	OUT	OUT	"			"	CLK to QA	4	13	ns
		"	136	"	"					"	"	"	"	"	"					"			"	CLK to QB	"	"	"
		"	137	"	"					"	"	"	"	"	"					"			"	CLK to QC	"	"	"
		"	138	"	"					"	"	"	"	"	"					"			"	CLK to QD	"	"	"
	t_{PHL1}	"	139	"	"	GND	GND	GND	GND	"	"	"	"	"	"	OUT	OUT	OUT	OUT	"			"	CLK to QA	7	16	"
		"	140	"	"					"	"	"	"	"	"					"			"	CLK to QB	"	"	"
		"	141	"	"					"	"	"	"	"	"					"			"	CLK to QC	"	"	"
		"	142	"	"					"	"	"	"	"	"					"			"	CLK to QD	"	"	"
	t_{PLH2}	"	143	"	"	4.5 V	GND	GND	4.5 V	"	"	"	"	"	"					"		OUT	"	CLK to $\overline{RCO}$	12	28	"
	t_{PHL2}	"	144	"	"	GND	"	"	GND	"	"	"	"	"	"					"		OUT	"	CLK to $\overline{RCO}$	10	19	"

See footnotes at end of device type 05.

Source: https://assist.dla.mil -- Downloaded: 2026-02-21T14:34Z
Check the source to verify that this is the current version before use.

TABLE III. Group A inspection for device type 05 - Continued.
Terminal conditions (pins not designated may be high ≥2.0 V, or low ≤0.8 V, or open).

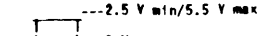
Subgroup	Symbol	MIL-STD-883	Cases R,S,2 Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	Measured terminal	Limits		Unit
				U/D	CLK	A	B	C	D	ENP	ACLR	SCLR	GND	LOAD	ENT	QD	QC	QB	QA	G	CCO	RCO	V_{CC}		Min	Max	
9 T_C = +25°C	t_{PLH3}	3003 Fig. 9	145	4.5 V	IN	4.5 V	GND	GND	4.5 V	GND	4.5 V	4.5 V	GND	GND	GND					GND	OUT		5.0 V	CLK to $\overline{CCO}$	5	13	ns
	t_{PHL3}	"	146	"	IN	"	"	"	"	"	"	"	"	"	"					"	OUT		"	CLK to $\overline{CCO}$	6	25	"
	t_{PLH8}	"	147	"		"	"	"		4.5 V	"	"	"	4.5 V	IN					"		OUT	"	ENT to $\overline{RCO}$	6	15	"
	t_{PHL8}	"	148	"						"	"	"	"	"	"					"		OUT	"	ENT to $\overline{RCO}$	4	13	"
	t_{PLH9}	"	149	"						"	"	"	"	"	"					"	OUT		"	ENT to $\overline{CCO}$	5	13	"
	t_{PHL9}	"	150	"						"	"	"	"	"	"					"	"		"	ENT to $\overline{CCO}$	9	23	"
	t_{PLH10}	"	151	"						IN	"	"	"	"	GND					"	"		"	ENP to $\overline{RCO}$	4	12	"
	t_{PHL10}	"	152	"						IN	"	"	"	"	"					"	"		"	ENP to $\overline{CCO}$	5	14	"
	t_{PLH11}	"	153	IN						GND	"	"	"	"	"					"		OUT	"	U/$\overline{D}$ to $\overline{RCO}$	9	23	"
	t_{PHL11}	"	154	IN						"	"	"	"	"	"					"		OUT	"	U/$\overline{D}$ to $\overline{RCO}$	9	19	"
	t_{PHL12}	"	155	4.5 V	1/	4.5 V	4.5 V	4.5 V		"	IN	"	"	GND	"					"			"	ACLR to QA	"	20	"
		"	156	"	"					"	"	"	"	"	"					"			"	ACLR to QB	"	"	"
		"	157	"	"					"	"	"	"	"	"					"			"	ACLR to QC	"	"	"
		"	158	"	"				4.5 V	"	"	"	"	"	"	OUT	OUT	OUT	OUT	"			"	ACLR to QD	"	"	"
	t_{PZH}	"	159	"	"	4.5 V	4.5 V	4.5 V		"	4.5 V	"	"	"	"					IN			"	G to QA	6	18	"
		"	160	"	"					"	"	"	"	"	"					"			"	G to QB	"	"	"
		"	161	"	"					"	"	"	"	"	"					"			"	G to QC	"	"	"
		"	162	"	"				4.5 V	"	"	"	"	"	"	OUT	OUT	OUT	OUT	"			"	G to QD	"	"	"
	t_{PZL}	"	163	"	"	GND	GND	GND		"	"	"	"	"	"					"			"	G to QA	11	24	"
		"	164	"	"					"	"	"	"	"	"					"			"	G to QB	"	"	"
		"	165	"	"					"	"	"	"	"	"					"			"	G to QC	"	"	"
		"	166	"	"				GND	"	"	"	"	"	"	OUT	OUT	OUT	OUT	"			"	G to QD	"	"	"
	t_{PHZ}	"	167	"	"	4.5 V	4.5 V	4.5 V		"	"	"	"	"	"					"			"	G to QA	1	8	"
		"	168	"	"					"	"	"	"	"	"					"			"	G to QB	"	"	"
		"	169	"	"					"	"	"	"	"	"					"			"	G to QC	"	"	"
		"	170	"	"				4.5 V	"	"	"	"	"	"	OUT	OUT	OUT	OUT	"			"	G to QD	"	"	"
	t_{PLZ}	"	171	"	"	GND	GND	GND		"	"	"	"	"	"					"			"	G to QA	3	13	"
		"	172	"	"					"	"	"	"	"	"					"			"	G to QB	"	"	"
		"	173	"	"					"	"	"	"	"	"					"			"	G to QC	"	"	"
		"	174	"	"				GND	"	"	"	"	"	"	OUT	OUT	OUT	OUT	"			"	G to QD	"	"	"

Source: https://assist.dla.mil -- Downloaded: 2026-02-21T14:34Z
Check the source to verify that this is the current version before use.

TABLE III. Group A inspection for device type 05 - Continued.
Terminal conditions (pins not designated may be high ≥2.0 V, or low ≤0.8 V, or open).

Subgroup	Symbol	MIL-STD-883	Cases R,S,X Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	Measured terminal	Limits Min	Limits Max	Unit
				U/D	CLK	A	B	C	D	ENP	ACLR	SCLR	GND	ENAU	ENT	QD	QC	QB	QA	G	CCU	RCU	V_{CC}				
10 T_C = +125°C	f_{MAX} 7/																								20		MHz
	t_{PLH1}	"																							4	17	ns
	t_{PHL1}	"																							7	18	"
	t_{PLH2}	"																							12	31	"
	t_{PHL2}	"																							10	22	"
	t_{PLH3}	"																							5	15	"
	t_{PHL3}	"																							6	30	"
	t_{PLH8}	"	Same tests, and terminal conditions as subgroup 9 except T_C = +125°C.																						6	17	"
	t_{PHL8}	"																							4	17	"
	t_{PLH9}	"																							5	15	"
	t_{PHL9}	"																							9	28	"
	t_{PLH10}	"																							4	14	"
	t_{PHL10}	"																							5	17	"
	t_{PLH11}	"																							9	25	"
	t_{PHL11}	"																							9	23	"
	t_{PHL12}	"																							9	22	"
	t_{PZH}	"																							6	21	"
	t_{PZL}	"																							11	29	"
	t_{PHZ}	"																							1	8	"
	t_{PLZ}	"																							3	19	"
11	Same tests, terminal conditions, and limits as for subgroup 10, except T_C = -55°C.																										

1/ Apply one clock pulse prior to test as follows:



2/ I_{IL} limits in μA are as follows:

Test	Circuit A	Circuit B	Circuit C
I_{IL}	0/-200	0/-100	0/-200

3/ Method 3011 shall be used, except the output voltage shall be as specified herein, and the output current shall be operating rather that short circuit current. The output conditions have been chosen to produce a current that closely approximates one-half of the true short circuit output current I_{OS}.
4/ Only a summary of attributes data is required.
5/ A = 3.0 V min; B = 0.0 V or GND.
6/ H >1.5 V; L <1.5 V.
7/ The f_{MAX} minimum limit specified is the frequency of the input pulse. The output frequency shall be one-half of the input frequency.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:34Z
Check the source to verify that this is the current version before use.

5. PACKAGING.

5.1 Packaging requirements. The requirements for packaging shall be as specified in MIL-M-38510.

6. NOTES

6.1 Notes. The notes specified in MIL-M-38510 are applicable to this specification.

6.2 Intended use. Microcircuits conforming to this specification are intended for original equipment design applications and logistic support of existing equipment.

6.3 Ordering data. The acquisition document should specify the following:

a. Complete part number (see 1.2).

b. Requirements for delivery of one copy of the quality conformance inspection data pertinent to the device inspection lot to be supplied with each shipment by the device manufacturer, if applicable.

c. Requirements for certificate of compliance, if applicable.

d. Requirements for notification of change of product or process to the contracting activity in addition to notification to the qualifying activity, if applicable.

e. Requirements for failure analysis (including required test condition of method 5003 of MIL-STD-883), corrective action and reporting of results, if applicable.

f. Requirements for product assurance options.

g. Requirements for special carriers, lead lengths, or lead forming, if applicable. These requirements shall not affect the part number. Unless otherwise specified, these requirements shall not apply to direct purchase by or direct shipment to the Government.

h. Requirements for "JAN" marking.

6.4 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein are defined in MIL-M-38510, MIL-STD-1331, and as follows:

GND	Ground zero voltage potential.
I_{IN}	Current flowing into an input terminal.
V_{IN}	Voltage level at an input terminal.

6.5 Logistic support. Lead materials and finishes (see 3.3) are interchangeable. Unless otherwise specified, microcircuits acquired for Government logistic support will be acquired to device class B (see 1.2.2), lead material and finish C (see 3.3). Longer length leads and lead forming shall not affect the part number.

6.6 Substitutability. The cross-reference information below is presented for the convenience of users. Microcircuits covered by this specification will functionally replace the listed generic-industry type. Generic-industry microcircuit types may not have equivalent operational performance characteristics across military temperature ranges or reliability factors equivalent to MIL-M-38510 device types and may have slight physical variations in relation to case size. The presence of this information shall not be deemed as permitting substitution of generic-industry types for MIL-M-38510 types or as a waiver of any of the provisions of MIL-M-38510.

Military device type	Generic-industry type
01	54ALS160
02	54ALS162
03	54ALS168
04	54ALS560
05	54ALS568

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:34Z
Check the source to verify that this is the current version before use.

6.6 Manufacturers' designators. Manufacturers' circuits which form a part of this specification are designated with an "X" as shown in table IV herein.

TABLE IV. Manufacturers' designations.

Device type	Circuit		
	A	B	C
	Texas Instrument	Motorola	National
01	X		
02	X		
03	X		
04	X		
05	X		

Custodian:
Air Force - 17

Review activities:
Air Force - 11, 19, 85, 99
DLA - ES

Agent:
DLA - ES

Preparing activity:
Air Force - 17

(Project 5962-F644)

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:34Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

DEPARTMENT OF THE AIR FORCE

Rome Air Development Center
Griffiss AFB, NY 13441

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO. 73236 WASHINGTON D. C

POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE AIR FORCE

Commander
Rome Air Development Center
Attn: RBE-2
Griffiss AFB, NY 13441

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:34Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-M-38510/381(USAF)	MICROCIRCUITS, DIGITAL, ADVANCED LOW-POWER SCHOTTKY, TTL, DECADE COUNTERS, MONOLITHIC SILICON

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION (Mark one)
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER (Specify): _____
b. ADDRESS (Street, City, State, ZIP Code)	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER (Last, First, MI) – Optional	b. WORK TELEPHONE NUMBER (Include Area Code) – Optional
c. MAILING ADDRESS (Street, City, State, ZIP Code) – Optional	8. DATE OF SUBMISSION (YYMMDD)

DD FORM 82 MAR 1426

PREVIOUS EDITION IS OBSOLETE.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:34Z
Check the source to verify that this is the current version before use.